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Fairness in Negative Influence Minimization in Social Networks

Unnikrishnan, Deepthy

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EINDHOVEN UNIVERSITY OF TECHNOLOGY

Department of Mathematics and Computer Science
Data Mining Research Group

Fairness in Negative Influence Minimization in Social Networks

Master's Thesis

Deepthy Unnikrishnan

Supervisors:
prof. dr. Mykola Pechenizkiy
dr. Akrati Saxena

Assessment committee:
prof. dr. Mykola Pechenizkiy
dr. Akrati Saxena
dr. Felix Mannhardt

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Abstract

The proliferation of fake news and misinformation campaigns in social networks has emerged as a pressing concern, inflicting substantial harm on the public at large. Truth campaigning, a popular technique to combat fake news, involves finding a set of users in the network to spread the truth or facts to counter the misinformation that is being spread. The Negative Influence Minimization (NIM) problem aims at finding an optimal set of truth campaign initiators in a network that will minimize the spread of negative information spread in the network when the misinformation sources are known. NIM is a well-studied topic in network science and data mining. However, existing solutions to the NIM problem often overlook individual fairness. Without a fairness consideration, NIM algorithms tend to ignore poorly connected nodes, benefiting only a subset of the users in the network.

To address this gap, this thesis focuses on individual fairness in NIM. Multiple fairness notions are examined, and it is determined that a fair solution should not overlook or neglect any user in the network, given the detrimental effects of fake news. Drawing from social welfare theory, fairness-aware truth campaigning techniques are proposed to identify a set of users that improves the likelihood of all users in the network not being influenced by misinformation, while ensuring that no user is overlooked or neglected.

Comparisons are made between the proposed fairness-aware methods, fairness-agnostic methods, and group-fairness-based NIM methods. Results obtained from real-world and synthetic networks demonstrate the unfairness of existing methods. The trade-off between fairness and effectiveness is analyzed, revealing that the proposed methods achieve superior fairness, while incurring minimal costs in certain cases. Additionally, comparisons with newly introduced individual fairness benchmarks indicate that the proposed methods outperform them in terms of both performance and computational efficiency in most of the cases.

Overall, this thesis contributes to the advancement of fair mitigation strategies against fake news and misinformation campaigns in social networks by highlighting the importance of individual fairness and providing novel fairness-aware truth campaigning techniques and fairness evaluation metrics.

Preface

I would like to begin this preface by expressing my deepest gratitude to my first supervisor, Dr. Akrati Saxena, for her invaluable guidance and unwavering support throughout my thesis journey. Her timely feedback and insightful suggestions have been instrumental in shaping my research and nudging me in the right directions while allowing me the freedom to explore my own ideas. I am truly grateful for her mentorship and dedication.

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I would like to take this opportunity to extend my heartfelt appreciation to my family and friends who have been my pillars of strength throughout this journey. My parents and sister have consistently been my biggest cheerleaders, and I am grateful for their constant encouragement. I am greatly thankful to my husband, whose support and belief in me have been a constant source of motivation. His encouragement and understanding have played a pivotal role in helping me overcome challenges and stay focused on my goals. I am truly blessed to have him by my side.

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Chapter 1

Introduction

This chapter serves as an introduction to the problem of fairness in Negative Influence Minimization (NIM) in social networks. It is organized as follows: In Section 1.1, we establish the context of the problem. Section 1.2 formally explains the NIM problem. The motivation behind pursuing fairness in NIM is discussed in Section 1.3, emphasizing the reasons driving this research endeavor. Section 1.4 focuses on the research questions that will guide our investigation and exploration of fair NIM. In Section 1.5, we outline the contributions made by this thesis, shedding light on the novel insights it brings to the field of NIM. Finally, in Section 1.6, we provide an overview of the thesis structure, outlining the organization of the subsequent chapters.

1.1 Context

In the year 2004, a significant milestone was achieved in the realm of digital connectivity as Facebook, an online platform, was established and swiftly amassed one million users within its initial year. However, it was only the beginning of a transformative era, as subsequent years witnessed the rise of several other influential social platforms. In the years that followed, other social platforms such as WhatsApp, YouTube, Instagram, Twitter, and Reddit emerged, quickly becoming an integral part of our daily lives. These platforms have enabled us to interact with individuals and communities across the globe, fostering a sense of shared experiences and collective identity.

With every one-in-three people in this world using social media [43], it has become the fastest means to spread information. This has made social media a favorite for companies to build their brand and advertise their products. The ability to reach a specific audience, engage with customers, and measure campaign performance in real-time are some of the advantages that make social media advertising an appealing choice for businesses. In 2022, the ad spending on social media was over 200 billion USD, and it is expected to exceed 300 billion dollars by 2024 [2].

Further, social media became an effective emergency management tool in times of natural calamities [44, 6, 69]. They were used to connect survivors with rescue teams, send updates about available food, shelter, and medical assistance, and reach out for help when traditional communication lines failed. The Earthquake in Nepal (2015) and Typhoon Haiyan (2013) are some examples. At the peak of the COVID-19 crisis, government and international organizations like WHO used social media to spread awareness about the disease and keep people updated about the known best practices at a pace faster than the spread of the epidemic.

The ease of spreading information has made social media an easy target for malicious users to spread misinformation. Buchanan [10] defines misinformation as false information shared by individuals who believe it to be true. Disinformation is the misinformation that is shared by individuals knowing that it is not true. We follow Ecker et al. [16] and use the term "misinformation" as a comprehensive term encompassing any information that is subsequently proven to be false.

Garimella et al. [21] and Quattrociocchi et al. [46] studied how echo chambers contribute to

the isolation of individuals from information that contradicts their existing beliefs or perspectives. Echo chambers, or filter bubbles, are social environments on online platforms where individuals are predominantly exposed to information and viewpoints that align with their own. These echo chambers emerge as a result of the algorithms employed by social networks, which prioritize content based on users' preferences and past engagement, as well as the inherent structure of the social network itself. Consequently, users are rarely exposed to diverse perspectives or dissenting opinions, fostering a sense of isolation from information that challenges their existing beliefs or perspectives. In parallel, confirmation bias, another influential factor, describes the tendency of individuals to actively seek out and interpret information that confirms their preconceived notions, while dismissing or downplaying contradictory evidence. This cognitive bias effectively acts as a filter, reinforcing individuals' existing opinions and making it arduous to entertain alternative perspectives. The interplay between confirmation bias and echo chambers significantly shapes the way people consume information, further exacerbating the tendency to gravitate towards reinforcing one's pre-existing beliefs.

During the COVID-19 pandemic, a copious amount of misinformation was circulating on social media. Studies show the profound impacts that fake news can have on individuals. Notably, Forati et al. [20] conducted a study focusing on the spread of COVID-19-related misinformation on Twitter. Their findings revealed a significant correlation between the geographic origin of misleading tweets and subsequent surges in COVID-19 infections and fatalities in those areas several weeks later. This connection highlights the consequences of misinformation, as it directly contributes to the worsening of public health outcomes. Furthermore, Greene et al. [24] conducted a study demonstrating the efficacy of misinformation in influencing individuals' intentions. Through exposing test subjects to COVID-19 misinformation, they were able to establish a direct link between the dissemination of false information and subsequent shifts in individuals' attitudes and behaviors. This study provides compelling evidence of the impact of misinformation on shaping public opinion and decision-making processes. In addition, Wilson et al. [71] investigated the relationship between online misinformation campaigns and COVID-19 vaccination rates, revealing a concerning trend of declining vaccination rates coupled with an increase in negative discussions surrounding the vaccine. These studies highlight the tangible negative impact of fake news on social networks, emphasizing the urgent need to address and counteract the spread of false information.

Moreover, real-world incidents such as the notorious Pizzagate episode[1] serve as stark reminders of the potential dangers inherent in misinformation. This alarming event, which unfolded in the United States, demonstrates how unfounded conspiracy theories propagated through social networks can escalate into community conflicts and even incite acts of violence. Such instances underscore the gravity of the situation and the critical importance of combating misinformation. Therefore, we see that the detrimental impacts of fake news extend to every individual who is exposed to it.

There are multiple aspects to combating fake news. Mainly, they are Fake News Detection and Fake News Mitigation [57]. Fake News detection is commonly modeled as a classification task that uses features such as the content, the source or origin of the news, the propagation patterns, and/or the users' engagement with the news to predict if the given news is real or fake [56]. Several approaches have been proposed which guarantee high precision. In this study, our focus is on how to limit the spread of misinformation once it has been detected.

1.2 Negative Influence Minimization problem

The propagation of information within a network follows a cascading process, wherein users share it with their connections, resulting in widespread dissemination. Fake News Mitigation deals with containing the spread of fake news once it has started spreading. This challenge is widely investigated in the field of Network Analysis, often referred to as the Influence Minimization problem or the Influence Blocking Maximization problem. We will refer to it here on as the Negative Influence Minimization (NIM) problem.

[1] https://shorturl.at/chjE0.

Truth campaigning is one of the various approaches to tackling NIM. Truth Campaigning involves selecting a set of nodes to spread the truth to combat the spread of fake news. The goal is to minimize the number of users that are influenced by misinformation. Given a network and the set of users in the network that start spreading misinformation, the NIM problem aims at identifying an optimal set of users who can initiate the dissemination of truth, thereby reducing the overall impact of fake news.

The NIM problem can be formally defined as a constrained optimization problem on graphs. Consider a directed graph $G = (V, E)$ representing a social network, where V represents the set of users in the network and E represents the edges denoting interactions between users. Let $M \subset V$ be the initial set of users that start the spread of misinformation and are detected after a time delay r. If a user in the network gets infected by the misinformation campaign started by M, we say that the node is influenced by M. Let $\pi(M)$ be the set of users influenced by M. The goal is to find a set $T \subset V \backslash M$ of k nodes to start spreading truth so that the expected number of users influenced by M is minimized, where k is the available budget for T and $\pi(M|T)$ is the users in G infected by misinformation in the presence of truth campaign T.

$$\begin{aligned} &\underset{T}{\text{minimize}} \quad \mathbb{E}[|\pi(M|T)|] \\ &\text{subject to} \\ &\quad T \subset V \setminus M, \\ &\quad |T| \leq k \end{aligned} \tag{1.1}$$

In other words, NIM involves finding the optimal combination of items to maximize the total value of the items while staying within a certain budget. This is similar to the classic knapsack problem, where nodes with different weights (in this case, effectiveness) must be selected such that they collectively maximize their total value while staying within a given capacity constraint (in this case, the budget).

Various techniques have been developed to solve the NIM problem, each focusing on specific aspects and settings. Some works have emphasized scalability, aiming to handle large-scale networks efficiently and effectively [72]. Others have addressed temporal aspects, taking into account the time delay associated with each node's sharing of a rumor as well as a deadline, beyond which the rumor naturally loses its effect [61]. Additionally, community-based approaches have been explored, considering the structure and dynamics of communities within the network [73].

1.3 Motivation for Fair NIM

Surprisingly, despite the vast amount of work in this area, there has been no prior work that considers individual fairness in NIM. Individual fairness is fundamental for ensuring fairness in decision-making processes that directly impact people's lives. In the context of NIM, where the goal is to protect individuals from fake news and misinformation, considering individual fairness becomes particularly significant.

An important aspect to take into account while considering fairness is the risk faced by individuals who are influenced by misinformation. In the case of NIM, this would be considering the detrimental effects that an individual may have if they are influenced by misinformation. For instance, when misinformation about COVID-19 vaccines spreads on social media, individuals who believe the misinformation and decide not to take the vaccine put themselves at an increased risk of infection, potentially harming both themselves and others. Thus, the likelihood of a person in the network being shielded from misinformation, which we refer to as the save probability, becomes an important measure to consider fairness of a solution.

The NIM problem, in its traditional form, does not take this likelihood into account. We will see in section 2.3 that the NIM objective could widen disparities in the network, with its solution only improving the save probability of a subgroup of the network. This has motivated the research on fairness-aware fake news mitigation.

While the actual risk faced by the person may be impacted by various factors, we consider here as risk the chances of the person being influenced by misinformation. Other factors are beyond the scope of this study and are left for future work.

1.4 Research Questions

The main research question that will guide our investigation of individual fairness in negative influence minimization is:

RQ1: Can we develop an algorithm to achieve fairness in Negative Influence Minimization and measure the fairness of the algorithm?

To answer this question, we study the below sub-questions:

- In the context of NIM, what is a fair solution?
- Are the existing solutions unfair in this regard?
- How can fairness be achieved for NIM?
- What are the appropriate fairness metrics to measure the fairness of the proposed solution?

The following research question helps us recognize the effectiveness of the algorithm in terms of limiting the number of users infected compared to fairness-agnostic methods and answer the question of whether fairness comes at a cost.

RQ2 : What is the trade-off between the fairness and effectiveness of the algorithm?

The next research question helps us explore the influence of different topological features on the performance of the proposed method. This investigation will provide insights into the method's adaptability and effectiveness in diverse network structures.

RQ3 : Does the performance of the algorithm vary with different network structures?

With these questions, we aim to explore and address key aspects of individual fairness in NIM. By investigating the concept of fairness in the context of NIM, evaluating the trade-off between fairness and effectiveness, and examining the influence of network structures on algorithm performance, we can gain a comprehensive understanding of the challenges and opportunities in achieving fairness in NIM.

1.5 Contribution

This thesis addresses the pressing issue of fairness in fake news and misinformation mitigation on social networks. We define individual fairness for NIM problem and propose the **Maximin Protection problem** as a fairness-aware approach to NIM. We prove that the Maximin Protection problem is NP-Hard and not submodular. We propose a heuristic: PROTECT: Proximity-based Maximin Protection Heuristic to solve the problem efficiently. This heuristic uses the shortest distance between all pairs of nodes to identify the set of nodes that can maximize the minimum save probability of the network without widening disparities in the network. We also propose an alternative heuristic with lesser complexity that works well in certain networks. We call it Naive PROTECT. The performance of the proposed method is compared with existing fairness-agnostic methods and group-fairness-aware NIM methods. As individual fairness-based methods are not available for comparison, we implement the NIM alternatives to the Greedy Heuristic and Myopic

Heuristic proposed by Fish et al. in [19] for Fair Influence Maximization problem and use these as fairness benchmarks.

Though our objective optimizes for Maximin fairness, we also propose fairness metrics to measure Equity as well as Inequality of NIM solutions. For Equity fairness, we define similarity between individuals for the NIM problem and propose a similarity metric. We then use this metric to compute Consistency, which is a measure of equity fairness. For measuring inequality, we make use of the Gini coefficient, a metric commonly used in economics to compare income inequalities. We compare the fairness of the methods using these fairness metrics.

We evaluate our proposed methods on real-world and synthetic datasets. We experiment with datasets from social networks (Facebook, Twitter) as well as communication and collaboration networks. These networks vary in their network characteristics such as size, average degree, clustering coefficient, degree assortativity, etc. Synthetic networks of varying characteristics are generated using the HICH-BA algorithm [7]. Our experimental results demonstrate the effectiveness and fairness of our proposed approach compared to existing methods.

Overall, this paper aims to highlight the significance of individual fairness in the context of the NIM problem, present a logical approach to address fairness concerns and showcase the theoretical and experimental results of our proposed methodology.

In summary, the contributions of this thesis are as follows. We highlight the significance of individual fairness in the context of the NIM problem. A fairness objective is proposed, and two heuristics are proposed to solve this problem. Fairness metrics are proposed to measure the fairness of the solution. Experimental results demonstrate the unfairness of fairness-agnostic methods and highlight the superiority of our proposed approach in terms of fairness, as measured by the minimum save probability. Additionally, we analyze the trade-off between fairness and effectiveness, quantifying the price of fairness by looking at the *saved fraction*. Our proposed approach performs well in terms of both fairness and effectiveness. Not only does it achieve superior fairness outcomes, but it also demonstrates competitive performance in terms of the number of nodes saved, even surpassing fairness-agnostic methods in certain scenarios.

The implementation of the proposed methods, as well as the fairness-agnostic and group-fairness benchmarks mentioned, along with newly introduced benchmarks and the diffusion model, can be accessed in the following repository: https://github.com/DeepthyU/master-project. This repository also contains the code for the diffusion model used, as well as the code that was used to generate the synthetic networks. This would enable researchers and practitioners to reproduce the experiments and further explore the proposed methods.

1.6 Outline

The structure of this document is outlined as follows. Chapter 2 provides a comprehensive overview of the background knowledge that forms the foundation of this research. This is followed by Chapter 3, in which the problem is formally defined. Chapter 4 presents a review of relevant literature and related work. Chapter 5 introduces the proposed methods, explaining the intuition and proofs for the effectiveness of the PROTECT method and Naive PROTECT method, along with their algorithms. The experimental design is detailed in Chapter 6. This chapter elaborates the datasets, benchmarks, and evaluation metrics that are used. Chapter 7 presents the results of the experiments. Finally, Chapter 8 concludes the thesis by summarizing the key findings, discussing limitations, and suggesting directions for future research.

Chapter 2

Preliminaries

This chapter presents the concepts that served as the foundation for this work. It introduces information diffusion models, negative influence minimization using truth campaigning, definitions relevant to the study, and explores the notion of fairness in the context of NIM using social welfare theory. Additionally, this chapter explores different definitions of individual fairness, including equality, equity, and maximin. By establishing a solid understanding of these preliminaries, we lay the groundwork for the subsequent chapters.

2.1 Information Diffusion models

Information dissemination happens in a network over interconnected relationships. In a social network, when one individual becomes aware of new information, they may share it with their immediate acquaintances. Subsequently, those who perceive the information as credible or intriguing pass it along to their own network of connections. This process continues, resulting in the widespread diffusion of the information throughout the entire network.

Social network analysis utilizes information diffusion models to simulate the processes of information propagation within a network. Numerous propagation models have emerged to simulate how information spreads on a network. We will explain the two most commonly used propagation models that simulate the spread of influence in a network when there is only one campaign propagating. We then explain the extensions made to them to fit the scenario where there are two campaigns in the network, and present our model of choice for this study.

2.1.1 Single Campaign propagation models

Two widely recognized and extensively used propagation models are the **Independent Cascade Model (ICM)** and the **Linear Threshold Model (LTM)**. They were proposed by Kempe et al., in [32]. They simulate the spread of influence in a network when there is only one campaign propagating.

In ICM, all nodes in the network can be in either of the two states: “active” or “inactive”. The model operates in discrete time steps, with the spread of information occurring gradually. Initially, only a specific set of nodes known as campaign starters are active, while all other nodes remain inactive. During each time step, active nodes attempt to activate their immediate neighbors. The probability of influence is determined by the edge weight of the directed edge connecting the start node to the end node. Hence, ICM is a probabilistic model. Each node has a single opportunity to influence its neighbors. If successful, the neighbor becomes active; otherwise, it remains inactive. Once a node becomes active, its state remains unchanged throughout the simulation. In the subsequent time step, newly activated nodes proceed to influence their immediate neighbors. This process continues until no new nodes are activated in a time step. At this point, the diffusion

process is considered complete. The expected number of active nodes at the end of the diffusion process represents the outreach of the chosen campaign starters.

In LTM, each node possesses a threshold value that reflects its level of receptivity or openness to receiving information. Similar to ICM, the nodes in the network can be in either of the two states: "active" or "inactive". In this model, an inactive node becomes active if the sum of the edge weights from its active neighbors exceeds the node's threshold. This implies that the cumulative influence from neighboring active nodes must surpass the node's personal threshold for it to become active and engage in information propagation.

2.1.2 Two campaign Propagation models

Budak et al., in [11], extended ICM to model the spread of misinformation in a network, in the presence of a truth campaign. They proposed the **Multi-Campaign Independent Cascade Model (MCICM)** and **Campaign-Oblivious Independent Cascade Model (COICM)**. In MCICM and COICM, nodes within the network can exist in three possible states: "positively active", "negatively active", or "inactive". A node is considered positively active if it is influenced by the truth campaign, while a negatively active node is influenced by the misinformation campaign. A positively active node tries to activate its inactive neighbors positively, while a negatively active node tries to activate its inactive neighbors negatively. The initial set of negatively active nodes are called the negative seeds, while the initial set of positively active nodes are called the positive seeds.

If two or more nodes from different campaigns try to activate a node at the same time step, we assume that the node becomes positively activated. This approach is consistent with the practices followed in other recent studies [11, 60], where the positive influence takes precedence in such situations.

The key difference between MCICM and COICM lies in the propagation probabilities of the campaigns. In MCICM, both the campaigns propagate with different propagation probabilities, while in COICM, both the campaigns propagate with the same propagation probability. In other words, if C_M and C_T denote the misinformation and truth campaigns respectively for the directed graph $G = (V, E)$, and p_M and p_T are the respective functions for pair-wise propagation probabilities between the nodes for C_M and C_T, in COICM, $p_M = p_T$, while in MCICM $p_M \neq p_T$.

One particular variant of MCICM worth mentioning is the **MCICM with high effectiveness property** from [11]. In this model, the truth campaign exhibits a propagation probability of 1, denoted as $p_T = 1$. This implies that the truth is guaranteed to propagate to all neighboring nodes of a positively active node. We will leverage this model in Chapter 5 to demonstrate the effectiveness of our proposed heuristic.

Two common variants for the propagation probabilities are Uniform Cascade and Weighted Cascade. Under the Uniform Cascade, the propagation probability function is a constant. Therefore, all pairs of nodes share the same propagation probability. In Weighted Cascade, the propagation probability is a function of the in-degree of the destination node.

For the purposes of this work, we adopt **COICM with uniform cascade** as our model of choice. This model is closest to the Independent Cascade Model with Uniform Cascade, which was utilized in the study conducted by Fish et al. [19] that focused on investigating fairness in Influence Maximization. Our fairness baselines are built upon the foundations laid by this research, making the COICM with uniform cascade an appropriate and relevant choice for our study.

2.2 Negative Influence Minimization using Truth Campaigning

To effectively minimize the spread of negative information, two distinct categories of strategies are employed. The first category focuses on identifying specific nodes or edges that should be temporarily blocked, thereby impeding the further dissemination of misinformation [45, 29]. This approach, commonly referred to as immunization or blocking, aims to disrupt the flow of false

narratives within the network. On the other hand, the second strategy, known as truth campaigning, involves the propagation of true information to counteract ongoing misinformation campaigns [11, 27]. By promoting accurate and reliable information, truth campaigning aims to diminish the influence and impact of false narratives.

From a fairness perspective, the truth campaigning approach is more appropriate, as the former method relies on withholding information from users. Blocking or censoring any information, regardless of its truthfulness, can be seen as a violation of an individual's right to access information [72]. Studies have demonstrated the effectiveness of truth campaigning in countering misinformation. For example, Linden et al. [68] investigated the effectiveness of inoculation in reducing misperceptions about climate change and found that inoculation can effectively counter misinformation. Tanaka et al. [65] showed that when individuals are exposed to criticisms before rumors, they are more likely to take action to prevent the spread of rumors compared to when they are exposed to rumors first and then criticisms. Given the effectiveness and fairness of truth campaigning, this study adopts this approach to address the problem of fair Negative Influence Minimization.

2.3 Definitions

In this section, we define the key terms that will be frequently used in the coming sections.

misinformation probability: The misinformation probability of an individual in the network represents the likelihood that the individual becomes infected by a misinformation campaign initiated by M. We denote the misinformation probability of an individual u as $mp(u)$. The computation of this value will be discussed in Chapter 5.

save probability: The save probability of an individual represents the likelihood that the individual is not infected or is saved from the misinformation campaign initiated by M. We denote the save probability of an individual u as $sp(u)$. It is calculated using the equation:

$$sp(u) = 1 - mp(u) \tag{2.1}$$

Vulnerable set: The vulnerable set, denoted as $vulnerable$, consists of individuals in the network who have a non-zero chance of being influenced by the misinformation campaign.

$$vulnerable = \{v | mp(v) > 0 \ \forall v \in V \setminus M\} \tag{2.2}$$

2.4 Fairness in NIM using Social welfare theory

Welfare economics is a branch of economics that studies how to allocate resources and distribute goods and services in a way that maximizes social welfare, or the well-being of members of a society. Its primary objective is to identify principles and mechanisms that can lead to the highest level of well-being for individuals within a society.

Social welfare functions are used to quantify social welfare. They are used to study the balance between equity and efficiency. A choice of resource allocation that maximizes the social welfare is called Pareto efficient or Pareto optimal when it becomes infeasible to reallocate inputs and outputs in a manner that improves the well-being of any individual without simultaneously worsening the well-being of at least one other individual [8]. Such an allocation may or may not be equitable based on metrics used to measure well-being (called utility) and the welfare function used.

The application of social welfare theory to examine fairness is becoming increasingly prominent across various domains. It has been successfully employed in tasks such as classification and regression [28] and has also found relevance in the field of Influence Maximization [19, 47]. Notably, the work by Fish et al. [19] specifically delves into individual fairness within the context

of the Influence Maximization problem, aligning closely with the focus of our research. Drawing motivation from these notable contributions, we derive inspiration from social welfare theory to investigate the aspect of individual fairness in the domain of Negative Influence Maximization. By leveraging the principles and insights from social welfare theory, we aim to develop a deeper understanding of fairness in NIM and contribute to the advancement of fair NIM strategies.

In the upcoming section, we present the properties that a social welfare function must possess.

2.4.1 Welfare functions

Social welfare functions are mathematical functions that aggregate individual well-being or utility levels into an overall measure of social welfare. The social welfare function of a society of n individuals whose utility is in the range $[0, 1]$ can be expressed as :

$$\mu : [0, 1]^n \to \mathbb{R}$$

Social welfare functions are used to compare the desirability of different resource allocations or interventions. Let μ be a welfare function and x_i indicate the utility of an individual v_i in the network. To ensure a reasonable preference ordering, μ should possess certain properties [19, 47]. We explore the following key properties:

- **Symmetry** : A function is considered symmetric if its outcome remains unchanged regardless of the order in which the individual utilities are presented. In other words, the social welfare function is independent of the permutations of utilities.

  $$\mu(x_i, ..., x_m) = \mu(x_{\sigma(1)}, ..., x_{\sigma(m)})$$

  for all permutations of σ

- **Monotonicity** : A monotonically increasing function ensures that if all individual utilities increase or remain the same, the overall welfare also increases or remains the same.

  $$\mu(x_1, ..., x_m) \geq \mu(x'_1, ..., x'_m) \ \texttt{if} \ \ x_i \geq x'_i \ \ \forall i$$

- **Independence of unconcerned agents** : This property states that the social welfare function should be independent of the utilities that do not change.

  $$\begin{gathered} \text{If } \mu(x_1, x_2, ..., x_m) \geq \mu(x_1, x'_2, ..., x'_m) \\ \text{then } \mu(y, x_2, ..., x_m) \geq \mu(y, x'_2, ..., x'_m) \ \ \forall y \end{gathered}$$

- **Affine Invariance**: The social welfare function should remain unchanged when all utilities are multiplied by a non-zero constant and added to another constant. This property ensures that the ordering is preserved under such linear transformations. For any $a > 0$ and b,

  $$\begin{gathered} \text{if } \mu(x_1, ..., x_m) < \mu(x'_1, ..., x'_m) \\ \text{and } f(z) = az + b \\ \text{then } \mu(f(x_1), ..., f(x_m)) < \mu(f(x'_1), ..., f(x'_m)) \end{gathered}$$

In the following section, we present the utility and explore potential welfare functions for fair NIM.

2.4.2 Utility and Welfare function for NIM

Utility in social welfare refers to a measure of individual well-being or satisfaction within a society. In the context of NIM, the save probability of a node (sp) is indicative of the well-being of an individual in the network. Hence, we use sp as the utility for the welfare function for fair NIM.

The welfare of a social network G, with an intervention T can be expressed as:

$$\mu_G(T, vulnerable) = \mu(sp(v_1), ..., sp(v_{|vulnerable|})) \tag{2.3}$$

An optimal intervention T would be one that maximizes the welfare, i.e. maximize $\mu_G(T, vulnerable)$.

The fairness-agnostic NIM problem aims to minimize $\mathbb{E}[|\pi(M|T)|]$, which is the expected number of individuals influenced by M (refer to Eq:1.1, in Section 1.2). $\mathbb{E}[|\pi(M|T)|]$ is equivalent to the average probability with which the vulnerable nodes are infected by M.

$$\mathbb{E}[|\pi(M|T)|] = \frac{1}{|vulnerable|} \sum_{v \in vulnerable} mp(v) \tag{2.4}$$

Therefore, the NIM problem can be viewed as minimizing the average misinformation probability. As save probability is the complement of misinformation probability, this is equivalent to maximizing the average save probability. Consequently, the welfare function for the NIM problem can be expressed as:

$$\mu_{saved}(T, vulnerable) = \frac{1}{|vulnerable|} \sum_{v \in vulnerable} sp(v) \tag{2.5}$$

The welfare function μ_{saved} satisfies all the aforementioned properties necessary for a welfare function (refer to section 2.4.1). Thus, it qualifies as a suitable welfare function for the NIM problem.

Fish et al. demonstrated that using average as the welfare function in the context of the Influence Maximization problem leads to the widening of disparities. They referred to this phenomenon as the "rich getting richer" phenomenon. μ_{saved} is an average function. With the following proposition, we will prove that the rich getting richer phenomenon can occur when using average as the welfare function for the NIM problem as well and hence μ_{saved} is not an optimal choice for a welfare function.

PROPOSITION 2.1: The phenomenon of the rich getting richer under optimal intervention can occur when using the welfare function μ_{saved}

PROOF: Consider the following network where $M = \{m1, m2, m3, m4\}$. All nodes are at a distance of one from M. α represents the propagation probability of the network. Given $T = \emptyset$ and a budget of $k = 2$, our objective is to add two nodes to T that maximize the welfare μ_{saved}.

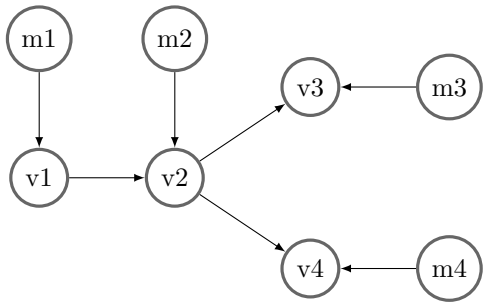


Figure 2.1: Example for *Rich getting richer*. When the optimal T is selected, the disparity between partitions {v1,v2} and {v3,v4} widens.

The save probability of each node when $T = \emptyset$ is as follows:

$$sp(v1) = 1 - \alpha$$
$$sp(v2) = 1 - (\alpha + \alpha^2) = 1 - \alpha - \alpha^2$$
$$sp(v3) = sp(v4) = 1 - (\alpha + \alpha^2 + \alpha^3) = 1 - \alpha - \alpha^2 - \alpha^3$$

If the network is partitioned such that $P1 = \{v1, v2\}$ is one partition and $P2 = \{v3, v4\}$ is another partition, the welfare μ_{saved} of each partition is:

$$\mu_{saved}(\emptyset, P1) = 1 - \alpha - \frac{\alpha^2}{2}$$
$$\mu_{saved}(\emptyset, P2) = 1 - \alpha - \alpha^2 - \alpha^3$$

The disparity in welfare between the two partitions is :

$$\begin{aligned} disparity(\emptyset) &= \mu_{saved}(\emptyset, P1) - \mu_{saved}(\emptyset, P2) \\ &= \frac{\alpha^2}{2} + \alpha^3 \end{aligned}$$

The intervention that maximizes the welfare is $T = \{v1, v2\}$. With this intervention, the save probability of each node becomes:

$$sp(v1) = sp(v2) = 1$$
$$sp(v3) = sp(v4) = 1 - (1 - \alpha)\alpha = 1 - \alpha + \alpha^2$$

The disparity in the welfare of the two partitions with $T = \{v1, v2\}$ is:

$$\begin{aligned} disparity(T) &= \mu_{saved}(T, P1) - \mu_{saved}(T, P2) \\ &= \alpha - \alpha^2 \end{aligned}$$

We can observe that, with any $\alpha < 0.5$, the intervention that maximizes the total welfare leads to an increase in the disparity, as $disparity(T)$ is greater than $disparity(\emptyset)$ for any $\alpha < 0.5$. Hence, we have proven that for the welfare function μ_{saved} under optimal intervention, the rich get richer, widening the disparity in the case of NIM as well.

As μ_{saved} is equivalent to the fairness-agnostic NIM objective, this proves that the fairness-agnostic NIM objective has the potential to widen the disparity in the network and hence can be unfair to individuals in the network.

Similar to μ_{saved}, we can also establish that the rich getting richer phenomenon occurs when using the generalized mean (μ_ϕ) as welfare function when $0 < \alpha < \frac{1}{2\phi}$ and $1 \leq \phi < \infty$. μ_ϕ also satisfies all the properties mentioned in section 2.4.1. The generalized mean welfare function can be expressed as:

$$\mu_\phi(T, vulnerable) = \left(\frac{1}{|vulnerable|} \sum_{v \in vulnerable} sp(v)^\phi \right)^{1/\phi}$$

PROPOSITION 2.2: The phenomenon of the rich getting richer under optimal intervention can occur when using the welfare function μ_ϕ for $0 < \alpha < \frac{1}{2\phi}$ and $1 \leq \phi < \infty$

PROOF: Consider the same network in Figure 2.1. With the welfare function μ_ϕ, welfare of the partitions under $T = \emptyset$ is as follows. The welfare of partition $P1 = \{v1,v2\}$ is:

$$\mu_\phi(\emptyset, P1) = \left(\frac{(1 - \alpha)^\phi + (1 - \alpha - \alpha^2)^\phi}{2} \right)^{1/\phi}$$

The welfare of partition $P2 = \{v3,v4\}$ is:

$$\mu_\phi(\emptyset, P2) = 1 - \alpha - \alpha^2 - \alpha^3$$

The disparity in this case can be calculated as:

$$disparity(\emptyset) = \mu_\phi(\emptyset, P1) - (1 - \alpha - \alpha^2 - \alpha^3)$$

The welfare of the partitions $P1$ and $P2$ under the intervention $T = \{v1, v2\}$ is:

$$\mu_\phi(T, P1) = 1$$
$$\mu_\phi(T, P2) = 1 - \alpha + \alpha^2$$

The disparity in the welfare of the two partitions with $T = \{v1, v2\}$ is:

$$disparity(T) = \alpha - \alpha^2$$

If $disparity(T) > disparity(\emptyset)$, we can conclude that the disparity is increased by the intervention.

$$\begin{aligned} & disparity(T) > disparity(\emptyset) \\ \iff \quad & \alpha - \alpha^2 > \mu_\phi(\emptyset, P1) - (1 - \alpha - \alpha^2 - \alpha^3) \\ \iff \quad & \mu_\phi(\emptyset, P1) < 1 - 2\alpha^2 - \alpha^3 \\ \iff \quad & \frac{(1-\alpha)^\phi + (1-\alpha-\alpha^2)^\phi}{2} < (1 - 2\alpha^2 - \alpha^3)^\phi \end{aligned}$$

Since $\phi \geq 1$ and $0 < \alpha < \frac{1}{2\phi}$, it follows that

$$(1 - 2\alpha^2 - \alpha^3)^\phi \geq 1 - 2\phi\alpha^2 - \phi\alpha^3$$

Additionally, as $1 - \alpha > 0$, we can apply the Power Mean Inequality, leading to,

$$\frac{(1-\alpha)^\phi + (1-\alpha-\alpha^2)^\phi}{2} \leq \frac{(1-\alpha) + (1-\alpha-\alpha^2)}{2}$$

Thus, as

$$\frac{(1-\alpha) + (1-\alpha-\alpha^2)}{2} < 1 - 2\phi\alpha^2 - \phi\alpha^3$$

the inequality is true. Consequently, we have proven that interventions can result in the rich getting richer for all $\phi \geq 1$.

Therefore, we observe that disparity increases with a wide range of welfare functions. However, an especially undesirable case of this disparity increase is when the welfare of the worse-off group never improves. Fish et al. define **k-imbalance for Influence Maximization** as "a particularly egregious form of the rich getting richer", where optimal welfare that can be achieved for the least-benefited half of the network is less than the initial welfare of the other half. Here, initial welfare refers to the welfare before intervention. Analogous to this, we can define the conditions for **k-imbalance for Negative Influence Minimization** as follows:

Let there be a graph G with two partitions of its vertices: RP (rich partition) and WP (worst-off partition). With a budget of k, the optimal intervention for G is T^*, and the optimal intervention for WP is T_{WP}. Then, a welfare function μ is k-imbalanced if the following conditions are satisfied:

1. There exists a set of seeds that will improve the welfare of WP
   $\mu(\phi, WP) < \mu(T_{WP}, WP)$
2. The welfare WP can ever achieve is less than the initial welfare of RP.
   $\mu(T_{WP}, WP) \leq \mu(\phi, RP)$
3. The optimal solution for the network (T^*) improves the welfare of RP.
   $\mu(T^*, RP) > \mu(\phi, RP)$
4. The optimal solution for the network (T^*) does not improve the welfare of WP.
   $\mu(T^*, WP) \leq \mu(\phi, WP)$

Thus, we say that the welfare function is imbalanced if

$$\mu(T^*, WP) \leq \mu(\phi, WP) < \mu(T_{WP}, WP) \leq \mu(\phi, RP) < \mu(T^*, RP)$$

According to Fish et al. μ is $\Omega(n) - imbalanced$ if μ is symmetric and strictly increasing, where n is the number of nodes in the network. We prove the same for NIM.

PROPOSITION 2.3: If μ is symmetric and strictly increasing, μ is $\Omega(n) - imbalanced$ where n is the number of vulnerable nodes in the network

PROOF: Consider a disjoint network G of 10 nodes (see Figure 2.2), where $V1$ and $V2$ are equal partitions of G and the propagation probability is α.

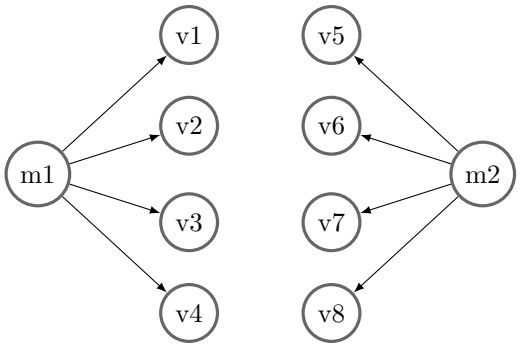


Figure 2.2: μ_{saved} is $k - imbalanced$

All the vertices of $V1$ are connected to $m1 \in M$ and all the vertices of $V2$ are connected to $m2 \in M$. Thus, all the nodes in the network have an equal save probability of $1 - \alpha$. Consider the case where $k = n/2 = 4$ and $T = \{v1, v2\}$. We need to add two more nodes to T such that welfare is maximized. As the welfare function is strictly increasing, adding a new seed should always increase the welfare. The ideal seed set of $V2$ would be any two nodes from $V2$. But by symmetry, the optimal seed set for G can be any four nodes from G, as the overall welfare will remain the same, for any two nodes selected. Hence, we can assume that these are from $V1$. Then the welfare of $V1$ increases, while the welfare of $V2$ remains the same. Thus, we prove that monotonic and strictly increasing functions are $k - imbalanced$.

Hence, μ_{saved} is also $\Omega(n) - imbalanced$. Fish et al. proved that $\mu_{-\infty}$ is balanced for the Influence Maximization problem (Proposition 3.3 [19]). By following the same line of reasoning, we can conclude that $\mu_{-\infty}$ is also balanced for the NIM problem. Consequently, we choose to utilize $\mu_{-\infty}$ as our welfare function.

The selection of the minimum function as the welfare function is justified by Rawls "veil of ignorance" thought experiment, which Heidari et al. elaborates in [28]. In this experiment, we are to imagine that we are one among the individuals in society and are unaware of our position in society, and we are to pick the most fair allocation among all available allocations. Then, any rationale individual would pick the allocation that is best for the worst-off individual. They argue this strategy is the fairest as it reduces the risk of all individuals in the society. Further, as noted by Fish et al., this objective is closely linked to the difference principle in Rawls Theory of justice. The difference principle states that social and economic inequalities are permissible if they work to the advantage of the least advantaged members of society. This principle aims to prioritize the well-being and improvement of those who are in the most disadvantaged positions. By focusing on this value, we ensure that no individual is left behind and that the most vulnerable individuals receive adequate protection.

When comparing μ_{saved} (mean) and $\mu_{-\infty}$ (minimum) from an ethical standpoint, we can observe that the former aims to enhance the save probability for all vulnerable nodes, while the latter prioritizes the most vulnerable nodes, thereby mitigating extreme vulnerabilities. In this regard, μ_{saved} takes into consideration the overall well-being of the population and strives to improve the save probability for a wider range of individuals, including both the privileged and disadvantaged. On the other hand, the latter objective strives to ensure that no individual, especially the most disadvantaged, is left behind, guaranteeing them a basic level of save probability.

Considering the detrimental impact of misinformation on people's lives, it can be argued that, under a limited budget, maximizing the minimum save probability is a more ethical approach. This

objective places greater emphasis on safeguarding those who are at the highest risk of being influenced by the misinformation campaign. By doing so, it addresses the urgent need to protect the most vulnerable users of the network and mitigate the potential harm caused by misinformation. We present the formal problem statement in Chapter 3.

Next, we present the definitions for fair NIM based on other popular fairness perspectives.

2.5 Individual Fairness Definitions

We define Individual fairness for NIM based on three popular perspectives: Equality, Equity and Maximin

2.5.1 Equality

Equality focuses on the fair allocation of resources. It refers to the principle of treating all individuals the same, without considering their specific needs or circumstances. It aims to distribute resources equally among all individuals, regardless of their vulnerabilities or requirements. In the context of NIM, equality fairness would imply that all nodes have an equal probability of being selected as truth campaign initiators. Mathematically, this can be expressed as:

$$\mathbb{P}(u \in T) = \mathbb{P}(v \in T)\ \forall u, v \in vulnerable$$

This is equivalent to randomly selecting positive seeds. Treating everyone the same disregards the varying influence and vulnerability of individuals. Therefore, while equality ensures a fair distribution of resources, it may not be ideal for NIM when there are constraints on the available budget.

2.5.2 Equity

Equity fairness focuses on fair treatment. It refers to the principle of addressing and reducing disparities, inequalities, or biases that exist by offering equal treatment for all. It aims at equalizing the outcomes or impacts for individuals who are similar in certain relevant aspects. In the context of NIM, equity fairness mandates that similar nodes in the network should have a similar chance of getting saved from misinformation. Hence, defining the similarity of individuals is an essential component of Equity fairness.

Similarity Metric

A similarity metric in social network algorithms is a measure of how similar two nodes are in terms of their attributes. This metric allows for an assessment of whether two individuals with similar attributes are treated the same, or whether one is treated more favorably than the other.

To comprehend which nodes can be regarded as similar for NIM, it is essential to understand the different motivations behind the creation of misinformation. Fake news or misinformation is often crafted with the intention to influence people's opinions and actions, either to serve specific agendas, discredit individuals or organizations, or manipulate public sentiment [10]. Financial motives can also drive the spread of fake news, as sensational or clickbait content generates web traffic and advertising revenue [58]. We see that fake news or misinformation can have either a local impact, targeting specific regions, or a global impact, affecting individuals regardless of their geographical location.

For instance, during elections, false information may be spread to influence voter behavior or discredit political opponents. This would primarily impact the voters in that region, whereas the same news may have little impact on individuals residing in another country. In such cases, considering individuals from the same region as similar can be appropriate. However, our approach adopts a feature-blind perspective and does not delve into specific regional considerations, which can be explored in future work.

In contrast, fake news with global relevance, such as misinformation about ideologies, famous personalities, or the COVID-19 pandemic, holds relevance for individuals worldwide. Consequently, we can consider two individuals in a network to be similar if they have an equal likelihood of being influenced by the misinformation campaign. It is important to note that factors like cognitive, social, and affective processes that influence the actual impact of misinformation on an individual are beyond the scope of this study and are not taken into consideration.

Hence, our similarity metric is based on the initial probability of an individual being infected by misinformation, when no intervention is applied (i.e. when $T = \phi$) which we will call *vulnerability*. The vulnerability of an individual u, denoted as $vulnerability(u)$, is :

$$vulnerability(u) = mp(u) \text{ when } T = \phi \tag{2.6}$$

Consequently, the similarity metric for fair NIM problem can be defined as:

$$s_{i,j} = 1 - (|vulnerability(i) - vulnerability(j)|) \tag{2.7}$$

Here, $s_{i,j}$ represents the similarity between any two nodes i and j in a network.

Equity fairness for fair NIM problem can then be expressed as:

$$sp(u) = sp(v) \text{ if } vulnerability(u) = vulnerability(v) \ \ \forall u, v \in vulnerable$$

2.5.3 Maximin

Maximin fairness also focuses on fair treatment or fairness in the outcome. It refers to the principle of maximizing the well-being of the worst-off individuals in a given context. It aligns closely with the concept of equity fairness in that it considers the disparities or inequalities present. Maximin fairness in NIM prioritizes the protection of the most vulnerable individuals in the network. It aims at maximizing the minimum save probability of the network. As described in section 2.4, this is the objective of our choice.

Chapter 3

Problem Definition

In this chapter, we formally define Fair NIM and explore its properties. In Section 3.1, we introduce the Maximin protection problem. Building upon this, in Section 3.2 we establish the key properties of the Maximin protection problem that guides our approach to achieving fairness in NIM.

3.1 Fair Negative Influence Minimization Problem

In the previous chapter, we took the social welfare perspective to fairness and established that using minimum as the welfare function is the optimal choice for fair NIM. Inspired by the Maximin Access problem proposed by Fish et al. in [19], we present the **Maximin Protection problem**.

The objective of the Maximin Protection problem is to maximize the minimum probability of a node being shielded from misinformation, thereby prioritizing the well-being of the least advantaged individuals in the network. It can be expressed as follows:

$$\begin{aligned} &\underset{T}{\text{maximize}} \quad \mu_{-\infty}(T, vulnerable) \\ &\text{subject to} \\ &\quad T \subset V \setminus M, \\ &\quad |T| \leq k, \end{aligned} \tag{3.1}$$

where

$$\mu_{-\infty}(T, vulnerable) = \min_{v \in vulnerable} sp(v) \tag{3.2}$$

We will study this problem under the Campaign Oblivious Independent Cascade Model (COICM) with Uniform Cascade (refer to section 2.1.2) to align with [19].

We demonstrate in the next section that the Maximin Protection problem is NP-hard and non-submodular, posing challenges in finding an optimal solution. To address this, we devise heuristics to approximate the solution efficiently, which will be discussed in Chapter 5.

3.2 Properties of Maximin Protection problem

In the preceding section, we introduced the Maximin Protection problem. In this section, we will present its properties with proofs.

PROPERTY 1: ***The Maximin Protection problem is NP-Hard.***

PROOF: We can demonstrate the NP-hardness of the Maximin Protection problem by reducing it to the well-known NP-hard problem called the set cover problem.

The set cover problem can be described as follows: Given a set of elements called the Universe and a collection S of m sets, whose union is the universe, the set cover problem is to find k sets from S whose union equals the universe.

Consider the Maximin Protection problem in the context of a network with a propagation probability of α. A maximin save probability of at least α can be guaranteed if all vulnerable nodes in the network are at a distance of one from T. This means that we need to find a set of nodes such that the union of their neighbors covers all the vulnerable nodes in the network. This is equivalent to the set cover problem, where the vulnerable nodes correspond to the elements in the Universe, and the sets in the set cover problem correspond to the set of neighbors of each node in the network. Since the set cover problem is known to be NP-hard, we can infer that the Maximin Protection problem is also NP-hard.

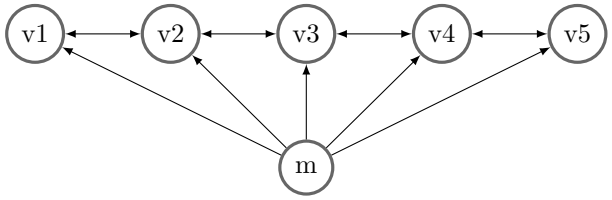


Figure 3.1: The Maximin Protection problem is not submodular.

PROPERTY 2: ***The Maximin Protection problem is not submodular.***

PROOF: A function $f(.)$ is sub-modular if it exhibits a diminishing returns property. In other words, adding an element to a smaller set yields a larger increase in the function value compared to adding the same element to a larger set.

$$f(A \cup \{v\}) - f(A) \geq f(\{v\}) - f(B) \;\; \forall v \notin B \text{ and } A \subset B$$

To prove that the Maximin Protection problem is not submodular, we show a counterexample where adding an element to a smaller set yields a smaller increase in the function value compared to adding the same element to a larger set.

Consider the network depicted in Figure 3.1 with a misinformation starter node m and a budget of $k = 2$. All campaigns propagate with a propagation probability α where $0 < \alpha < 0.5$.

Let $f(.)$ be the objective function, i.e., the minimum save probability of the network. $v3$ has the highest misinformation probability initially, as it has the most number of paths from m. Let $\mathbb{P}(v3(t))$ be the probability that $v3$ is infected at time step t

$$\begin{aligned} mp(v3) &= \mathbb{P}(v3(t=1)) \quad + \mathbb{P}(v3(t=2)) \quad + \mathbb{P}(v3(t=3)) \\ &= \alpha \qquad\qquad\quad + 2\alpha^2 \qquad\qquad + 2\alpha^3 \end{aligned}$$

Consequently, $v3$ has the least save probability in the network.

$$\begin{aligned} f(\emptyset) &= sp(v3) \\ &= 1 - mp(v3) \\ &= 1 - \alpha - 2\alpha^2 - 2\alpha^3 \end{aligned}$$

Suppose we start with an empty set A. $f(\{A\}) = f(\emptyset)$. We add the node $v4$ to form the set $A \cup \{v4\}$. Under $T = A \cup \{v4\}$, the misinformation probability of each node is the probability

that it is not influenced by T and is influenced by m. $v2$ would have the highest misinformation probability in this case.

$$\begin{aligned} mp(v2) &= \mathbb{P}(v2(t=1)) &&+\mathbb{P}(v2(t=2)) &&+\mathbb{P}(v2(t=3)) \\ &= \alpha &&+((1-\alpha^2)\times 2\alpha^2) &&+0 \end{aligned}$$

Then,

$$\begin{aligned} f(\{v4\}) &= sp(v2) \\ &= 1 - mp(v2) \\ &= 1 - \alpha - 2\alpha^2 + 2\alpha^4 \end{aligned}$$

Now, consider a larger set $B = \{v2\}$. $f(\{v2\}) = f(\{v4\})$. We add a second node $v4$ to form the set $B \cup \{v4\}$. Under $T = B \cup \{v4\}$, $v1$ and $v5$ are the nodes with the highest misinformation probability.

$$mp(v1) = mp(v5) = P(v1(t=1)) = (1-\alpha)\times\alpha$$

Then the minimum save probability becomes:

$$f(B \cup \{v4\}) = 1 - \alpha + \alpha^2$$

It can be observed that the gain $f(A\cup\{v4\}) - f(A) = 2\alpha^3 + 2\alpha^4$, while the gain $f(B\cup\{v4\}) - f(B) = 3\alpha^2 - 2\alpha^4$ and $A \subset B$. In this case, $2\alpha^3 + 2\alpha^4 < 3\alpha^2 - 2\alpha^4$, violating the submodularity property.

Thus, the Maximin protection problem is not submodular. Hence, the greedy hill climbing algorithm, which relies on the submodularity property, does not have guaranteed approximation bounds for this objective. Therefore, we have developed heuristics to solve the problem, which are presented in Chapter 5.

Chapter 4

Literature Review

We present here the literature review and related works for this project. In Section 4.1, we briefly discuss the works in Influence propagation problems. In section 4.2, we look into previous works in NIM. In Section 4.3, we present studies that explore fairness in Influence propagation problems. In Section 4.4, we present works on Social Welfare Based Fairness.

4.1 Influence propagation problems

In this section, we look at the existing work in this domain to see where this research can be positioned. According to Shirazipouraza et al. [59], the influence propagation problems are classified into three types based on the adversarial involved:

- Non-adversarial - A campaign is introduced in a network with the intention to influence maximum number of users in the network. Users are influenced by their neighbors in the network with a certain probability. This is the **Influence Maximization problem**. One of the initial works in Influence Maximization is [15] by Domingos et al. In order to take advantage of customer interactions, they compute the "network value" of each customer. This is the predicted revenue generated by a customer from sales to other customers that they might influence to buy, as well as other customers that those customers may influence, and so on in a recursive manner. They called this the Expected Lift in Profit (ELP) and proposed approximate methods to identify a set of nodes that would maximize the ELP. Influence maximization is commonly used for viral marketing. The objective of the influence maximization problem is to find the optimal seed set for a given budget, that can influence the maximum number of users in the network. Companies try to boost the reach of their products in a network by making use of the large number of connections of popular people in that network. This can be seen with companies hiring famous "Influencers" to promote their products on Instagram. Kempe et al. [31] build on [15] by studying the problem on two diffusion models: The Linear Threshold model and the Independent Cascade model (explained in section 2.1.1). They proved that this is an NP-hard problem that has a constant factor approximation (1 - 1/e) in both these models.

- Adversarial with active adversary - In this case, there are two simultaneous campaigns in the network, each competing with the other to maximize its influence. This is the **Competitive Influence Maximization problem**. It is an extension of the influence maximization problem. In [5], Bharathi et al. proposed a generalization of ICM to support multiple competing campaigns in a network. They studied the diffusion of competitive campaigns in a network using this model and demonstrated that even the last player (campaign) can find a seed set with a performance lower bound of $(1 - 1/e)$ of the optimal solution. Carnes et al. [12] studied the Competitive Influence Maximization problem and found that the highest degree heuristic was the optimal behavior for the first campaign.

- Adversarial with passive adversary - This is the **Negative Influence Minimization problem**. The NIM problem can be seen as another extension of the influence maximization problem. As we stated above, the network has a seed set propagating misinformation. The goal of this problem is to find an optimal set of nodes to counter this propagation such that the effect of the former is minimized. This is different from the Competitive Influence Maximization problem in that the goal of the second campaign is to minimize the spread of the first campaign and not to maximize its influence in the network.

4.2 Negative Influence Minimization

Budak et al. [11] were the first to study the problem of influence minimization, though there have been works prior to this on competing campaigns [5, 12]. They showed that choosing a set of nodes in the network to start a "good" campaign to limit the spread of the "bad" campaign is NP-Hard, and hence finding the optimal solution for large social networks is infeasible. They prove that the problem is submodular under MCICM with the high effectiveness property as well as COICM. Owing to the problem being submodular and monotone, the greedy hill climbing algorithm provides a solution that has an efficiency $(1 - 1/e)$ of the optimal solution. They also consider three heuristics (degree centrality, early infectees and largest infectees) as the greedy algorithm has a high complexity (O(k $|V|$ R $|E|$)).

He et al. [27] studied the problem under the Competitive Linear Threshold model (CLT) which is an extension of LTM. They call this the Influence Blocking Maximization problem. Similar to [11], they proved that this problem is NP-Hard and submodular and hence the greedy hill climbing algorithm can achieve an approximation ratio of $1 - 1/e$. Further, they proposed the CLDAG method that constructs local directed acyclic graphs to compute the influence of nodes in linear time. CLDAG method matches the efficiency of the greedy algorithm while being two magnitudes faster.

Wu et al. [72] studied the problem on two competitive propagation models and proposed separate heuristics (CMIA-H for the MCICM with high effectiveness and CMIA-O for COICM) that use the maximum influence arborescence (MIA) structures. These heuristics are faster than the greedy algorithm and matched the performance of the greedy algorithm in their experiments. Another path-based method is BIOG proposed by Lin et al. [38]. They make use of Maximum Influence Out-Graphs to compute the blocking score of each node.

The works of Song et al. [61] and Manouchehri et al. [40] consider delays and deadlines in the NIM problem. They call this the Temporal Influence Blocking (TIB) problem. Song et al. [61] studied the problem on MCICM and proposed TIB-Solver method. It involves computing a threat level for each node at each time step. This is followed by greedily selecting the nodes that can save the most number of nodes using Weighted Reverse Reachable sets. Manouchehri et al. [40] studied the problem on the Temporal COICM and proposed TIBM-M method. They use Temporal reverse-reachable sets (θ). First, they estimate the number of θ's to be generated to reach the estimated efficiency. They then generate that many θ's and greedily select the nodes with maximum gain. They provide theoretical guarantees for a $(1-1/e-\epsilon)$ approximate solution.

Simpson et al. [60] studied the problem under MCICM with high effectiveness and proposed the Reverse Prevention Sampling (RPS) method that gives a $(1-1/e-\epsilon)$ approximate solution. They used, for node selection, Random Reverse Reachability without Cutoff (Random RRC) Sets which avoid paths that can get obstructed by the misinformation campaigns. The node that appears in the maximum number of RRC sets are iteratively selected as the positive seeds.

Gong et al. [23] proposed the Conformity-Aware Competitive Independent Cascade (CCIC) model to simulate the spread of truth and rumor in a network. The RBF and CDD heuristics they proposed for their rumor-blocking problem are based on centrality measures [54]. RBF algorithm utilizes the number of shortest paths from M to all the nodes. CDD algorithm computes the number of truth-active neighbors each node requires so that it is not activated by rumor. Both these heuristics demonstrate speed and efficiency in blocking rumor in CCIC model.

Saxena et al. [55, 53] studied the NIM problem in social networks by considering the biases that

users may have towards different topics. In the proposed model [55], when a user holds a particular belief, their bias towards that belief becomes stronger as they receive news that aligns with their opinion. At the same time, their bias towards the opposing belief weakens. The probability that a user is influenced by misinformation is computed as the propagation probability times their bias. They proposed the k-Truth Score method to find the optimal solution from a set of "prospective" truth campaigners.

An alternative approach to the truth campaign is to restrict the spread of the negative campaign by blocking specific nodes or edges in the network. One strategy is to select nodes at crucial positions in the network and assign them with information filtering functionality [18] or monitoring functionality [35] or to block them altogether [29]. In these approaches, the negative seed set is unknown. A recent work that adopts this approach is [14]. In this work, Cheng et al. establish an equivalence between the influence minimization problem (or the targeted immunization problem as they call it) with the influence maximization problem. The former is minimizing the penalty (number of seeds affected) while the other is maximizing this value. They immunize the nodes with maximum spreading potential.

Another approach is to identify and block/immunize nodes or edges whose removal or immunization would result in the greatest reduction of negative influence spread when the negative seed set is known [45].

In addition, another set of solutions makes use of community information to counter the negative information spread. This either involves using the community information that is available in the dataset or using a community detection algorithm to identify communities in the network. This is followed by selecting the optimal set of nodes to minimize the spread within each community. Some examples are the FC_IBM method [4] proposed by Arazkhani et al., the CB_IBM method [39] proposed by Lv et al. and the CoF-MIDR method [73] proposed by Wu et al.

More recent works in this topic present solutions that make use of Machine Learning and Deep Learning techniques. He et al. [26] use Q-learning to select the optimal seed set, while Chen et al. [13] use deep learning models to predict the cascade size of the nodes.

We summarize the works discussed above in table 4.1. In this table, for each study, we present the names of the proposed methods. This is followed by the general strategy used. This column indicates if the method uses Truth Campaigning (TC) or Node/Edge Blocking. The third column specifies the category of the method. In the next column, we mention the novelty or the main focus of the study. Last, the diffusion model that was utilized, and the baselines used to compare the methods are specified.

4.3 Fairness in Influence Propagation Problems

Fairness considerations have not been extensively addressed in the context of NIM problems, with only a few notable works focusing on this aspect. One such work by Gutiérrez [7] introduces feature-aware group fairness and proposes the Fairness-aware Weighted Reversible reachable system (FWRRS) method. The FWRRS method aims to achieve community fairness while maintaining high efficiency. It employs a maximin fairness constraint that seeks to maximize the minimum proportion of users saved from each community. It should be noted that this fairness objective differs from our maximin objective, which aims to maximize the minimum save probability of the network.

Notably, there is a lack of feature-blind and/or individual fairness-based techniques proposed for fair NIM. In light of this, we can explore existing fair Influence Maximization solutions, given the similarity between the two problem domains. It is observed that most fairness-aware algorithms for the influence maximization problem primarily target group fairness.

Khajehnejad et al. [33] address the fair influence maximization problem using Adversarial Graph Embedding. Their approach involves employing a Variational Auto Encoder to generate a fair graph embedding, ensuring a similar distribution of nodes from different communities. The central nodes are then selected from this embedded space, while maintaining statistical parity of the seed set. Their results demonstrate increase in the diversity of the seed set with no drop in

outreach.

Another relevant work in this context is presented by Gong et al. [22]. Their work on group fairness for influence maximization uses the Gini coefficient to measure inequalities between communities. They argue that the maximin criterion for group fairness is an inaccurate fairness measure as it only looks at the extreme values in each community and not the actual distribution of values in them. The Gini coefficient is popularly used in economics to measure disparities in income, education, opportunities, etc. We will use the Gini coefficient to understand the inequality in save probability distribution in networks for our proposed method.

Stoica et al. [62] aim to achieve group fairness by enforcing statistical parity of the seed set. Recognizing that computing the most comprehensive seed set is computationally demanding, they propose that ensuring statistical parity in the seed set can be an effective approach to promote parity in outreach.

Rahmattalabi et al. [47] and Fish et al. [19] adopt social welfare theory to formulate fairness objectives in the context of influence maximization. Rahmattalabi et al. [47] focus on group fairness, while Fish et al. [19] explore individual fairness, which, to the best of our knowledge, is the only study on individual fairness for influence maximization. In the following section, we will delve into these works in more detail.

4.4 Social Welfare-Based Fairness in Social Networks

Social welfare theory is a branch of economics that focuses on studying societal well-being and the allocation of resources to maximize overall social welfare. As mentioned in Section 2.4, the social welfare function provides a means to rank different allocations of resources or policy choices based on their impact on social well-being.

Heidari et al. [28] proposed the use of cardinal social welfare to measure the fairness of automated decision-making systems. Their study aimed to balance fairness and accuracy in a supervised learning setting. They considered risk and welfare for evaluating fairness, using a normative theory based on the Rawlsian conception of "fairness behind a veil of ignorance". They presented a convex formulation for welfare-based measures of fairness for classification and regression problems.

Fish et al. [19] used a social welfare function to bring individual fairness in Influence maximization. This work focused on reducing the *access gap* in networks. *access* of an individual in a network is the likelihood that information propagating in the network reaches that individual. *access gap* is the difference in the minimum *access* of two random partitions of the network. As explained in section 2.3, they use minimum ($\mu_{-\infty}$) as their welfare function and *access* as the utility. By choosing the minimum probability value as the social welfare, they ensured symmetry and effectively prevented the optimal seed set from improving the *access* of only a subset of the network. They implement three heuristics: (i) Greedy heuristic, (ii) Myopic heuristic, and (iii) Naive Myopic heuristic, to tackle this NP-hard problem. These heuristics utilized Monte-Carlo estimation to compute the *access* of nodes. In their Greedy Heuristic, each positive seed is selected by computing the minimum potential *access* of nodes in the network when each node in the network is selected as the seed and finally selecting the node that results in the maximum minimum *access* in the network. As this heuristic has a high time complexity, they propose faster approaches (Myopic and Naive Myopic heuristics) that require a lesser number of Monte-Carlo estimations. We implement the equivalents of the Greedy heuristic and Myopic Heuristic for the fair NIM problem to use as benchmarks for our proposed solution. This is described in sections 5.3 and 5.4. We omit the Naive Myopic heuristic as they give poor results.

Another work that uses social welfare theory for fairness is that of Rahmattalabi et al. [47]. They used the welfare optimization approach for achieving Group Fairness in Influence Maximization. To ensure fairness, their welfare function follows strict monotonicity, symmetry, Independence of Unconcerned Individuals, Affine Invariance, and Transfer principle. They present two

functions that satisfy these conditions:

$$W_\alpha(\mathbf{u}) = \sum_{i=1}^{N} u_i^\alpha/\alpha \text{ for } \alpha < 1, \alpha \neq 0$$

and,

$$W_\alpha(\mathbf{u}) = \sum_{i=1}^{N} \log(u_i) \text{ for } \alpha = 0$$

where u_i is the utility value of the average utility value of the group node i belongs to, N is the total number of nodes in the network, and α is the Inequality aversion parameter. Both functions are monotone and submodular. Hence, they have a $(1-1/e)$ approximate solution. They showed that adjusting the inequality aversion parameter allows for trade-offs between fairness and efficiency.

4.5 Feature-blind Individual Fairness in Networks

Fairness is not well-explored in social network analysis (SNA) [50] problems with only a few recent works investigating fairness in various SNA problems, including influence maximization [63, 64, 66, 19], link prediction [51, 41, 52], and centrality ranking [67]. Most of these works adopt a feature-aware approach, while there are very limited works that adopt a feature-blind approach to achieve fairness in SNA. Borges et al. [9] developed a feature-blind fair recommender system that focused on refining recommendations by removing position bias and popularity bias without using sensitive attributes. They introduced the term "Feature-Blind Unfairness" to describe unfairness that may arise from not utilizing sensitive attributes. While their recommender system achieved a significant reduction in bias, it also experienced a drop in efficiency. The social welfare-based fairness approaches by Fish et al. [19] and Heidari et al. [28], mentioned in the previous section, also fall under the category of feature-blind fairness.

Utilizing protected attributes, such as community information, is often challenging due to their limited availability in data. Using proxy variables to approximate community membership may be suboptimal and, in fact, can be harmful to minority communities, as noted by Kallus et al. in [30]. Moreover, employing community detection techniques to identify communities without ground truth can be risky, as the algorithm may miss or overlook smaller communities, which can significantly impact fairness in downstream tasks [50]. Fair community detection itself lacks a well-defined framework [50]. Therefore, in this study, we explore a feature-blind approach to Fair NIM, which is novel to the best of our knowledge.

Table 4.1: Summary of works in Negative Influence Minimization

Method [Reference]	**Strategy**	**Method Type**	**Characteristic feature**	**Diffusion model**	**Baselines compared**
CMIA-O and CMIA-H [72]	TC	Path-based	Scalability	COCIM and MCICM with high effectiveness	Greedy, Proximity, degree, random
BIOG [38]	TC	Path-based	Scalability	COCIM	Greedy, CMIA-O, Proximity, random
TIB-Solver [61]	TC	Sampling-based	delay and deadline	MCICM	PageRank, LSMI, Largest infectees
TIBM-M [40]	TC	Sampling-based	delay and deadline	Temporal COICM	TIB-Solver, CMIA-O, community-based IBM, First Neighbor Selection degree, random
RPS [60]	TC	Sampling-based	obstructed nodes	MCICM with high effectiveness	Greedy
RBF and CDD [23]	TC	Centrality-based	Conformity-aware model	CCICM	degree, betweenness centrality, RIS
k-truth score[55]	TC	Path-based	considers user biases	CICBM	TIB-Solver, TBM
constraint generation, DEF-MILP[29]	Node Blocking	MILP	defense approach against attack	ICM and LTM	Degree, Betweenness, PageRank, Influence
HIGHPROB-LPR, APPROX-BENDERS[14]	Blocking	MILP	epidemic control model	LTM	highest weighted degree, Greedy, LDAG, SIMPATH
CB_IBM [39]	Node blocking	Community structure and path-based	community-based algorithm	COICM	degree, random, proximity, greedy
FC_IBM [4]	TC	Community structure and centrality-based	supports overlapping communities	MCICM	degree, betweenness, closeness
CoF-MIDR [73]	Node blocking	Community structure and Centrality-based	seeds are selected from key nodes and bridge nodes		DRIMUX, Positive CB, dynamic blocking
CCSQ[26]	TC	community detection and Q-learning	uses reinforcement learning	multistage competitive LTM	random, RIS, CDIM, unblocking
Ctl-diff [13]	Edge Blocking	Influence Prediction	gradient driven network topology modification	LTM	edge selection based on degree, betweenness, closeness, clustering centrality

Chapter 5

Proposed Methods

In this section, we present the methods that were developed to solve the Maximin Protection problem introduced in Chapter 3. As discussed in Section 3.2, this problem is NP-Hard and lacks submodularity properties, necessitating the development of effective heuristics. In Section 5.1, we introduce the primary proposed method, the Proximity-based Maximin Protection Heuristic (PROTECT), designed to address the objective efficiently. To provide a less complex, faster alternative, we present the Naive PROTECT approach in Section 5.2. Moreover, in Sections 5.3 and 5.4, we introduce individual fairness benchmarks, which are based on the individual fairness algorithms proposed for fair Influence Maximization [19]. These methods will serve as fairness baselines in the subsequent experiments.

5.1 PROTECT method

The PROTECT (Proximity-based Maximin Protection) Heuristic leverages the proximity of nodes to the initiators of the misinformation campaign (M) and to each other to identify nodes that maximize network protection. The fundamental idea behind PROTECT is that a node v in the network can **improve** the save probability of a node u, which has a shortest distance of d from nodes in M, if the shortest distance from node v to node u in the network is less than or equal to $d - r$, where r represents the delay in identifying M. This concept serves as the core principle for the PROTECT method. For simplification, we assume r to be 0.

In the following sections, we will validate this intuition initially in MCICM with the high effectiveness property (described in section 2.1.2). This will provide a better understanding before we validate it in COICM employed in this study.

5.1.1 Rationale behind the Proximity-Based Approach

In ICM (described in section 2.1), the propagation of campaigns follows a stochastic process. In [11], Budak et al. explain the activation of a neighboring node in a network $G = (V, E)$ by a positively (negatively) active node, under ICM as similar to flipping a coin with bias p_T (p_M). An edge is *live* if the starting node successfully activates the end node. Consequently, a specific instance of influence spread can be represented as a graph $G' = (V, E')$ where $E' \subseteq E$ denotes the set of the *live* edges.

In the MCICM with the high effectiveness property, where the propagation probability function of the truth campaign (p_T) is 1 and that of misinformation campaign is any $\alpha < 1$, the truth campaign (C_T) propagates through G, while the misinformation campaign (C_M) propagates through a G'. According to Budak et al. [11], under this diffusion model, a node w can be saved if and only if there exists a node $v \in T$ such that the shortest path from v to w in G, denoted as $SP_G(v, w)$, is shorter than or is of equal length as that of the shortest path from all nodes $m \in M$

to w in G', denoted as $SP_{G'}(M, w)$. That is, for w to be saved, the condition is

$$\exists v \in T \text{ s.t } |SP_G(v, w)| \leq |SP_{G'}(M, w)| \tag{5.1}$$

In other words, a node w can be saved by a node v if the shortest distance from v to w in G is less than or equal to the shortest distance from all $m \in M$ to w in G'.

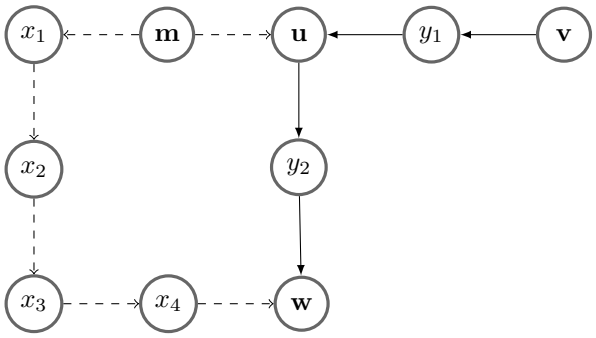


Figure 5.1: Example for obstructed path. The dotted lines indicate E'. $\mathbf{m} \in M$. $|SP_G(\mathbf{v}, \mathbf{w})| < |SP_{G'}(\mathbf{m}, \mathbf{w})|$. But $|SP_G(\mathbf{m}, \mathbf{u})| < |SP_G(\mathbf{v}, \mathbf{u})|$. u obstructs the path from $\mathbf{v}$ to $\mathbf{w}$ and hence $\mathbf{v}$ cannot save $\mathbf{w}$

Now consider the network in figure 5.1 where $M = \{\mathbf{m}\}$ and $T = \{\mathbf{v}\}$. The dashed lines indicate the live edges for the misinformation campaign C_M. The shortest path from $\mathbf{m}$ to $\mathbf{w}$ for C_M is $\{\mathbf{m}, x_1, x_2, x_3, x_4, \mathbf{w}\}$ making the length of the shortest path from M to $\mathbf{w}$ in G', $|SP_{G'}(M, \mathbf{w})|$ equal to 5. Node $\mathbf{v}$ satisfies condition 5.1 as $|SP_G(\mathbf{v}, \mathbf{w})|$ is 4.

In this network, the node $\mathbf{u}$ is at a distance 1 from M in G' ($|SP_{G'}(M, \mathbf{u})| = 1$) and at a distance 2 from node $\mathbf{v}$ in G ($|SP_G(\mathbf{v}, \mathbf{u})| = 2$). Thus, $\mathbf{m}$ will infect $\mathbf{u}$ in the first time step blocking the shortest path from $\mathbf{v}$ for $\mathbf{w}$: $\{\mathbf{v}, y_1, u, y_2, \mathbf{w}\}$. It is easy to see from this example that condition 5.1 is necessary, but not sufficient, for a node to save another node.

Simpson et al. noted the same in [60] and introduced the concept of "obstructed" nodes. Any node v satisfying the condition 5.1 can still be obstructed by M if, in every path from $v \in T$ to w, there exists a node u in the network such that $|SP_{G'}(M, u)| < |SP_G(v, u)|$. u is called an obstructed node. In other words, only if there exists a shortest path from a node v to w that satisfies the condition 5.1 and does not contain any node u that is closer from M in G' than from v in G can save w.

If we consider the shortest path on G for both campaigns, we can ensure that the case of obstructed nodes does not arise. That is, if there exists a node $v \in V \setminus M$ such that $|SP_G(v, w)| \leq |SP_G(M, w)|$, obstructed nodes cannot occur and such a node can save w. We prove the same below.

PROPOSITION 5.1: In MCICM with the high effectiveness property, if $|SP_G(v, w)| \leq |SP_G(M, w)|$, then node v can save node w.

PROOF: Consider a network $G = (V, E)$ with misinformation starters M. $w \in V \setminus M$ is a node to be saved and v is any non-seed node in the network such that $|SP_G(v, w)| \leq |SP_G(M, w)|$. Suppose there exists a node u in the shortest path from v to w in G, such that u is closer to M than to v. That is,

$$\exists u \in V \setminus T \text{ s.t. } |SP_G(M, u)| < |SP_G(v, u)|$$

Since u lies on the path from v to w, we can express the shortest path $SP_G(v, w)$ as:

$$SP_G(v, w) = SP_G(v, u) + SP_G(u, w)$$

From the above two equations, we can deduce that the shortest path from M to w in G is:

$$SP_G(M, w) = SP_G(M, u) + SP_G(u, w)$$

Then, it follows that

$$|SP_G(v, w)| \leq |SP_G(M, w)| \text{ only if } |SP_G(v, u)| \leq |SP_G(M, u)|$$

which contradicts our assumption that u is closer to M. Therefore, if w is closer to v than to M, there cannot exist a node u that is closer to M than to v. Thus, we have proven that obstructed nodes cannot occur when we consider the shortest distances in the graph G for both campaigns. Consequently, if $|SP_G(v, w)| \leq |SP_G(M, w)|$, we can conclude that node v saves node w.

Hence, when the campaign C_T propagates with high effectiveness, the condition

$$\exists v \in T s.t. |SP_G(v, w)| \leq |SP_G(M, w)| \tag{5.2}$$

becomes a sufficient criterion for the node w to be saved. In other words, a node v can save a node w in the network if it is either closer to w than or at the same distance from w as all nodes $m \in M$.

In COICM, both campaigns C_T and C_M propagate with the same probability. That is, both the campaigns propagate on graph G'. In this case, Proposition 5.1 does not hold true. This is because G' may not have all the edges in G. As a result, the shortest distance between any nodes v and w in G' may be greater than the shortest distance between them in G. That is, it is possible that,

$$\exists v, w \in V \setminus M \text{ s.t. } |SP_G(v, w)| \leq |SP_G(M, w)|$$
$$\text{but } |SP_{G'}(v, w)| > |SP_{G'}(M, w)|$$

In other words, a node that is closer to v than M in G could be farther from v than M in G' making condition 5.2 insufficient for the COICM.

In our diffusion model, we make the assumption that truth takes precedence over misinformation. This means that when a node is simultaneously influenced by both truth (C_T) and misinformation (C_M), the node will be influenced by the truth (C_T). This assumption implies that positively active nodes attempt to influence their neighbors first, followed by the negatively active nodes attempting to influence their remaining inactive neighbors.

Furthermore, in this study, we adopt the uniform cascade framework as our model of choice. This means that all edges in the network have the same weight, implying an equal likelihood of influence propagation along each edge.

Owing to these two characteristics of the diffusion model, we can prove that, for any nodes $w, v \in V \setminus M$, v can **reduce** the probability of w being infected by M if v satisfies condition 5.2. We will hereon use $D_M[u]$ ($D_T[u]$) to denote the shortest distance possible from all $m \in M$ ($t \in T$) to any vulnerable node u in the network.



Figure 5.2: Figure to demonstrate the effect of proximity of T

PROPOSITION 5.2: In COICM with Uniform Cascade, if $|SP_G(v, w)| \leq |SP_G(M, w)|$, then node v can improve the save probability of node w.

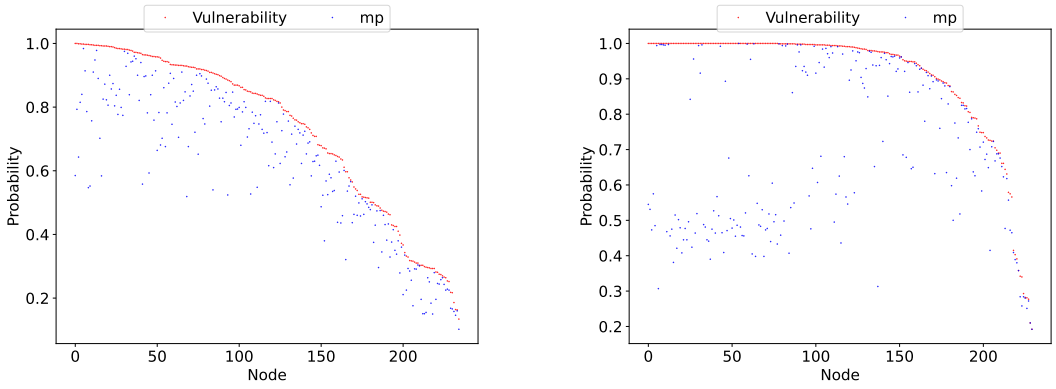
PROOF: Consider the network $G = (V, E)$ in Figure 5.2 where misinformation starters $M = \{m\}$ and u is the node to be saved. When α is the propagation probability of the campaigns, and $T = \emptyset$, m infects u with a probability, $mp(u) = \alpha^2$. Nodes $u1$, $u2$ and $u3$ are at distances 1, 2 and 3 respectively from node u.

If $T = \{u_1\}$, T will influence u at time step 1 with a probability α. Then, $mp(u)$=probability that T does not influence u × probability that m influences $u = (1 - \alpha) \times \alpha^2$.

Alternatively, if $T = \{u_2\}$, T will still influence u before m (because truth has precedence), this time with a probability α^2. Then, $mp(u) = (1 - \alpha^2) \times \alpha^2$. Thus, when a node u is closer to T or at the same distance from T than M (i.e. $D_M(u) \geq D_T(u)$), the probability of it getting infected by M is depreciated by the probability that it is not influenced by T.

On the other hand, if u is closer to m, then T cannot impact the probability that m infects u. For example, consider $T = \{u_3\}$. The earliest time step at which u_3 can influence u is 3. But m can infect u at time step 2. Thus, T cannot impact $mp(u)$ when $D_M(u) < D_T(u)$.

Therefore, under COICM with uniform cascade, any node $v \in V \setminus M$ can reduce the probability of a node w being infected by M if the length of the shortest path from v to w is less than or equal to the length of the shortest path from all nodes $m \in M$ to w. We further support this claim through empirical evidence. In Figure 5.3, the misinformation probability of nodes when $T = \emptyset$ is represented by the red line for the Arenas Email dataset and the Facebook dataset (details of these datasets are provided in Section 6.1.1). This value is known as the *vulnerability* of the nodes. The blue dots indicate the misinformation probability of all nodes that are either closer to or at the same distance from a node $t \in V \setminus M$ when t is added to T. We observe that t reduces the misinformation probability of all these nodes, thereby validating the proposition.



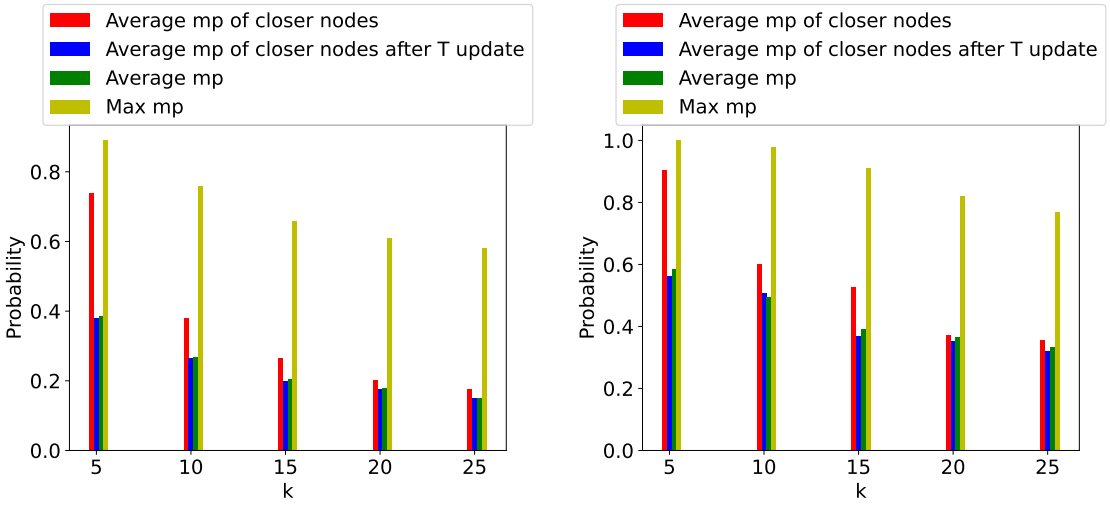
(a) Arenas Email dataset (b) Facebook dataset

Figure 5.3: misinformation probability reduces when $|SP_G(T, w)| \leq |SP_G(M, w)|$.

Our social welfare-based fairness objective is focused on improving the utility of the least well-off nodes, specifically by increasing the save probability of the most vulnerable nodes in the network. A greedy approach would involve selecting the node that maximizes the minimum save probability in each iteration. However, since the objective is not submodular, such an approach does not guarantee an optimal solution. Instead, we adopt an approach that allows us to improve the minimum save probability of the network while also enhancing the save probability of the maximum number of nodes in the network with each selected seed. This approach maximizes the number of nodes benefiting from the intervention, while ensuring that the worst-off nodes also receive benefits. Experimental results corroborate this observation.

Figure 5.4 demonstrates the effectiveness of the PROTECT heuristic in reducing the maximum misinformation probability of the network, while also decreasing the average misinformation probability of the network. The bars at different values of k (5, 10, 15, 20, and 25) indicate the change in probabilities as more nodes added are added to T. For each k value, we identify the nodes that are closer to or at the same distance from k^{th} seed than M. The first bar (red) and the second bar (blue) respectively represent the average misinformation probability of these closer nodes before and after T is updated. The third bar (green) corresponds to the average misinformation probability of the vulnerable nodes in the network, while the fourth bar (yellow) represents the maximum

misinformation probability of the vulnerable nodes in the network. We can see that addition of nodes to T reduces the misinformation probability of closer nodes and decreases the average and maximum misinformation probability of the network.



(a) Arenas email dataset

(b) Facebook dataset

Figure 5.4: Effectiveness of the PROTECT heuristic in reducing the maximum mp as well as the average mp of the network.

In summary, if a node u is the most vulnerable, adding a node t to T, where t is any node in $V \setminus M$ at a distance $d \leq D_M[u]$ to u, will reduce the vulnerability of u by at least a factor of $(1 - \alpha^d)$, when $p_T = \alpha$. Therefore, any node t at a distance $d \leq D_M[u]$ will improve the save probability of the network. Moreover, t will also increase the save probability of all the nodes in the "vulnerable" category that can be reached by t before M. Thus, an optimal choice for t is one that increases the minimum save probability of the network and reaches the maximum number of vulnerable nodes before M. Therefore, we utilize distance information to determine the selection of nodes for T.

5.1.2 Save probability estimation

In each iteration, we estimate the current save probability of all the nodes in the network to identify the nodes that are most vulnerable to misinformation. We use the Monte-Carlo technique to estimate the vulnerability of each node in the network and find the most vulnerable nodes. This is detailed in Algorithm 1.

This algorithm takes 5 parameters as inputs. They are: the graph (G) which is the network of interest, the misinformation starters (M), the truth starters (T), the propagation probability (α), and the number of simulations to be executed (R). To estimate the *sp*, we run R simulations of the COICM diffusion model. A simulation of the diffusion model returns the negatively influenced nodes in that simulation for the given set of parameters. We count the number of times each node gets infected by M over R simulations. *mp* is calculated as the average number of times a node is infected in R simulations. *sp* is computed using the formula in equation 2.1.

5.1.3 PROTECT algorithm

The proposed PROTECT algorithm is designed to identify the optimal set of seed nodes (T) that maximize the protection of the network against misinformation campaigns. Algorithm 2 as well as the flowchart in Figure 5.5 outline the steps involved in the method, which takes six parameters

Algorithm 1: Estimate_sp

input : $G(V, E), M, T, \alpha, R$
output: sp
Initialize $influence_count[i] \leftarrow 0 \forall i \in V$
for R *iterations* **do**
  $negatively_influenced_nodes \leftarrow$COICM(G, M, T, α)
  for i *in* $negatively_influenced_nodes$ **do**
    $influence_count[i] \leftarrow influence_count[i] + 1$
  end
end
for $i \in V$ **do**
  $mp[i] \leftarrow influence_count[i]/R$
  $sp[i] \leftarrow 1 - mp[i]$
end
return sp

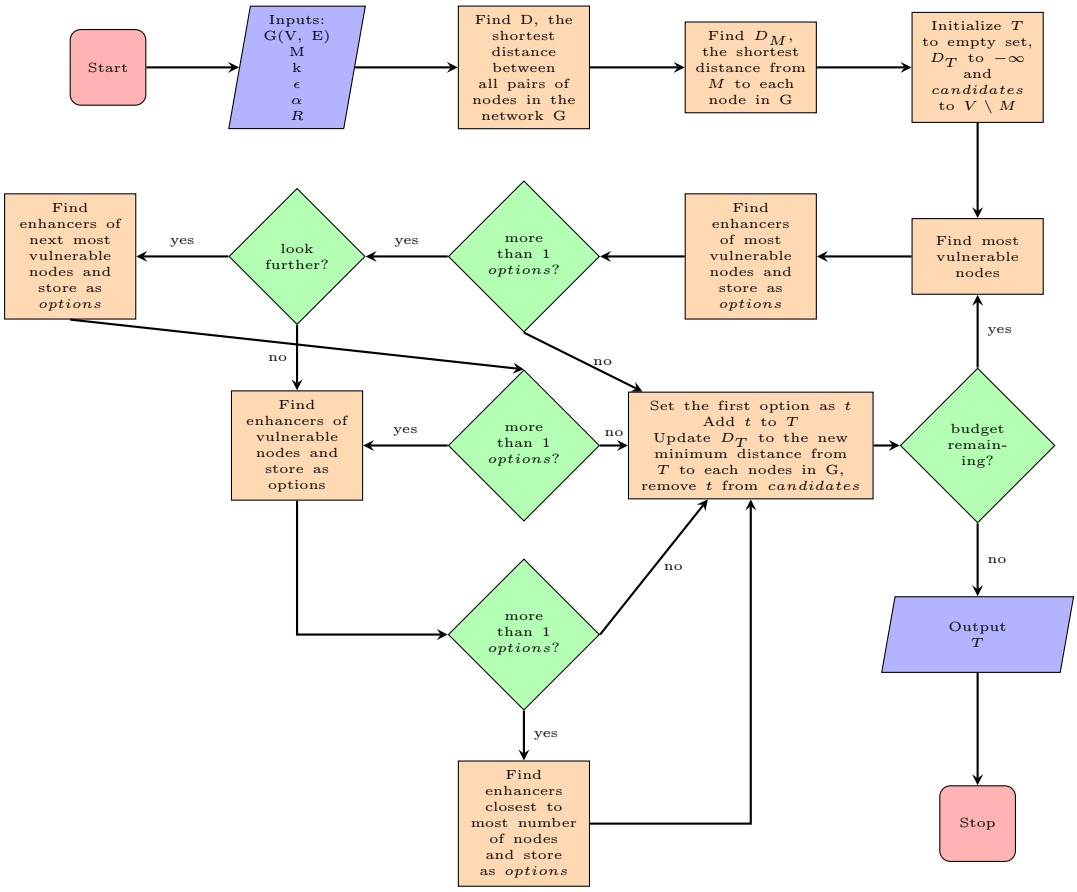


Figure 5.5: Flow chart of the PROTECT algorithm

as inputs: the network (G), the misinformation starters (M), the budget (k), a parameter ϵ which will be used for a $look - ahead$ function, the propagation probability (α), and the number of simulations (R) used to approximate the save probability.

To begin, the algorithm computes the shortest distance between all pairs of nodes in the network G. The adjacency matrix of a graph raised to a power l will give the number of paths

of length l from each node to every other node in the network [42]. We find the shortest distance between all pairs of nodes up to a length $max(D_M)$ as any shortest paths greater than this length is not useful for the algorithm. Additionally, we are only interested in the paths that do not cross any $m \in M$. Hence, we compute the distances on the graph after removing all incoming edges of M. To maintain simplicity, these specific details are omitted from Algorithm 2. The algorithm to compute the shortest paths is provided in the Appendix.

Next, the algorithm initializes the D_M array with the minimum distance possible from all $m \in M$ to each nodes in the network. The set of nodes reachable from M is identified and stored in the *vulnerable* set. The shortest distance from all nodes to T is initially set to $-\infty$ since T is empty at the beginning. The *candidates* set is initialized with all non-misinformation starter nodes, from which the T will be selected.

Then the algorithm proceeds to find T. To find each node $t \in T$, the current *most_vulnerable* nodes are identified. Algorithm 1 is used to estimate the save probability of all the nodes in the network for the current T. The *most_vulnerable* nodes are the nodes with minimum save probability (*min_sp*).

Once the *most_vulnerable* nodes are determined, the algorithm finds the nodes that can improve their save probability. We call such nodes hereon as *enhancers* and will use the term enhancing to denote "improving the save probability". These *enhancers* are all the nodes $v \in V \setminus M$ that are either closer to or at the same distance from *most_vulnerable* nodes as M is. These *enhancers* form the *options* set.

To ensure optimal utilization of each Monte-Carlo estimation, a $look - ahead$ feature is incorporated. If multiple nodes exist in the *options* set, the algorithm selects nodes that can also enhance the save probability of the next most vulnerable nodes. The *next_most_vulnerable* nodes are those with a save probability $\leq min_sp + \epsilon$. This allows for selecting options that have a wider spread if the *next_most_vulnerable* nodes do not share the same descendants with the *most_vulnerable* nodes.

From the remaining *options*, the algorithm selects the node that enhances the highest number of vulnerable nodes in the network. This criterion prioritizes options that benefit the largest number of individuals. In case of further ties, the algorithm chooses the node with the minimum average distance from the *vulnerable*. This will further help truth reach these nodes early, improving their save probability.

Finally, the first node from the *options* set is added to T, and D_T is updated to reflect the shortest distances from T to each node in the network. The selected node is removed from the candidate set to avoid its selection in subsequent iterations. This process is repeated k times to find k nodes for the truth starter set T

The PROTECT algorithm has a time complexity of $O(c|V|^3)$ for distance computation, where c is the diamter of the network in the worst case. We use numpy's matmul function in Python to implement distance computation, which is highly optimized. PROTECT runs k iterations to find k seed nodes. In each iteration, one Monte-Carlo estimation is done. This estimation has a time complexity $O(R|E|)$.

Algorithm 2: PROTECT heuristic

input : $G(V, E), M, k, \epsilon, \alpha, R$
output : T
$D \leftarrow$ shortest distance between all pairs of nodes in the network
$D_M[i] \leftarrow \min_{m \in M} D[m][i] \forall i \in V$
$vulnerable \leftarrow \{v | D_M[v] \neq \infty\}$
Initialize $D_T[v] \leftarrow -\infty \forall v \in vulnerable$
Initialize $candidates \leftarrow V \setminus M \cup T$
for k *iterations* **do**
  $sp \leftarrow Estimate_sp(G, M, T, \alpha, R)$
  $min_sp \leftarrow min_{v \in vulnerable} sp$
  $most_vulnerable \leftarrow \{v \in vulnerable | sp[v] = min_sp\}$
  $saved_count[c] \leftarrow |\{v \in most_vulnerable | D[c][v] \leq D_M[v]\}| \forall c \in candidates$
  $nodes_with_max_save_count \leftarrow \{c \in candidates | saved_count[c] = max(saved_count)\}$
  $options \leftarrow nodes_with_max_save_count$
  if $|options| > 1$ *and* $max(saved_count) = |most_vulnerable|$ **then**
    $next_most_vulnerable \leftarrow \{v \in vulnerable | sp[v] \leq min_sp + \epsilon\}$
    $saved_count[c] \leftarrow |\{v \in next_most_vulnerable | D[c][v] \leq D_M[v]\}| \forall c \in options$
    $nodes_with_max_save_count \leftarrow \{c \in options | saved_count[c] = max(saved_count)\}$
    $options \leftarrow nodes_with_max_save_count$
  end
  if $|options| > 1$ **then**
    $D_{T \cup c}[v] \leftarrow min(D[c][v], D_T[v]) \forall v \in vulnerable \forall c \in options$
    $saved_count[c] \leftarrow |\{v \in vulnerable | D_{T \cup c}[v] \leq D_M[v]\}| \forall c \in options$
    $nodes_with_max_save_count \leftarrow \{c \in options | saved_count[c] = max(saved_count)\}$
    $options \leftarrow nodes_with_max_save_count$
  end
  if $|options| > 1$ **then**
    $average_D[c] \leftarrow mean_{v \in vulnerable}(D_{T \cup c}) \forall c \in options$
    $nodes_with_min_avg_distance \leftarrow \{c \in options | average_D[c] = min(average_D[c])\}$
    $options \leftarrow nodes_with_min_avg_distance$
  end
  $t \leftarrow options[0]$
  $T \leftarrow T \cup [t]$
  $D_T \leftarrow min(D[t], D_T)$
  $candidates \leftarrow candidates \setminus t$
end
return T

5.2 Naive PROTECT method

This is a naive proximity-based method. In this method, we assume that the nodes closest to misinformation are the most vulnerable nodes. Further, among these nodes, we consider that nodes that have more number of misinformation starters close to them as more vulnerable.

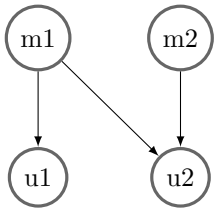


Figure 5.6: Figure to demonstrate the effect of number of closest Ms

Consider the network depicted in Figure 5.6 where $M = \{m1, m2\}$. Within this network, there are two nodes at a distance of 1 from M. Specifically, node $u1$ has one misinformation starter at a distance of 1, while $u2$ has two misinformation starters at this distance. It is evident that $u2$ has a higher likelihood of being infected by misinformation compared to $u1$ due to having more misinformation starters at a distance of 1. Assuming α represents the propagation probability of campaigns in this network, the misinformation probability induced on $u1$ will be α, whereas $u2$ will have a misinformation probability of 2α.

This observation can be generalized by considering the variable $n_M[u]$, which represents the number of misinformation starters at a distance of 1 from node u in the network. In such cases, the misinformation probability affecting node u due to all nodes in M at a distance of 1 can be expressed as $\alpha^{n_M[u]}$. Similarly, if $n_T[u]$ denotes the number of truth starter nodes at a distance of 1 from u, the probability of them influencing u at the first time step is $\alpha^{n_T[u]}$. Consequently, when the number of closest misinformation starters to a node u matches the number of truth starters at the same distance, the probability of saving the node at time step $D_M[u]$ will be equal to the initial vulnerability of that node. Therefore, minimizing the discrepancy between n_M and n_T will enhance the save probability of the nodes. Additionally, since we consider the number of closest instances of M rather than solely the distance to the closest M, we believe that this approach can effectively identify the most vulnerable nodes.

However, in this approach, we only consider the closest M's of a node, ignoring the total number of paths from all the $m \in M$ to the node, to avoid complex computations. Hence the algorithm is unaware of the actual infection probabilities and hence may underperform in networks with large number of paths.

5.2.1 Naive PROTECT algorithm

The steps involved in Naive PROTECT algorithm is presented in Algorithm 3 as well as the flowchart in FIgure 5.7. It takes four input parameters: the graph (G), the misinformation starters (M), the budget (k), and a parameter (ϵ) for the $look - ahead$ function.

During the algorithm execution, it maintains several variables, including the set of unprotected nodes ($target_nodes$), the shortest distances of misinformation starters to vulnerable nodes (D_M), the shortest distances of truth starters to vulnerable nodes (D_T), the number of the closest misinformation starters influencing each vulnerable node (n_M), and the number of the truth starters at distance D_M influencing each vulnerable node (n_T).

The algorithm proceeds in several steps. Firstly, it calculates the shortest distances between all pairs of nodes in the network. It then initializes $target_nodes$ which are the nodes at distance 1 from misinformation. It then compute n_M and n_T for each node in $target_nodes$. Based on these counts, it identifies the $most_vulnerable$ nodes.

The algorithm operates in two modes: the "Basic Protection" mode and the "Maximal Protection" mode. In the Basic Protection mode, the goal is to ensure that each node has at least one

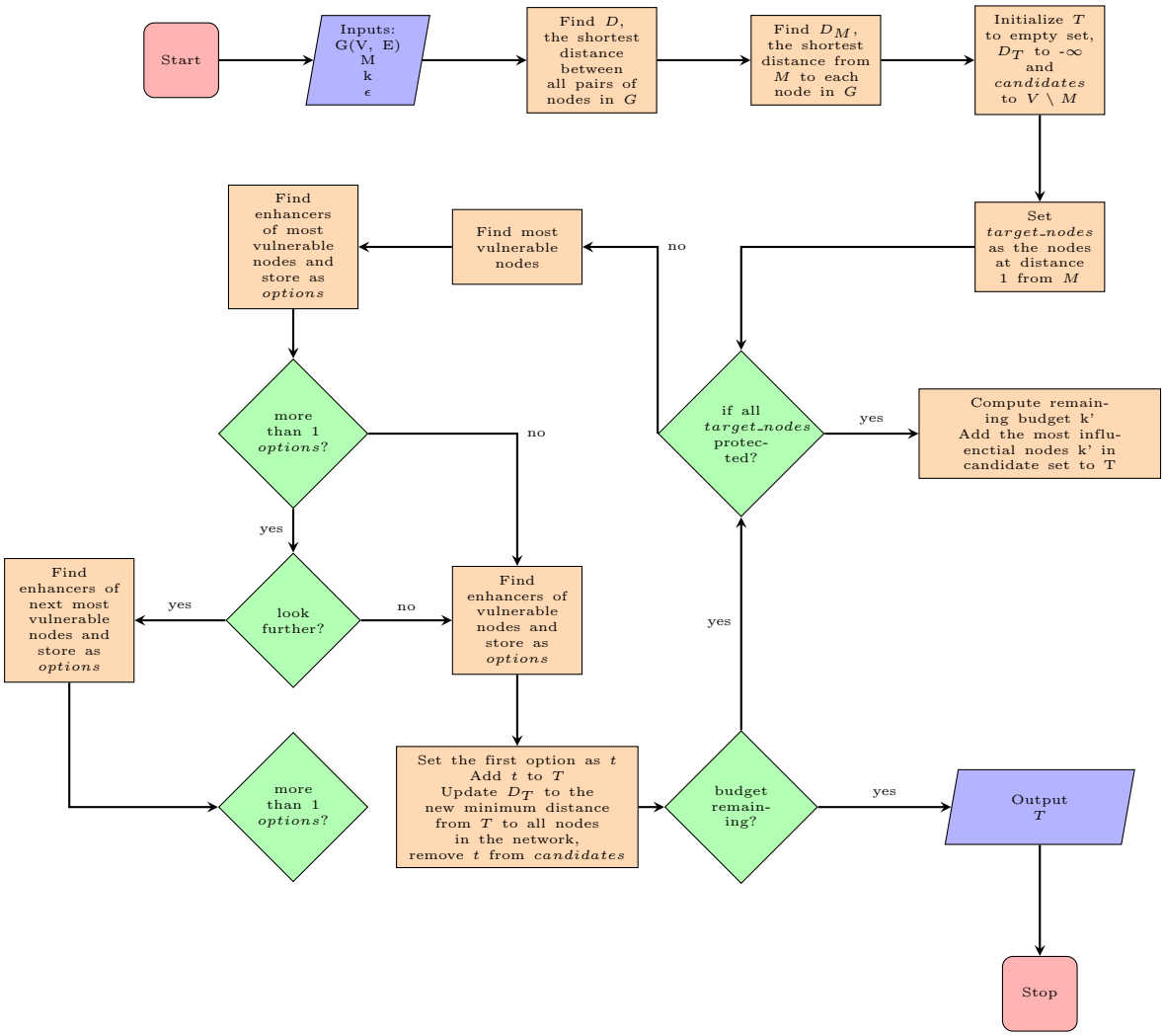


Figure 5.7: Flow chart of the Naive PROTECT algorithm

truth starter node ($t \in T$) that is closer to it or at the same distance to it than any misinformation starter. Once all nodes have achieved basic protection, the algorithm switches to the Maximal Protection mode. In this mode, the algorithm aims to increase the number of truth starter nodes closer to each vulnerable node, until the condition $n_M[u] \leq n_T[u]$ is satisfied for all vulnerable nodes u in the network.

The algorithm executes k iterations. In each iteration, the mode is selected based on the protection level of the nodes. A $diff$ value is computed for each *target_node* according to the selected mode. This value is used to identify the *most_vulnerable* nodes for an iteration. Specifically, the nodes with maximum value of $diff$ are chosen as the *most_vulnerable* nodes. For the Basic Protection mode, the value of $diff$ is set to n_M for the vulnerable nodes that do not yet have any protected nodes ($n_T = 0$).

$$diff[i] \leftarrow \begin{cases} n_M[i] & if\ i \in \{v \in target_nodes | n_T[v] = 0\} \\ -\infty & otherwise \end{cases} \tag{5.3}$$

This allows to prioritize enhancing the save probability of the nodes that have the most number of M closest to it. For the Maximal Protection mode, the value of $diff$ is computed as the relative

difference between n_M and n_T for each vulnerable node.

$$diff[i] \leftarrow \frac{n_M[i] - n_T[i]}{n_M[i]} \ \forall i \in target_nodes \tag{5.4}$$

This allows prioritizing enhancing the save probability of the nodes that have a proportionally lower sheilding from M.

Once the *most_vulnerable* nodes are identified, the algorithm determines which *candidate* node can enhance the save probability for the maximum number of *most_vulnerable* nodes. If multiple *candidate* nodes can achieve this, the algorithm looks further to identify the nodes that can also enhance the save probability for the *next_most_vulnerable* nodes. The *next_most_vulnerable* nodes are the nodes that have a $diff$ value close to that of the current *most_vulnerable* nodes. The input parameter ϵ controls the range of values to be considered.

If there are still multiple candidate nodes satisfying the above condition, the algorithm selects the node with the maximum influence on the *vulnerable* set, as this node benefit the most number of *vulnerable* nodes.

After each seed node is chosen, the variables D_T and n_T are updated. The new seed is also removed from the *candidates* set. For this method, we consider a node to be *protected* if the number of closest misinformation starters (n_M) is less than or equal to the number of closest truth starters for a node(n_T). That is, any node $u \in vulnerable$ is *protected* if

$$n_M[u] \leq n_T[u] \tag{5.5}$$

Thus, when a new node is added to T, *protected* nodes are removed from the *target_nodes*.

If all *target_nodes* are protected and there is still a budget, the algorithm adds to T the nodes that have the maximum influence on the vulnerable set. These are the nodes that can improve the save probability for the most number of vulnerable nodes.

In summary, the Naive PROTECT algorithm aims to select an optimal set of seed nodes for protection in order to minimize negative influence in a network by prioritizing the nodes with the most number of closest misinformation starters. By *protecting* all the *target_nodes* in the network, it aims at enhancing the save probability for the most vulnerable nodes.

Distance computation is the only time-consuming task in this algorithm. As it does not use Monte-Carlo simulations it is much faster than the PROTECT method.

Algorithm 3: Naive PROTECT Heuristic

input : $G(V, E), M, k, \epsilon$
output: T
$D \leftarrow$ shortest distance between all pairs of nodes in the network
$D_M[i] \leftarrow \min_{m \in M} D[m][i] \forall i \in V$
$n_M[i] \leftarrow |\{m \in M | D[m][v] = D_M[v]\}| \forall i \in V$
$vulnerable \leftarrow \{v | D_M[v] \neq \infty\}$
Initialize $D_T[v] \leftarrow -\infty \forall v \in vulnerable$,$n_T[i] \leftarrow 0 \forall i \in vulnerable$
Initialize $candidates \leftarrow V \setminus M$
Initialize $target_nodes \leftarrow \{v \in vulnerable | D_M[v] = 1\}$
for k *iterations* **do**
  if $|target_nodes|=0$ **then**
    $all_protected \leftarrow True$
    break
  end
  $min_n_T \leftarrow min_{v \in target_nodes} n_T$
  if $min_n_T = 0$ **then**
    Set $diff$ using equation 5.3
    $\epsilon' \leftarrow 1$
  else
    Set $diff$ using equation 5.4
    $\epsilon' \leftarrow \epsilon$
  end
  $max_diff \leftarrow \max(diff)$
end
$most_vulnerable \leftarrow \{v \in target_nodes | diff[v] = max_diff\}$
$saved_count[v] \leftarrow |\{v \in most_vulnerable | D[c][v] \leq D_M[v]\}| \forall c \in candidates$
$nodes_with_max_save_count \leftarrow \{c \in candidates | saved_count[c] = max(saved_count)\}$
$options \leftarrow nodes_with_max_save_count$
if $|options| > 1$ *and* $max(saved_count) = |most_vulnerable|$ **then**
  $next_most_vulnerable \leftarrow \{v \in vulnerable | diff[v] \geq max_diff - \epsilon'\}$
  $saved_count[v] \leftarrow |\{v \in most_vulnerable | D[c][v] \leq D_M[v]\}| \forall c \in options$
  $nodes_with_max_save_count \leftarrow \{c \in options | saved_count[c] = max(saved_count)\}$
  $options \leftarrow nodes_with_max_save_count$
end
if $|options| > 1$ **then**
  $saved_count[c] \leftarrow |\{v \in vulnerable | D[c][v] \leq D_M[v]\}| \forall c \in options$
  $nodes_with_max_save_count \leftarrow \{c \in options | saved_count[v] = max(saved_count)\}$
  $options \leftarrow nodes_with_max_save_count$
end
$t \leftarrow options[0]$
$T \leftarrow T \cup [t]$
$candidates \leftarrow candidates \setminus t$ $D_T[v] \leftarrow min(D[t], D_T) \forall v \in vulnerable$
$n_T[v] \leftarrow |\{t \in T | D[t][v] = D_M[v]\}| \forall v \in vulnerable$
$target_nodes \leftarrow target_nodes \setminus \{v \in vulnerable | n_T[v] \geq n_M[v]\}$
if $all_protected$ *and* $|T| < k$ **then**
  $saved_count[c] \leftarrow |v \in vulnerable | D[c][v] \leq D_M[v]| \forall c \in candidates$
  $options \leftarrow \{c \in options | saved_count[v] = max(saved_count)\}$
  $k' \leftarrow |T| - k$
  $top_options \leftarrow$ k' nodes with the highest saved_count $T \leftarrow T \cup top_options$
end
return T

5.3 Baselines

In the context of Fair Influence Maximization, Fish et al. [19] proposed three algorithms in their work. In this study, we consider two of them: Greedy heuristic and Myopic heuristics, as baselines to address the NIM problem.

5.3.1 Greedy Welfare Maximin Method

This algorithm greedily selects the nodes that maximize the minimum save probability of the network. The procedure followed is outlined in Algorithm 4.

This algorithm takes as input 5 parameters: the network (G), the misinformation starters (M), the budget (k), the propagation probability (α), and the number of simulations (R) used to approximate the save probability.

The algorithm (Algorithm 4) runs for k iterations. In each iteration, for each candidate node added to T, the save probabilities of all the nodes are estimated. The minimum save probability obtained with each candidate node c ($min_sp[c]$) is noted. $maximin_sp$ is computed as the maximum value achieved for min_sp across all candidate nodes. The candidate node whose min_sp is equal to $maximin_sp$ is the optimal seed.

To further improve the results, we introduce additional optimizations to the greedy algorithm. If there are multiple optimal seeds, we compute for each of these nodes min_count, which is the number of nodes that these candidate nodes save with a save probability of $maximin_sp$. The node with the minimum min_count is selected. If multiple nodes have the same min_count, we consider their average save probabilities (avg_sp) and choose the one with the highest average save probability. This ensures that each node added to T provides the best possible benefit to nodes in the network.

Algorithm 4: Greedy Welfare Maximin Heuristic

input : $G(V, E), M, k, \alpha, R$
output: T
$vulnerable \leftarrow V \setminus M$
$candidates \leftarrow V \setminus M$
for k *iterations* **do**
  for c *in candidates* **do**
    $sp \leftarrow Estimate_sp(G, M, T \cup [c], \alpha, R)$
    $min_sp[c] \leftarrow min_{v \in vulnerable} sp$
    $avg_sp[c] \leftarrow mean_{v \in vulnerable} sp$
    $min_count[c] \leftarrow |\{v \in vulnerable | sp[v] = min_sp[c]\}|$
  end
  $maximin_sp \leftarrow max(min_sp)$
  $candidates \leftarrow \{c \in candidates | min_sp[c] = maximin_sp\}$
  if $|candidates| > 1$ **then**
    $min_count_nodes \leftarrow \{c \in candidates | min_count[c] = min_{i \in candidates}(min_count)\}$
    $candidates \leftarrow min_count_nodes$
  end
  if $|candidates| > 1$ **then**
    $max_avg_prob_nodes \leftarrow \{c \in candidates | avg_sp[c] = max_{i \in candidates}(avg_sp)\}$
    $candidates \leftarrow max_avg_prob_nodes$
  end
  $t \leftarrow candidates[0]$
  $T \leftarrow T \cup [t]$
  $candidates \leftarrow candidates \setminus t$
end
return T

However, save probability estimation can be computationally expensive, requiring $k \times |candidates| \times R$ simulations of the diffusion model. Therefore, this method is not suitable for large networks.

5.3.2 Myopic Welfare Maximin Heuristic

The Myopic Welfare Maximin algorithm (Algorithm 5) provides a faster alternative. This method requires estimating the save probability k times for selecting k truth starters. At the beginning of each iteration, the save probability of the nodes with the current T is estimated. The node with the minimum save probability is then added to T. As a node is added to T, its save probability becomes 1 since it will always be saved.

The Myopic Welfare Maximin algorithm selects nodes that are least likely to be saved. These nodes are typically located farthest from the current set of nodes in T and closest to M. By selecting such nodes, the algorithm aims to increase the spread of T and benefit nodes irrespective of their position in the network. This approach benefits the most vulnerable node in the network, which aligns with our objective of achieving social welfare fairness.

However, in scenarios where multiple nodes have the same minimum save probability, the Myopic algorithm randomly selects one node among them, which may not yield the optimal solution. Additionally, if the nodes with the minimum save probability are poorly connected, the benefit of each node added to T on other worst-off nodes may be minimal. Consequently, this method exhibits poor worst-case performance when there are a large number of nodes with the minimum save probability and/or when these nodes are poorly connected.

Algorithm 5: Myopic Welfare Maximin Algorithm

input : $G(V, E), M, k$
output: T
Initialize $vulnerable \leftarrow V \setminus M$
for k *iterations* **do**
  $sp \leftarrow Estimate_sp(G, M, T)$
  $t \leftarrow argmin_{v \in vulnerable} sp$
  $T \leftarrow T \cup [t]$
  $vulnerable \leftarrow vulnerable \setminus t$
end
return T

PROTECT algorithm is an extention of the Myopic Welfare Maximin algorithm. It addresses the drawbacks of the Myopic approach by selecting nodes T based on their ability to reach the most vulnerable nodes, rather than directly selecting the most vulnerable nodes themselves. Moreover, PROTECT algorithm incorporates multiple tie-breaking strategies to select nodes that provide the best benefit to the highest number of nodes in the network, surpassing the random selection approach used by the Myopic Welfare Maximin Heuristic. As a result, PROTECT is expected to have a better performance than the Myopic Heuristic, while maintaining a similar running time.

Chapter 6

Experimental Setup

In this chapter, we outline the experimental setup for evaluating the performance and fairness of our proposed methods. We begin by describing the datasets used in our experiments, including both real-world and synthetic datasets, in Section 6.1. Subsequently, in Section 6.2, we present the baselines that serve as benchmarks for comparison. To assess the effectiveness of our approaches, we define evaluation metrics in Section 6.3, encompassing both performance and fairness evaluation.

6.1 Datasets

We use real-world datasets as well as synthetic datasets of different characteristics. These datasets provide diverse network structures and characteristics to assess the performance and generalizability of our approaches.

6.1.1 Real world datasets

In this section, we introduce the real-world datasets used in the experiments. The characteristics of these datasets are presented in Table 6.1. We carefully select representative real-world datasets that capture different type of online social interactions.

To simplify the experimental setup and maintain computational feasibility, we focus on the largest connected component of the graphs rather than considering the entire graphs. This decision is motivated by the fact that each seed node can only improve the save probability of the nodes it can reach. Consequently, if there are disconnected components in the graph, it would require adding seeds to each individual component. This could potentially necessitate a substantial number of seeds, especially when dealing with numerous disconnected nodes. Thus, we opt to run the experiments on the largest strongly connected component of the graphs.

The following are the real-world datasets used in our experiments:

- **socfb-Caltech36**: This is a snapshot of the online social network Facebook, where nodes in the network represent users, and edges represent *friend* relationships between users. It is a small dataset containing 796 nodes and 16,656 undirected edges. It is a disconnected graph. We use the largest connected component comprising 769 nodes and 33,302 directed edges. For this component, the average out-degree is 43.7, the clustering coefficient is 0.41, the average shortest path is 2.3 and the diameter is 6. The degree distribution is plotted in Figure 6.1a. The degree distribution does not follow a power-law distribution, which suggest that this network does not exhibit the characteristic hierarchy seen in scale-free networks. This dataset has been used in influence maximization research [75] and is available in Network Data Repository [48].

- **Arenas Email**: This is a network capturing the email communication network at the University Rovira i Virgili. Nodes of the network are users and each edge of the network represents email communication. This dataset was published in Guimera et al. [25]. It consists of 1133 nodes and, 5451 undirected edges. This is a fully connected network. In its directed version, it comprises 10,902 directed edges. The average out-degree is 9.6, the clustering coefficient is 0.22, the average shortest path is 3.6, and the diameter is 8. The degree distribution is plotted in Figure 6.1b. This distribution lacks the characteristic long tail typical to social networks, suggesting that the network is not scale-fee. This dataset is available in KONECT datasets [3]. It has been used in fair influence maximization research [19].

- **Facebook**: This dataset also represents a snapshot of the social network Facebook and was created by McAuley et al. [37]. Its directed version consists of 4039 nodes and, 176468 edges. This network has been extensively used in research in influence propagation problems : [19, 7, 73]. The average out-degree is 43, the clustering coefficient is 0.60, the average shortest path is 3.69 and the diameter is 8. The degree distribution is plotted in Figure 6.1c. The high clustering coefficient and low average shortest path demonstrate the small-world property. The degree distribution has a heavy tail, which is characteristic of scale-free networks. This dataset is available in SNAP datasets [36].

- **CA-AstroPh**: This dataset represents the arXiv Astro Physics collaboration network, which captures scientific collaborations between authors' papers submitted to the Astro Physics category in the e-print arXiv [1]. It is an undirected graph whose largest connected component has 18772 nodes. In its directed version, it has 396160 edges. The average out-degree is 21, the clustering coefficient is 0.63, the average shortest path is 4.17 and the diameter is 14. The degree distribution is plotted in Figure 6.1d. It displays the small-world property with small average shortest path and high clustering coefficient. The degree distribution also resembles the power-law distribution. This dataset is used in [7] and is available in SNAP datasets [36].

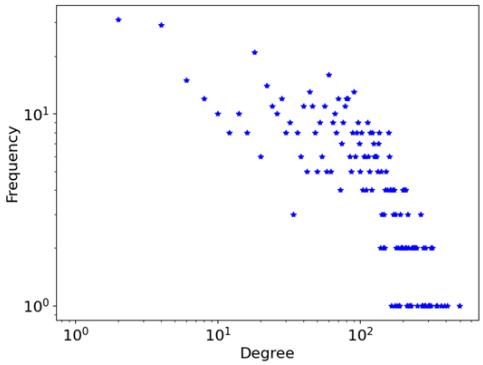
- **Twitter**: This dataset represents a snapshot of the online social network Twitter, where nodes represent Twitter users, and directed edges indicate the *following* relationship. It was also created by McAuley et al. [36] and comprises 81,306 nodes and 1,768,149 directed edges. The largest connected component consists of 68,413 nodes and 1,685,152 directed edges. For its largest connected component, the average out-degree is 21, the clustering coefficient is 0.63, the average shortest path is 4.13 and the diameter is 7. The degree distribution is plotted in Figure 6.1e. This dataset has been used in the following studies: [7, 70]

Dataset	citation	nodes	edges	Average Degree	Clustering Coefficient	Average shortest path	diameter	Degree Assortativity
socfb-Caltech36	[48]	769	33302	43.7	0.41	2.3	6	-0.066
Arenas email	[3]	1133	10902	9.6	0.22	3.6	8	0.078
Facebook	[36]	4039	176468	43	0.60	3.7	8	0.063
CA-AstroPh	[36]	18772	396160	21	0.63	4.17	14	0.20
Twitter	[36]	68413	1685152	24.6	0.43	4.13	7	-0.015

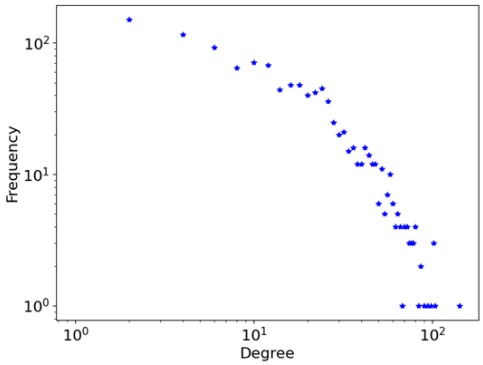
Table 6.1: Summary of the characteristics of Real-World datasets used. The properties mentioned are for the largest connected component of the graphs when the graph contains multiple components

We see that these networks exhibit diverse topologies and sizes, ensuring a comprehensive evaluation of our algorithms.

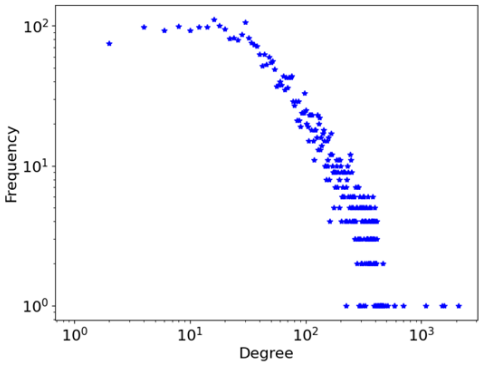
Next, we introduce the synthetic networks generated using the HICH-BA[7] algorithm.



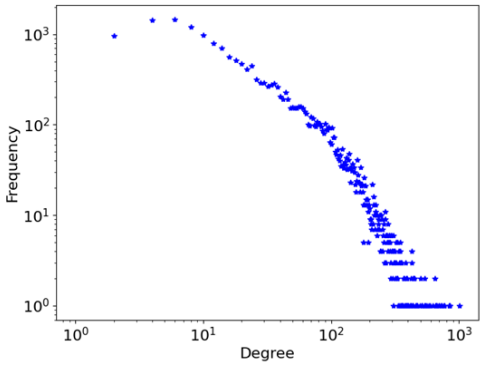
(a) socfb-Caltech36 Dataset



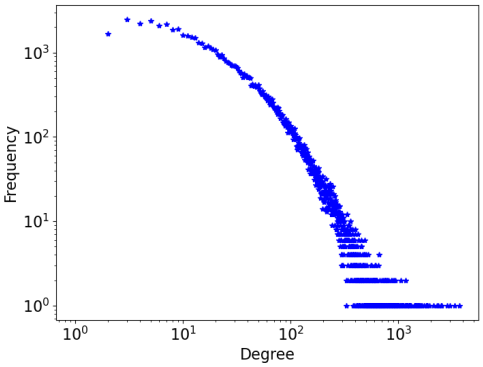
(b) Arenas Email Dataset



(c) Facebook Dataset



(d) CA-AstroPh Dataset



(e) Twitter Dataset

Figure 6.1: Degree distribution of the real-world datasets

6.1.2 Synthetic Networks

We introduce a set of synthetic networks generated using the HICH-BA algorithm proposed by Gutierrez in [7]. HICH-BA is short for HIgh Clustering Homophily Barabási-Albert. The HICH-BA model is specifically designed to capture key properties observed in real-world networks, including the small-world property, scale-free property, and community structure.

The small-world property refers to the phenomenon where a network exhibits both high clus-

tering coefficient (indicating a tendency for nodes to form clusters) and short average path lengths (indicating efficient communication between nodes). The HICH-BA algorithm incorporates a parameter, denoted as p_T, which controls the probability of triadic closure. Triadic closure refers to the likelihood of a node connecting to the neighbor of its neighbor, promoting clustering in the network.

The scale-free property is characterized by a highly skewed degree distribution, where most nodes have a low degree while a few nodes have a significantly higher degree. The HICH-BA algorithm includes a parameter, denoted as p_{PA}, which governs the probability of preferential attachment. Preferential attachment refers to the tendency of nodes to connect to already well-connected nodes, leading to the emergence of highly connected hubs in the network.

The community structure property signifies the presence of distinct groups or communities within the network, where nodes within the same community are more closely connected to each other compared to nodes in different communities. The HICH-BA algorithm incorporates a homophily parameter, h to control the tendency of nodes to connect to others from the same community, reinforcing the community structure.

We have generated five synthetic networks using different combinations of these parameters. Each network consists of five communities, with population proportions of [0.54, 0.3, 0.15, 0.005, 0.005]. These configurations are taken from [7]. The specific configurations and characteristics of the networks generated are summarized in the table below. :

Dataset	**nodes**	**edges**	p_T	p_{PA}	h	**Average Degree**	**Clustering Coefficient**	**Average shortest path**	**diameter**	**Degree Assortativity**
SN1	10000	200000	0.9	0.9	0.9	20	0.57	2.95	7	-0.39
SN2	10000	40000	0.9	0.9	0.9	4	0.31	4.35	13	-0.10
SN3	10000	100000	0.9	0.1	0.9	10	0.48	3.92	10	0.11
SN4	10000	100000	0.2	0.9	0.9	10	0.13	3.45	9	-0.09
SN5	10000	100000	0.9	0.9	0.2	10	0.40	3.35	11	-0.35

Table 6.2: Summary of the characteristics of Synthetic Datasets

Each configuration of parameters produces a network with distinct characteristics. The degree distribution of the synthetic networks is illustrated in Figure 6.2.

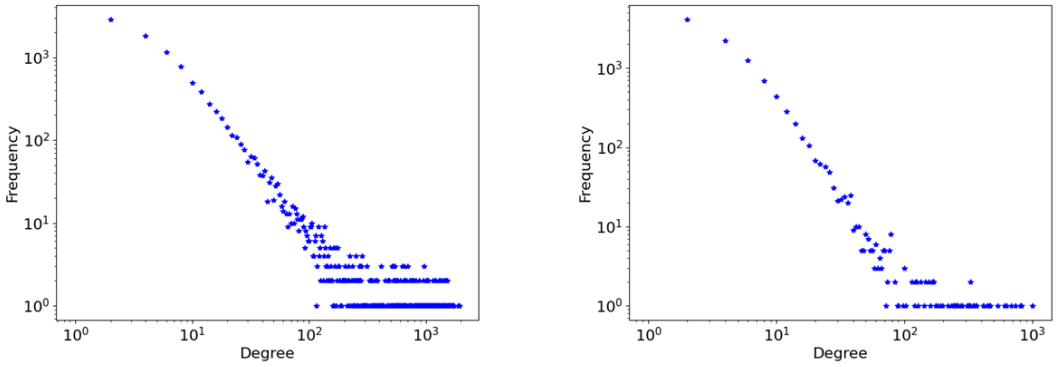
For the first two networks, we keep the p_T, p_{PA} and h parameters high and altered the number of edges. In the first configuration (SN1), we set the number of edges to 200000. This gives us a dense network with strong connections between neighboring nodes, resulting in a higher likelihood of clustering. The network exhibits a high clustering coefficient and a low average shortest path length.

In contrast, in the second configuration, we set the number of edges to 40000. This gives us a network with lesser number of connections. This network a has a longer diameter, low clustering coefficient and longer average shortest paths, indicating less efficient communication between nodes.

In the subsequent three configurations, we maintain the same number of nodes and edges while modifying one parameter at a time. In one configuration, we keep the homophily (h) and clustering (p_T) parameters high while eliminating preferential attachment (p_{PA}). As a result, these networks demonstrate a high clustering coefficient but longer average shortest paths. This suggests a network structure where nodes tend to cluster together, but the overall distance between nodes is increased.

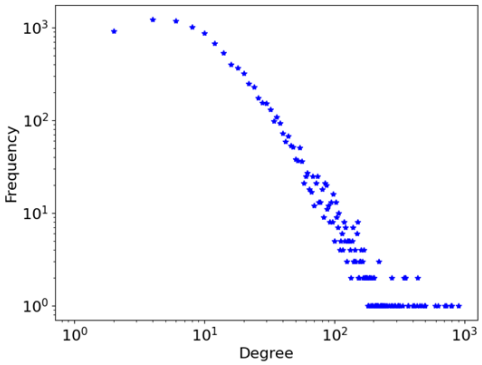
In the next configuration, the clustering parameter (p_T) is reduced while the homophily (h) and preferential attachment (p_{PA}) parameters remain high. This particular configuration yields a network with a low clustering coefficient and a moderate average shortest path length.

Finally, the last configuration minimizes the homophily parameter (h) while keeping the other parameters high. The resulting network exhibits a moderate clustering coefficient and average shortest path length.

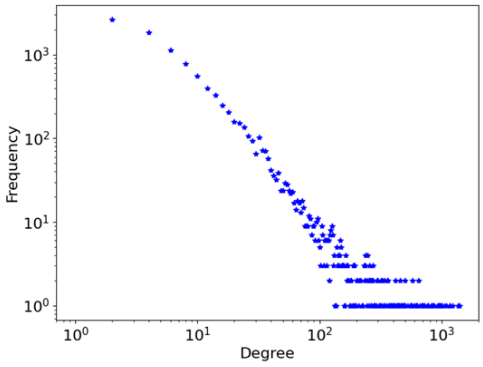


(a) SN1

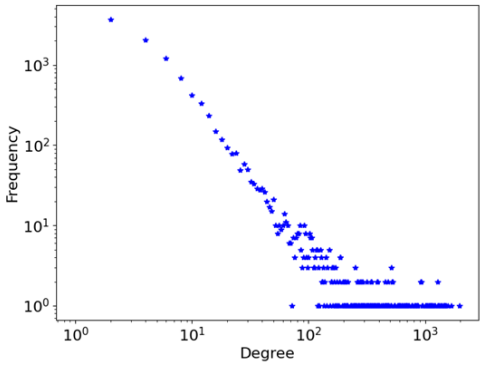
(b) SN2



(c) SN3



(d) SN4



(e) SN5

Figure 6.2: Degree distribution of the synthetic datasets

These different network configurations allow us to investigate the impact of each parameter on the performance of our proposed methods, providing insights into the effectiveness of our approach under varying network structures and properties.

6.2 Baselines

In this section, we present the fairness-agnostic and fairness-aware baselines used in our study.

6.2.1 Fairness-Agnostic Baselines

The following fairness-agnostic baselines are used:

- **Highest out-degree** (degree): Node with the highest out-degree in the network are selected. This heuristic is commonly used as a baseline for Influence Maximization [31] and NIM [11, 74, 40] problems.
- **PageRank** (pagerank): Nodes with the highest PageRank scores are selected. This heuristic has also been employed in other negative influence minimization studies, such as : [61, 74].
- **RBF** (rbf) [23]: Rumor-aware betweenness values (RBVs) are computed for each non-seed node in the network by counting the number of paths from nodes in M that pass through v. Nodes with the highest RBVs are chosen as T.
- **BIOG** (biog) [38]: This method computes the blocking score of each node on Maximum Influence Out-Graph (MIOG) structures. Nodes with the maximum blocking scores are selected as T.
- **CMIA-O** (cmia-o) [72]: In this method, MIA (Maximum Influence Arborescence) structures are used to estimate the blocking potential of nodes. This method is scalable and fast.
- **TIB-Solver** (tib) [61]: The *threat* and *leverage* are computed for all the nodes, and the nodes that maximally block the negative influence before a deadline are selected. We have set the *deadline* to the length of the longest shortest path from M to all nodes in the network.
- **TIBM-M** (tibmm) [40]: They use temporal RR sets similar to TIB-Solver. However, they provide theoretically guaranteed and have a higher time complexity. A parameter ϵ is used to limit the desired efficiency of the output. This parameter controls the number of samples. We have set this value to 0.2 for the algorithm to end in a reasonable amount of time. The deadline is set the same as TIB-Solver.
- **RPS** (rps) [60]: This method also utilizes a form of Reverse Reachable Sets. ϵ and the *deadline* are configured as in TIBM-M.

6.2.2 Fairness-Aware Baselines

The following fairness-aware baselines are considered. The first two (FWRRS and Fair CMIA-O) focus on group fairness, while the last two (Greedy Welfare Maximin and Myopic Welfare Maximin) consider individual fairness:

- **FWRRS** (fwrrs) [7]: In this group-fairness aware method, Weighted Reverse Reachable Sets are used to estimate the blocking power of each node. The nodes that maximize the minimum save fraction across all communities are selected as T. They use Louvain community detection method to detect communities. This community information is then used by the algorithm to achieve group fairness.
- **Fair CMIA-O** (fair-cmia-o): This is the (group) fairness-aware version of CMIA-O. It was proposed by Gutierrez in [7] to compare its performance with her proposed method. This method also uses the Louvain community detection method to detect communities when communities are not specified in the dataset.
- **Greedy Welfare Maximin** (greedy_maximin): This is the Heuristic explained in section 5.3

- **Myopic Welfare Maximin** (myopic_maximin): This is the Heuristic explained in section 5.4

6.3 Evaluation metrics

Performance evaluation metrics are used to assess the effectiveness and efficiency of the algorithms in achieving their objectives. In our study, we consider the following performance evaluation metrics:

6.3.1 Performance Evaluation metrics

- **Saved Fraction**: The saved fraction is used to measure the efficiency of a method in terms of the expected fraction of vulnerable nodes that are saved. It provides insights into how effective the algorithm is in its negative influence minimization objective.

$$\text{saved fraction} = \frac{\mathbb{E}[|\pi(M|\emptyset)|] - \mathbb{E}[|\pi(M|T)|]}{\mathbb{E}[|\pi(M|\emptyset)|]} \tag{6.1}$$

  where $\mathbb{E}[|\pi(M|\emptyset)|]$ represents the expected number of vulnerable nodes in the absence of any truth starters, and $\mathbb{E}[|\pi(M,T)|]$ represents the expected number of vulnerable nodes that are saved by T. The saved fraction ranges from 0 to 1, with a higher value indicating a higher proportion of vulnerable nodes being saved.

- **Average Save probability**: The average save probability is an evaluation metric that measures the effectiveness of a method in terms of the average probability of nodes in the network being saved. The average save probability can be expressed as:

$$\text{Average sp} = \frac{\sum_{v \in V} sp}{|V|} \tag{6.2}$$

  A higher average save probability indicates that, on average, the nodes in the network have a higher likelihood of being saved.

These performance evaluation metrics provide insights into the effectiveness of the algorithms in mitigating negative influence.

6.3.2 Fairness evaluation metrics

Fairness evaluation metrics are employed to assess the fairness of the algorithms in their decision-making process. In our study, we consider the following fairness evaluation metrics:

- **Maximin sp**: Maximin sp, or the minimum save probability, is a fairness evaluation metric that focuses on the welfare of vulnerable nodes in the network. It measures the lowest probability of a node being protected from negative influence in a network.

$$\text{Maximin sp} = min_{v \in vulnerable} sp \tag{6.3}$$

  Since it represents a probability, the value ranges from 0 to 1, with a higher value indicating that nodes in the network have a higher guarantee of being saved.

- **Consistency**: Consistency is a metric that has been used to evaluate equity fairness in network problems [76, 34]. This metric will be used in this study to measure how consistent the method is in providing similar protection to similar nodes. The similarity between nodes is measured in terms of their initial vulnerabilities (refer to Section 2.5.2 Similarity Metric).

$$\text{Consistency} = 1 - \frac{\sum_i \sum_j |sp(i) - sp(j)|.s_{ij}}{\sum_i \sum_j s_{ij}} \tag{6.4}$$

Here, s_{ij} is the similarity between any two nodes i and j in the network. It is computed using Equation 2.7.

A higher consistency value indicates that the proposed solution exhibits a higher level of equity fairness by providing comparable protection, in terms of save probability, to nodes that are similar.

- **Gini coefficient**: The Gini coefficient is a measure of inequality and is commonly used to assess fairness in economic contexts. Gong et al. uses Gini coefficient to measure fairness for the influence maximization problem in [22]. In our study, we adapt the Gini coefficient to evaluate the inequality in save probabilities across the network.

$$\text{gini coefficient} = \frac{\sum_i \sum_j |sp(i) - sp(j)|}{2|vulnerable| \sum_j sp(j)} \tag{6.5}$$

A lower Gini coefficient indicates a more equitable distribution of save probabilities. By using the Gini coefficient, we can assess the fairness of the method in terms of how evenly it distributes protection to the vulnerable nodes in the network.

Chapter 7

Results

In this chapter, we evaluate the performance of our proposed methods with fairness-agnostic and fairness-aware baselines discussed in the previous chapter. Experiments are conducted on real-world and synthetic datasets and we present the results for the performance and fairness evaluation metrics. This chapter is organized as follows: The experimental setting is described in Section 7.1. Subsequently, Section 7.2 examines the results and analysis of experiments conducted on the real-world networks discussed in Section 6.1.1, specifically focusing on evaluation with the highest degree misinformation starters (Section 7.2.1) and evaluation with random misinformation starters (Section 7.2.2). Section 7.3 presents the outcomes of experiments performed on real-world networks using an alternate diffusion model, providing insightful analysis. Finally, in Section 7.4, we present the results and analysis of experiments conducted on synthetic datasets presented in Section 6.1.2. The findings obtained from these experiments contribute to a comprehensive understanding of the effectiveness and performance of our proposed fair NIM solutions.

7.1 Experiment Setting

In this study, we employ the Campaign Oblivious Independent Cascade Model with Uniform Cascade. Under this model, each active node in the network attempts to activate its neighbors with a constant probability. The probability of a node being activated by a campaign increases as more of its neighbors become activated by the same campaign. This mechanism reflects the real-life phenomenon where individuals are more inclined to believe information if many of their friends also believe it. The uniform cascade model has been widely used in previous research on influence maximization [19, 49] and NIM [17].

For our experiments, we select a set of 10 nodes as the misinformation starters M. To determine the appropriate propagation probability (α) for the real-world datasets, we conduct experiments with different values. The outreach of M is defined as the percentage of the nodes M can influence by the end of the diffusion process, assuming there are no truth spreaders present in the network. To ensure a significant outreach for M, we choose α such that M's outreach is at least 70%.

Table 7.1 presents the selected propagation probabilities and the corresponding outreach values for each dataset. This ensures a consistent and controlled diffusion process across the experiments, allowing for meaningful comparisons and evaluations.

In the case of synthetic networks, we conduct experiments with multiple propagation probabilities to investigate their impact on the performance of the methods. By exploring a range of propagation probabilities, we can gain insights into how different levels of information diffusion affect the effectiveness of the different methods.

To comprehensively evaluate the performance of our proposed method under different selection strategies for the campaign initiators M, we consider two distinct settings. In [12], Carnes et al. reported that the Highest Out-degree Heuristic is the best strategy for the first campaign when there are two competing campaigns. Hence, in the first setting, we experiment with selecting

Dataset	α	outreach
socfb-Caltech36	0.1	77%
Arenas email	0.3	74%
Facebook	0.2	78%
CA-AstroPh	0.25	70%
Twitter	0.25	70%

Table 7.1: Propagation probability (α) and corresponding outreach for each data

M using this strategy. In the second setting, we examine the methods' performance when M is randomly chosen, disregarding any specific selection policies that may be employed by the misinformation campaign.

In our experimental setup, we set the budget k to 25, representing the maximum number of truth spreaders that can be selected to counter the misinformation campaign initiated by M. We believe that a budget of this magnitude provides a good assessment of the methods' performance in combating misinformation while allowing us to complete the computations in a reasonable amount of time. To capture the effectiveness of the methods at different budget levels, we measure and analyze the evaluation metrics at specific intervals. Specifically, we evaluate the metrics at k values of 5, 10, 15, 20, and 25.

The implementation of all algorithms in our study has been carried out in Python, taking advantage of its versatility and extensive libraries for network analysis and computations. The experiments were conducted on a high-performance server with the following specifications:

- CPU: 2x Intel Xeon Broadwell-EP 2683v4 @ 2.1GHz (64 hyperthreads)
- RAM: 1024GB
- GPU: 8x ASUS Turbo GeForce GTX 1080 Ti (RAM, 3584 CUDA cores, compute capability 6.1)

7.2 Experiments on Real-world datasets

7.2.1 Evaluation with the highest degree misinformation starters

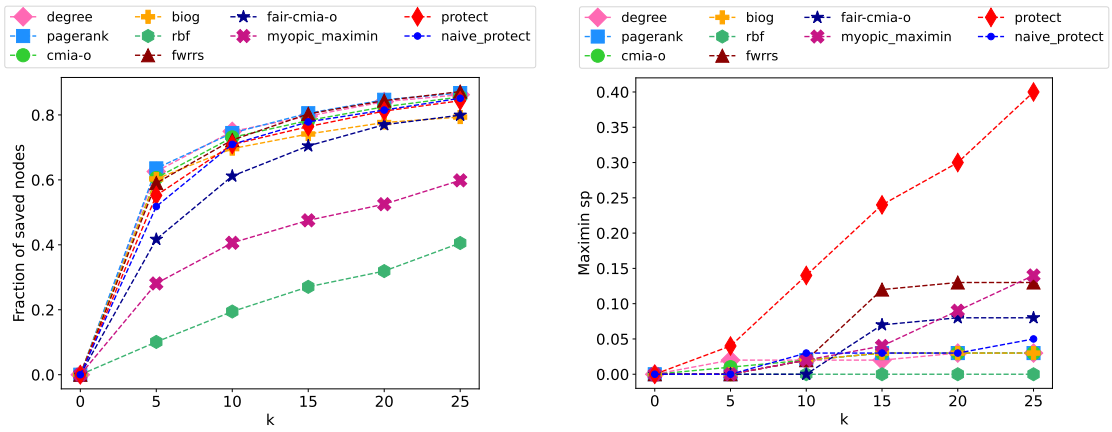
In this section, we present the results of the experiments on real-world datasets with the highest out-degree nodes as misinformation starters. The Saved Fraction and Maximin sp achieved on these datasets are visualized in Figure 7.1, Figure 7.2, Figure 7.3, Figure 7.4 and Figure 7.5

The results obtained from the experiments strongly demonstrate the inherent unfairness present in the current methods. Notably, we see that the Maximin sp of the fairness-agnostic methods is either zero or extremely close to zero, even when a significant proportion of the nodes are saved. This indicates that certain nodes have an exceedingly low probability of being saved when utilizing these methods. On the other hand, the fairness-aware approaches consistently outperform the fairness-agnostic methods in terms of Maximin sp when $k \geq |M|$. This consistent improvement highlights that the fairness-aware methods offer a higher level of assurance to all nodes within the network, ensuring a greater likelihood of being saved.

The disparity in Maximin sp between fairness-aware and fairness-agnostic methods is particularly evident in the larger datasets (CA-AstroPh dataset (Figure 7.1), Facebook dataset (Figure 7.2) and Twitter dataset (Figure 7.3)).

In the CA-AstroPh dataset, all fairness-agnostic methods yield a Maximin sp close to zero even with a seed set size (k) of 25. However, most of them demonstrate the ability to save 80% of the vulnerable nodes with 20 truth starters. This indicates that despite achieving satisfactory performance in terms of saved fraction, these methods fail to provide protection to certain nodes.

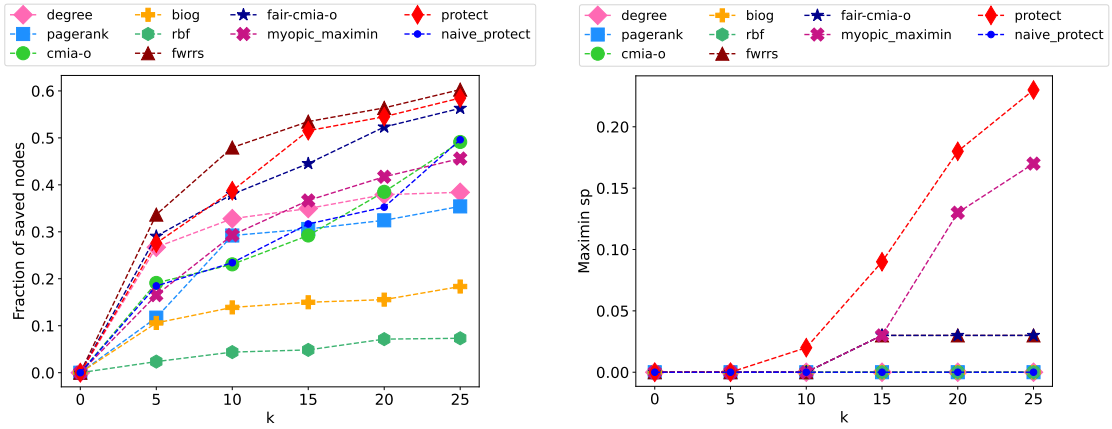
This network is characterized by a large diameter, a high clustering coefficient, and a mediocre average degree. The combination of a high clustering coefficient and moderate average degree



(a) Evaluation on Saved Fraction, with varying k values

(b) Evaluation on Maximin value, with varying k values

Figure 7.1: Comparison of results on the CA-AstroPh dataset with Highest Degree M and propagation probability 0.25



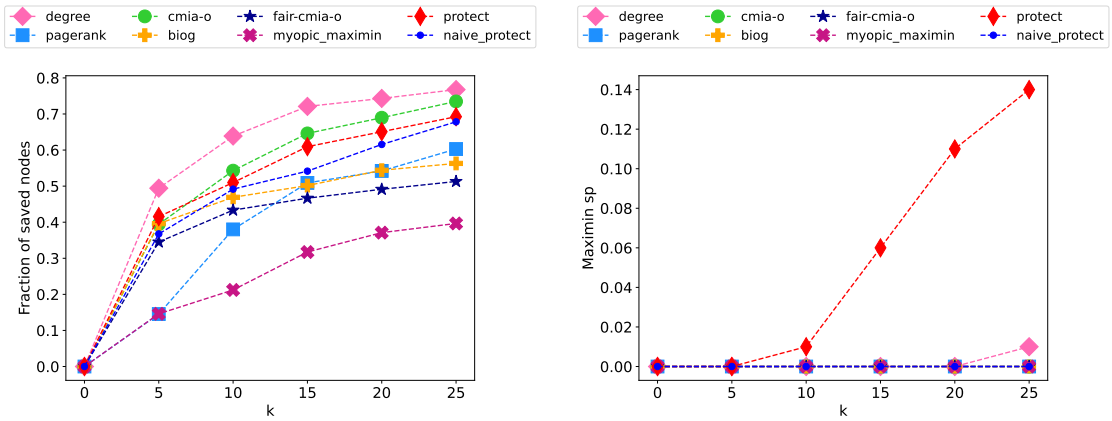
(a) Evaluation on Saved Fraction, with varying k values

(b) Evaluation on Maximin value, with varying k values

Figure 7.2: Comparison of results on the Facebook dataset with Highest Degree M and propagation probability 0.2

offers a wide selection of seed nodes that can maximize influence in the network. However, the large diameter necessitates the careful selection of seed nodes to ensure coverage of all areas of the network. Our PROTECT method, which aims to save the most vulnerable nodes, directs us to the less-covered areas of the network that may be overlooked by fairness-agnostic methods. Thus, with a slightly lower saved fraction (less by 0.01 units compared to the best-performing method, CMIA-O), PROTECT outshines all the fairness-agnostic algorithms in terms of Maximin sp. PROTECT achieves a Maximin sp value of 0.4, while the fairness-agnostic methods show only minimal improvement (0.03) in this dataset. In terms of Maximin sp, the performance of Naive PROTECT is only marginally better than that of the fairness-agnostic methods. This can be attributed to the large number of paths in this network, where estimating vulnerability based on the number of closest misinformation starters is ineffective. However, by selecting the nodes that can *enhance* the most number of nodes, Naive PROTECT achieves a good saved fraction.

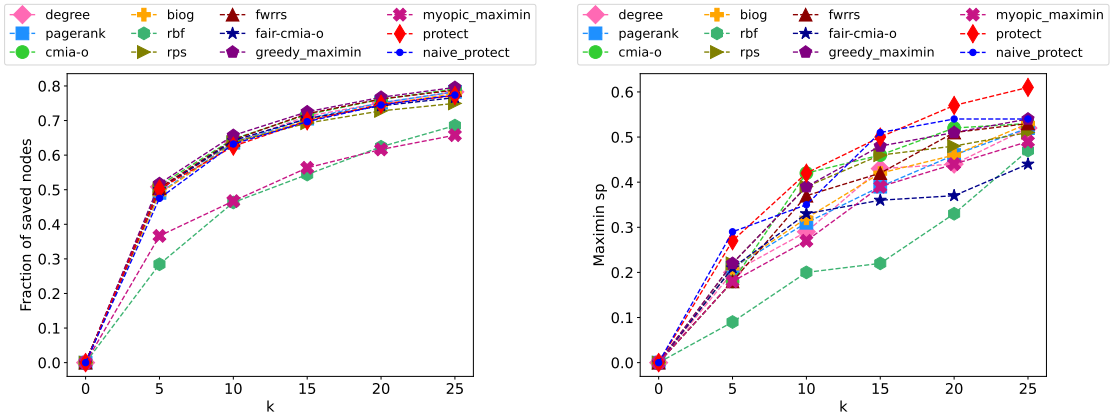
The fairness-agnostic methods in the Facebook dataset exhibit a zero Maximin sp demon-



(a) Evaluation on Saved Fraction, with varying k values

(b) Evaluation on Maximin value, with varying k values

Figure 7.3: Comparison of results on the Twitter dataset with Highest Degree M and propagation probability 0.25



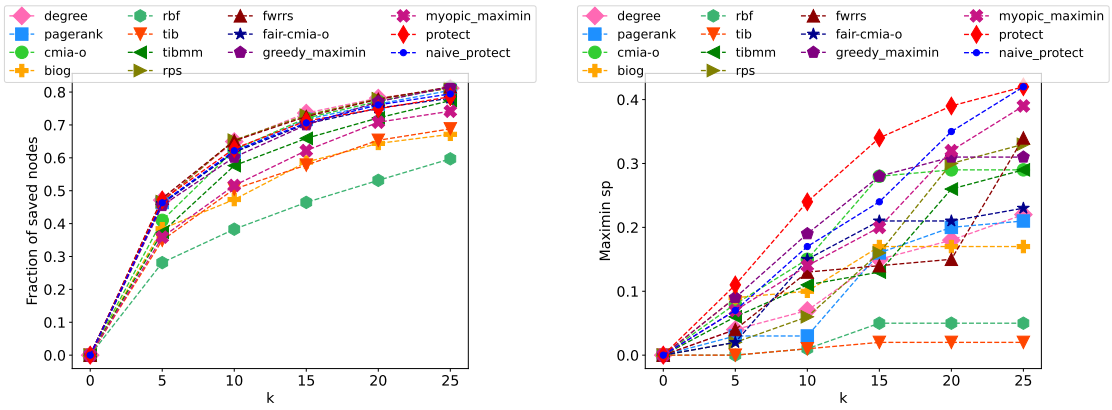
(a) Evaluation on Saved Fraction, with varying k values

(b) Evaluation on Maximin value, with varying k values

Figure 7.4: Comparison of results on the socfb-Caltech36 with Highest Degree M and propagation probability 0.1

strating their unfairness towards certain nodes in the network. Only the fairness-aware methods succeed in achieving a non-zero Maximin sp. Further investigation (refer to Figure 7.6) reveals that fairness-aware methods like PROTECT, Myopic Welfare Maximin, FWRRS, and Fair CMIA-O successfully eliminate nodes with zero *sp* when using 10 positive seeds ($k = 10$). Naive PROTECT reduces the number of zero *sp* nodes to 22, while CMIA-O is the only fairness-agnostic method that significantly reduces it (49 zero *sp* nodes). CMIA-O employs a path-based approach that utilizes maximum probable paths, which would be the shortest paths in the case of the Uniform Cascade setting. In Chapter 5, we established that these paths are effective in improving save probability due to the truth precedence policy. This effectiveness of CMIA-O can also be observed in other datasets, validating its performance.

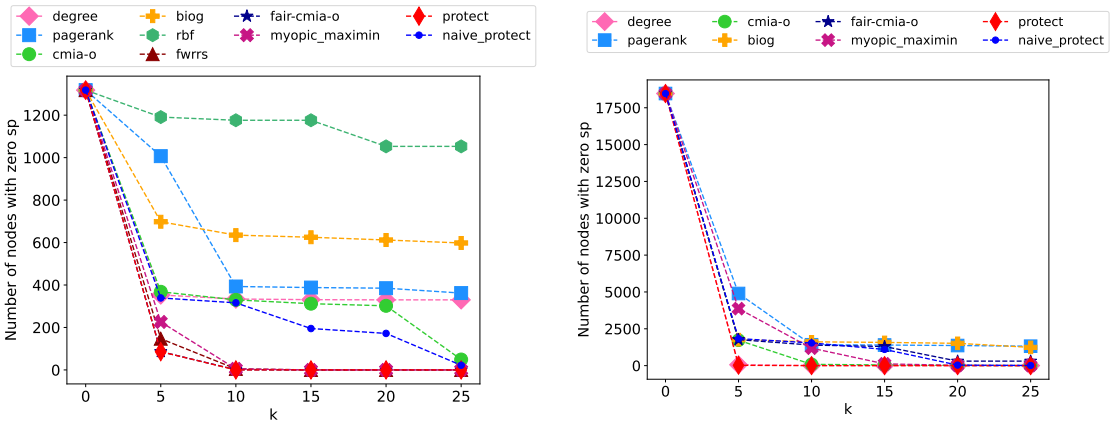
In the Twitter dataset, we observe a similar trend to the previous dataset. While the number of nodes with zero *sp* decreases rapidly for CMIA-O and the degree heuristic, PROTECT stands out by completely eliminating nodes with zero save probability, and it accomplishes this with a smaller seed set size of 10. This highlights the effectiveness of PROTECT in providing protection



(a) Evaluation on Saved Fraction, with varying k values

(b) Evaluation on Maximin value, with varying k values

Figure 7.5: Comparison of results on the Arenas email dataset with Highest Degree M and propagation probability 0.3



(a) Facebook dataset

(b) Twitter dataset

Figure 7.6: Number of vulnerable nodes that have a zero save probability at each value of k

to a wider range of nodes and thereby ensuring fairness. Despite the fairness-agnostic methods achieving a high saved fraction of 0.76, PROTECT surpasses them in terms of Maximin sp by providing a 14 times better Maximin sp value, while achieving a saved fraction of 0.69. This indicates that targeting the most vulnerable nodes in the network is an effective strategy to save a significant proportion of the vulnerable nodes.

In the previously mentioned datasets, we observe high clustering coefficients and mediocre to high average degrees. Additionally, the campaign initiators are chosen as the nodes with the highest degrees, which implies that information spreads through a large number of paths in these networks. The Naive PROTECT heuristic lacks awareness of the number of paths and only considers the number of closest misinformation starters to identify the most vulnerable nodes. Its lack of consideration for the number of paths could be the reason for its comparatively poor Maximin sp value in these datasets. However, despite this drawback, the Naive PROTECT heuristic demonstrates its effectiveness in reducing the number of nodes with zero save probability, as depicted in Figure 7.6. By targeting the nodes closest to the misinformation starters, it is successful in providing protection for a large proportion. It also achieves a saved fraction close to that of PROTECT in the above datasets.

We evaluate the performance of our proposed method in smaller datasets (socfb-Caltech36 dataset (Figure 7.4) and Arenas Email dataset (Figure 7.5) with eight fairness-agnostic methods. In the socfb-Caltech36 dataset, when k is 25, most methods achieve a saved fraction above 0.75, with CMIA-O reaching the highest saved fraction of 0.79. PROTECT heuristic, with a saved fraction of 0.77, surpasses the Maximin sp achieved by all fairness-agnostic methods. Though this network has a low clustering coefficient, it has a high average degree and a very low average shortest path. These factors contribute to the ease of reaching all nodes in the network, leading to high Maximin sp values achieved by all methods.

The Arenas Email dataset on the other hand has a low clustering coefficient, low average degree the a longer average shorter path. This longer shortest path makes the case of obstructed nodes [60] more probable. This could be the reason for the initial better performance of RPS method compared to CMIA-O in terms of the saved fraction. RPS also achieves the highest Maximin sp among fairness-agnostic methods, which is 0.33. However, both our proposed methods, PROTECT and Naive PROTECT, perform better, attaining a Maximin sp of 0.42. They also have similar saved fractions, with RPS at 0.81, PROTECT at 0.78, and Naive PROTECT at 0.79.

It is evident that our Naive PROTECT heuristic demonstrates suboptimal performance in larger networks, but it proves to be highly effective in networks characterized by lower clustering coefficients. Remarkably, it matches the Maximin sp achieved by the PROTECT method itself in the Arenas Email dataset. In networks with lower clustering coefficients, the dissemination of information is constrained, leading to a limited number of pathways through which information can spread. Consequently, the vulnerability computation based on the proximity to misinformation starters closely approximates the actual vulnerability of the nodes, resulting in its superior performance in such network settings.

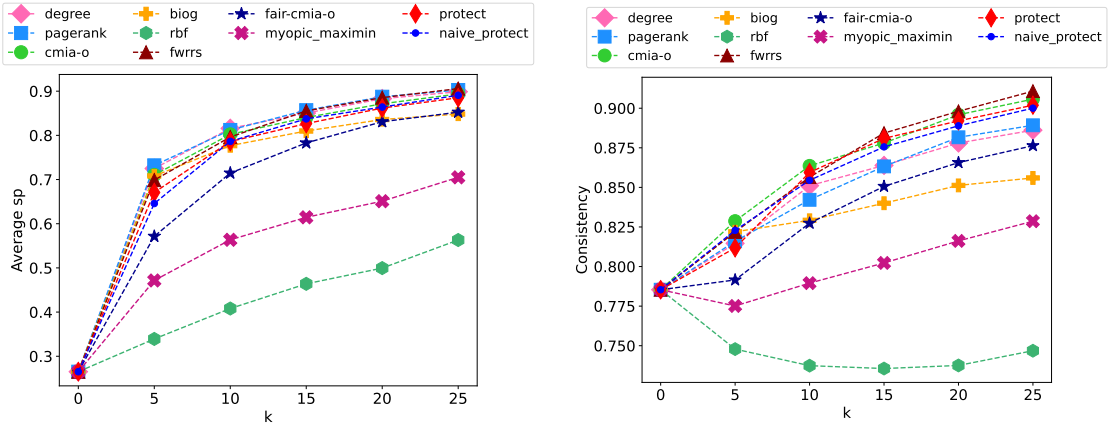
The group-fairness-based methods succeed in improving the Maximin sp in most cases. Their superior performance to fairness-agnostic methods could be due to their ability to guarantee a minimum saved fraction even for the smallest community, thereby reaching more nodes. However, our proposed methods surpass the Maximin sp they achieve in all cases. This dominance in performance becomes even more pronounced in larger datasets. For example, in the CA-AstroPh network, PROTECT provides a Maximin sp three times higher than FWRRS, with only a 0.028 unit drop in the saved fraction when the seed set size is 25. Similarly, in the Facebook dataset, PROTECT heuristic achieves a seven times better Maximin sp than the group fairness methods, with a 0.017 decrease in the saved fraction when the seed set size is 25. These results clearly demonstrate that optimizing the minimum fraction of nodes saved within each community (the goal of group fairness) does not guarantee the best protection of all nodes in the network. Individual fairness, as achieved by our PROTECT method, proves to be more effective in providing comprehensive node protection across the entire network.

Moreover, our PROTECT method consistently outperforms the individual fairness baseline, Myopic Welfare Maximin Heuristic, across all scenarios, demonstrating superior performance in both saved fraction and Maximin sp. The Greedy Maximin Heuristic was only evaluated for smaller datasets due to its high time complexity. While it achieves the highest saved fraction among all methods in the socfb-Caltech36 dataset at a seed set size of 25, it falls short in terms of Maximin sp compared to PROTECT. Similarly, in the Arenas Email dataset, the Greedy Maximin Heuristic matches the performance of the best-performing methods in terms of saved fraction but fails to achieve the same level of Maximin sp as PROTECT. Consequently, our proposed method outshines the individual fairness baselines in terms of fairness across all evaluated cases.

Figures 7.7, Figure 7.8, Figure 7.9, Figure 7.10 and Figure 7.11 show the average save probability and consistency of the datasets.

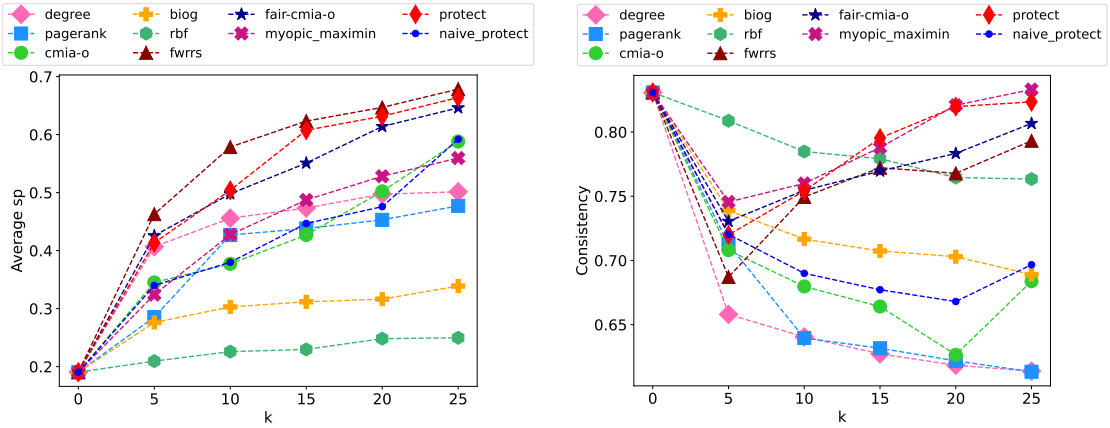
In general, we observe a steady relationship between the average save probability (sp) and the saved fraction, further validating the equivalence between the two measures established in Section 2.4.2.

We see that all the methods evaluated display high consistency, indicating that the NIM objective delivers equity fairness. The PROTECT heuristic provides a high consistency (over 0.8) in all cases. Except for the Facebook dataset, the Naive PROTECT heuristic also provides high consistency, exceeding the Consistency provided by the group fairness methods in the smaller



(a) Evaluation of Average sp, with varying k values (b) Evaluation of Consistency, with varying k values

Figure 7.7: Comparison of Average sp and Consistency on the CA-AstroPh dataset with Highest Degree M and propagation probability 0.25



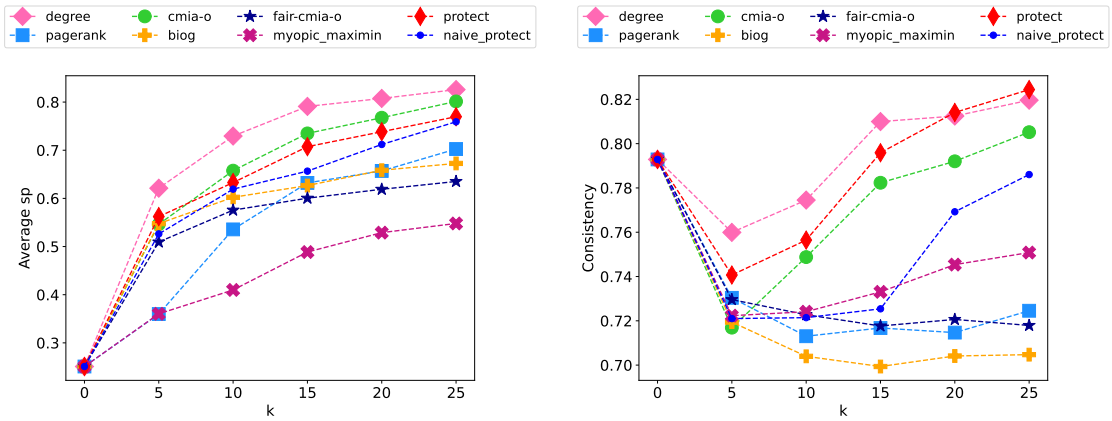
(a) Evaluation of Average sp, with varying k values (b) Evaluation of Consistency, with varying k values

Figure 7.8: Comparison of Average sp and Consistency on the Facebook dataset with Highest Degree M and propagation probability 0.2

datasets.

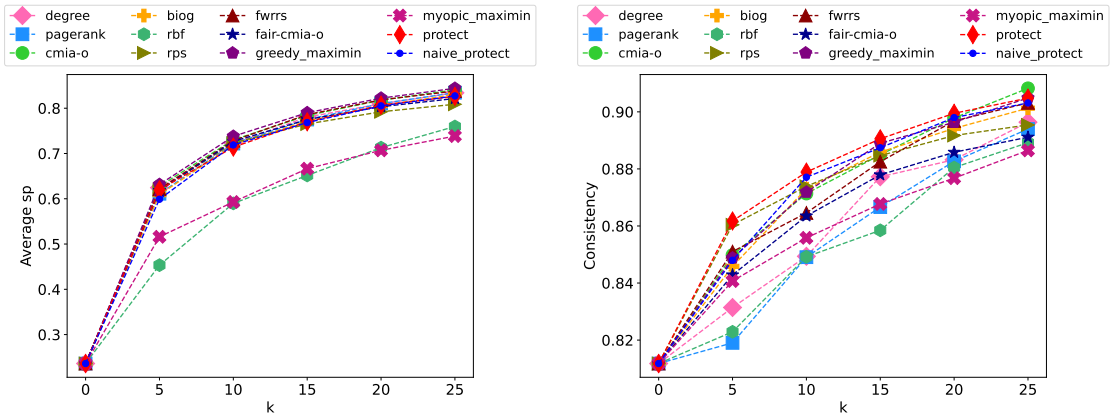
Further, we see in general that methods with high average *sp* (and high saved fraction) tend to exhibit high consistency. This could be because, as more nodes are saved, the majority of the nodes have a higher *sp*. This reduces the difference between *sp* among the nodes. Consistency is inversely related to the difference in *sp* (see equation 6.4). Therefore, as the difference in *sp* decreases, the consistency value increases.

When comparing the consistency of methods with similar average sp, it becomes evident that some methods provide a more equitable distribution than others. For instance, in the CA-AstroPh dataset (refer to Figure 7.7), FWRRS, Degree heuristic, and PageRank heuristic exhibit very similar average sp. However, FWRRS exhibits higher consistency than these methods, indicating a more balanced allocation of resources. Therefore, assessing the consistency of methods with similar average save probabilities (or saved fractions) provides insight into their ability to achieve equitable resource distribution. Notably, PROTECT demonstrates higher consistency than other fairness-agnostic methods with comparable average save probabilities. However, CMIA-O stands as an exception, displaying higher consistency in smaller networks.



(a) Evaluation of Average sp, with varying k values (b) Evaluation of Consistency, with varying k values

Figure 7.9: Comparison of Average sp and Consistency on the Twitter dataset with Highest Degree M and propagation probability 0.25



(a) Evaluation of Average sp, with varying k values (b) Evaluation of Consistency, with varying k values

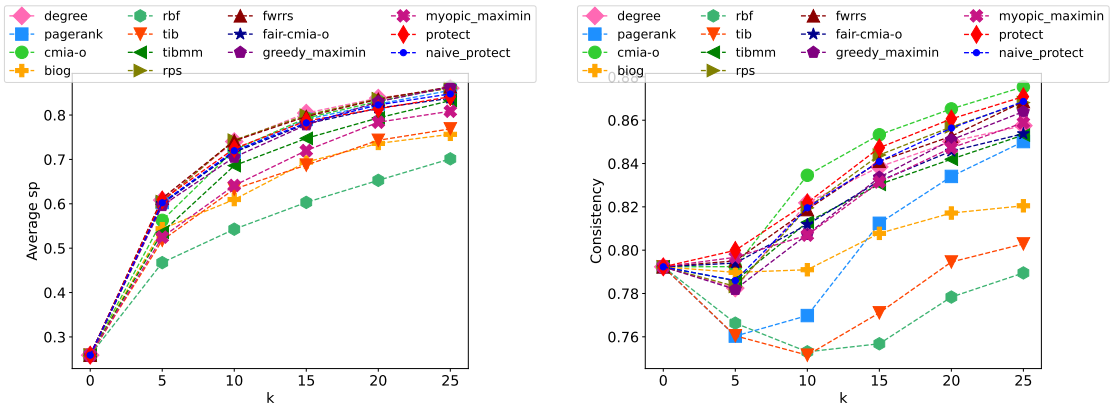
Figure 7.10: Comparison of Average sp and Consistency on the socfb-Caltech36 dataset with Highest Degree M and propagation probability 0.1

The Gini coefficient values for the different datasets are presented in the figure 7.12

When comparing the Gini coefficient values of the methods, we observe that PROTECT consistently achieves a very low Gini coefficient, indicating a significantly reduced level of inequality in the distribution of save probabilities across the networks. Specifically, in the Facebook dataset, PROTECT attains the lowest Gini coefficient value, closely followed by FWRRS, Fair CMIA-O, and the Myopic Welfare Heuristic. This suggests that these methods accomplish a more balanced and equitable distribution of save probabilities among the nodes in the network.

While the differences in Gini coefficient values are less pronounced in smaller networks (socfb-Caltech36 dataset and Arenas Email dataset), it is worth noting that in these networks, PROTECT initially demonstrates the lowest Gini coefficient value, particularly at smaller budget sizes (e.g., k=5). Further, both PROTECT and Naive PROTECT consistently exhibit better equality in save probability distribution compared to the Myopic Welfare Maximin Heuristic across all the datasets and budget sizes.

In summary, we see that our proposed method shows superior performance in guaranteeing protection for all nodes in the network. These carefully designed distance-based approaches effectively enhance the save probability of the most vulnerable nodes. Though we do observe a drop



(a) Evaluation of Average sp, with varying k values (b) Evaluation of Consistency, with varying k values

Figure 7.11: Comparison of Average sp and Consistency on the Arenas Email dataset with Highest Degree M and propagation probability 0.3

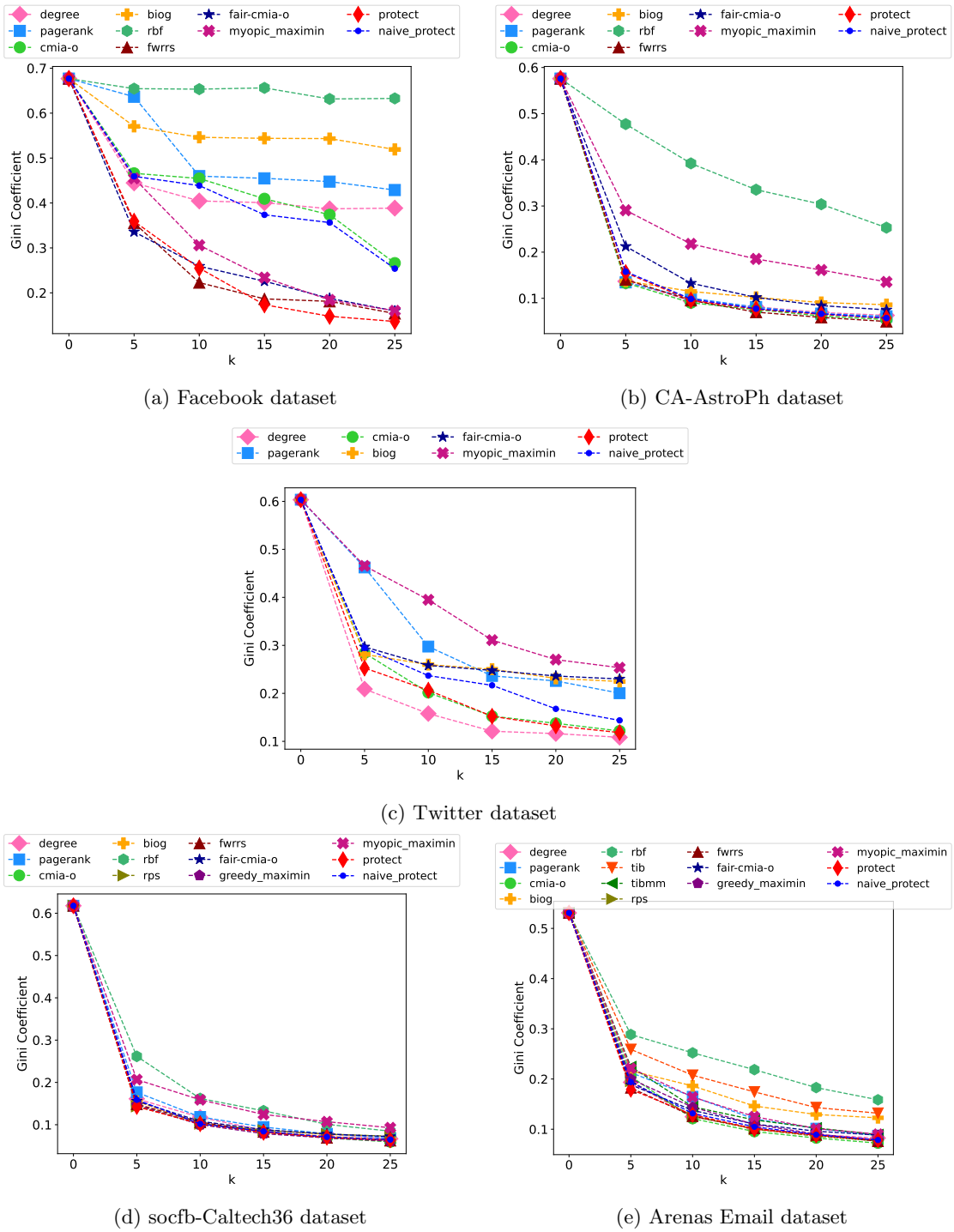
in saved fraction in some cases, this drop is very small. The simulation-based approach employed to identify the most vulnerable nodes enhances the adaptability of our method to networks with varying characteristics. Notably, we find that the Naive PROTECT heuristic performs particularly well in networks with low clustering coefficients, showcasing its effectiveness in such contexts. Furthermore, when evaluating additional fairness metrics such as Consistency and Gini coefficient, we see that our heuristic provides Equity fairness and less inequality in general.

7.2.2 Evaluation with random misinformation starters

In this section, we present the results of experiments on real-world datasets with random misinformation starters. This is a simpler NIM problem, as the most influential seeds are available to be picked as truth spreaders. The results are shown in Figure 7.13, Figure 7.14, Figure 7.15, Figure 7.16, and Figure 7.17.

Under random misinformation starters, we see that the PROTECT heuristic provides the best value for Maximin sp in all datasets. The saved fraction as high as 1 is reached with 25 seeds in most cases. In the CA-AstroPh network, when k=5, all methods achieve a saved fraction close to 1, with PROTECT, Naive PROTECT, and the degree heuristic obtaining the highest values. At k=5, PROTECT achieves a Maximin sp of 0.36, which is twice the best value achieved by a fairness-agnostic method with similar saved fractions. This demonstrates the superiority of the selection mechanism of our approach. On the other hand, the Myopic Welfare Maximin Heuristic, shows suboptimal performance in terms of saved fraction, although it achieves a higher Maximin sp compared to fairness-agnostic methods. This is because the Myopic Welfare Maximin Heuristic randomly selects nodes with the highest vulnerability as seeds. While this selection strategy may reduce the network's vulnerability by choosing seeds that are farther apart, it may not be the optimal choice for all the nodes in the network. This is evident from the results of our PROTECT heuristics, which achieve a high save probability of 0.98 while attaining a high Maximin sp of 0.73 (PROTECT) and 0.68 (Naive PROTECT). In comparison, the best Maximin sp achieved by a fairness-agnostic algorithm (CMIA-O) is 0.51.

In the case of the Facebook dataset, we see that PROTECT and Naive PROTECT method achieve saved fractions of 0.98 and 0.96, respectively, with just 5 seeds. PROTECT outperforms other baselines in terms of Maximin sp by achieving a value of 0.81. However, the Maximin sp of Naive PROTECT plateaus at 0.64, indicating its limitation in accurately estimating save probabilities. Sampling and simulation-based fairness-aware methods (Fair CMIA-O and Myopic Welfare Maximin) surpass its Maximin sp. A similar trend is observed in the Twitter dataset, which also exhibits a high clustering coefficient and a smaller diameter. It is important to note that



(a) Facebook dataset

(b) CA-AstroPh dataset

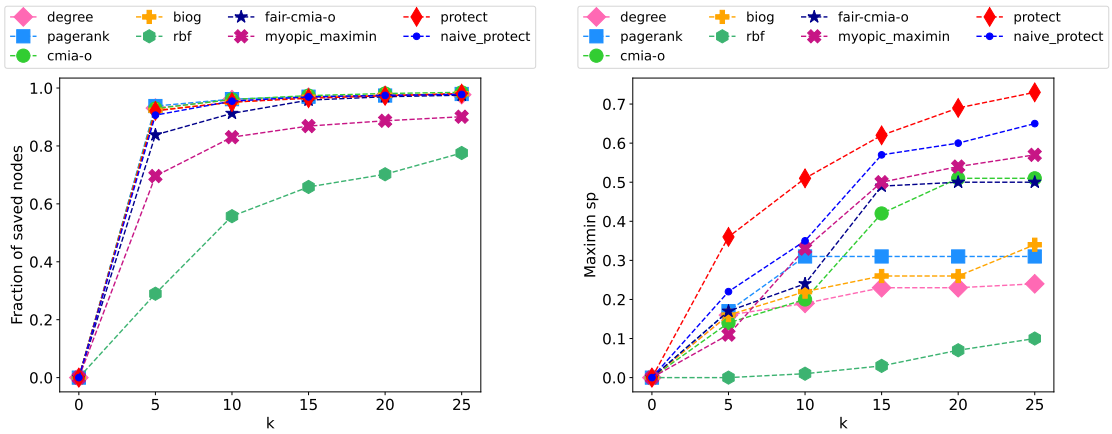
(c) Twitter dataset

(d) socfb-Caltech36 dataset

(e) Arenas Email dataset

Figure 7.12: Comparison of Gini coefficient on the real-world datasets with Highest Degree M

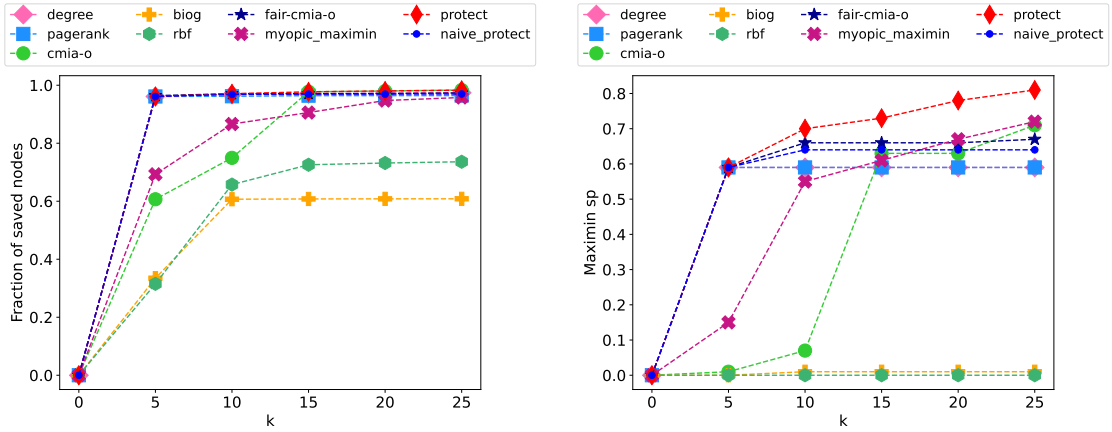
Naive PROTECT does not account for the number of paths from misinformation to each node, which becomes crucial in networks with a high clustering coefficient. On the other hand, in the CA-AstroPh network, which also has a high clustering coefficient, Naive PROTECT's effectiveness in this network could be due to its ability to carefully select seed nodes to ensure coverage of all areas which is important for the network's large diameter.



(a) Evaluation on Saved Fraction, with varying k values

(b) Evaluation on Maximin value, with varying k values

Figure 7.13: Comparison of results on the CA-AstroPh dataset with Random M and propagation probability 0.25



(a) Evaluation on Saved Fraction, with varying k values

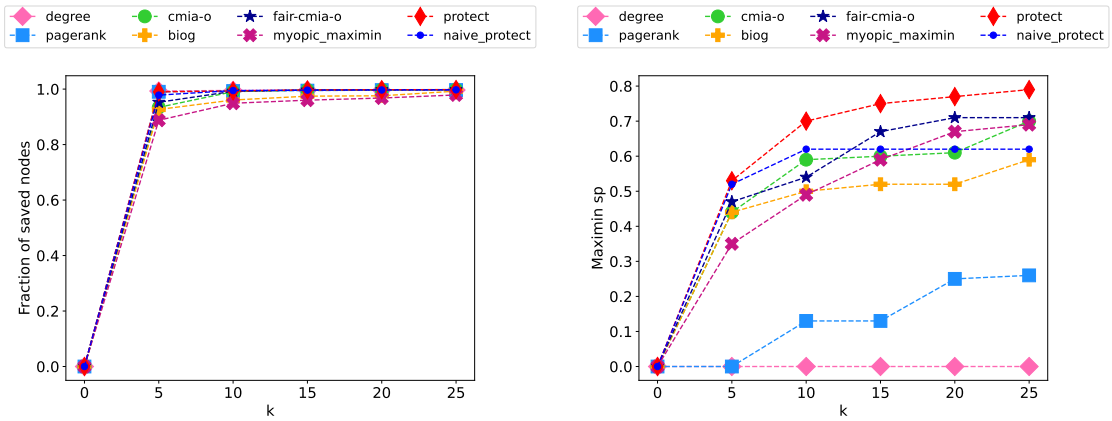
(b) Evaluation on Maximin value, with varying k values

Figure 7.14: Comparison of results on the Facebook dataset with Random M and propagation probability 0.2

Similarly to the highest degree M setting, we observe that Naive PROTECT performs well in networks with low clustering coefficient in this setting as well. The Arenas email network (Figure 7.17), characterized by a low clustering coefficient and low average degree, demonstrates this pattern. Naive PROTECT achieves a Maximin sp equal to that of PROTECT in this network.

In the case of the socfb-Caltech36 dataset (Figure 7.16), we observe that the Greedy Welfare Maximin method performs well in this setting. It achieves the highest Maximin sp at k=5, closely followed by PROTECT and FWRRS. However, as the value of k increases, PROTECT surpasses the Maximin sp achieved by the Greedy Welfare Maximin method. Moreover, Naive PROTECT, which requires significantly less computation time, outperforms the Greedy Welfare Maximin method at k=20. Thus, the PROTECT and Naive PROTECT methods provide better effectiveness in maximizing Maximin sp compared to other fairness baselines in this setting as well.

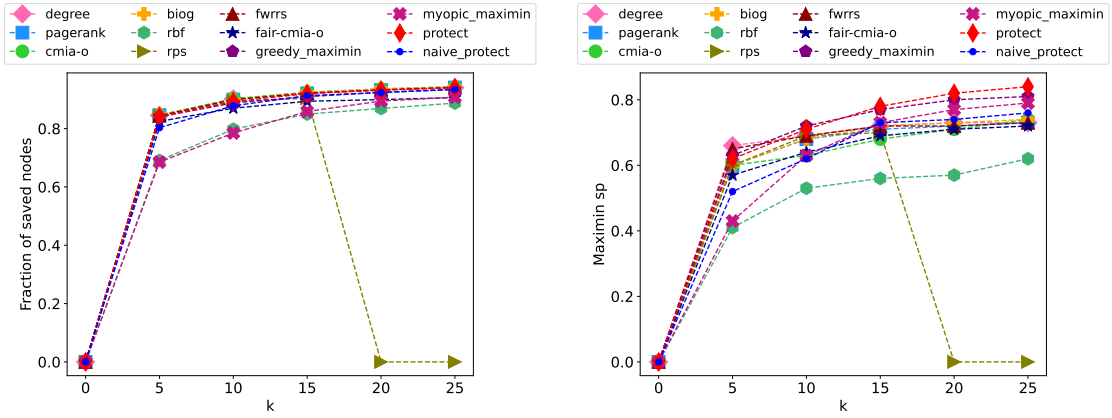
In summary, even in this setting, we see that PROTECT performs superior in terms of Maximin



(a) Evaluation on Saved Fraction, with varying k values

(b) Evaluation on Maximin value, with varying k values

Figure 7.15: Comparison of results on the Twitter dataset with Random M and propagation probability 0.25

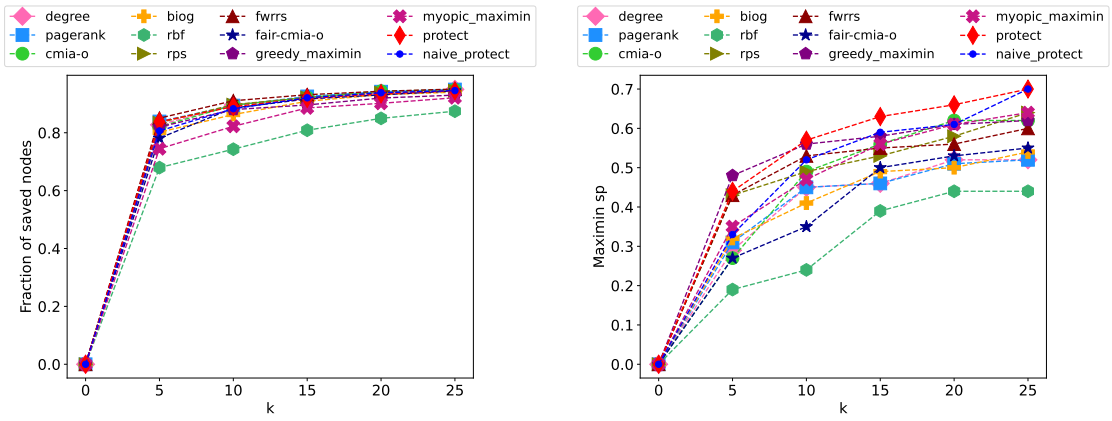


(a) Evaluation on Saved Fraction, with varying k values

(b) Evaluation on Maximin value, with varying k values

Figure 7.16: Comparison of results on the socfb-Caltech36 with Random M and propagation probability 0.1

sp for all the datasets. We note that Naive PROTECT is more efficient in this setting. This could be because M is chosen at random, and the number of paths from M that are to be blocked is lesser compared to the previous setting.



(a) Evaluation on Saved Fraction, with varying k values

(b) Evaluation on Maximin value, with varying k values

Figure 7.17: Comparison of results on the Arenas email dataset with Random M and propagation probability 0.3

7.3 Experiments on alternate diffusion model

In this section, we present the results of our experiments on an alternate diffusion model known as the Multi Campaign Independent Cascade Model (MCICM) with high effectiveness property. This model is commonly used in studies related to NIM [11, 60].

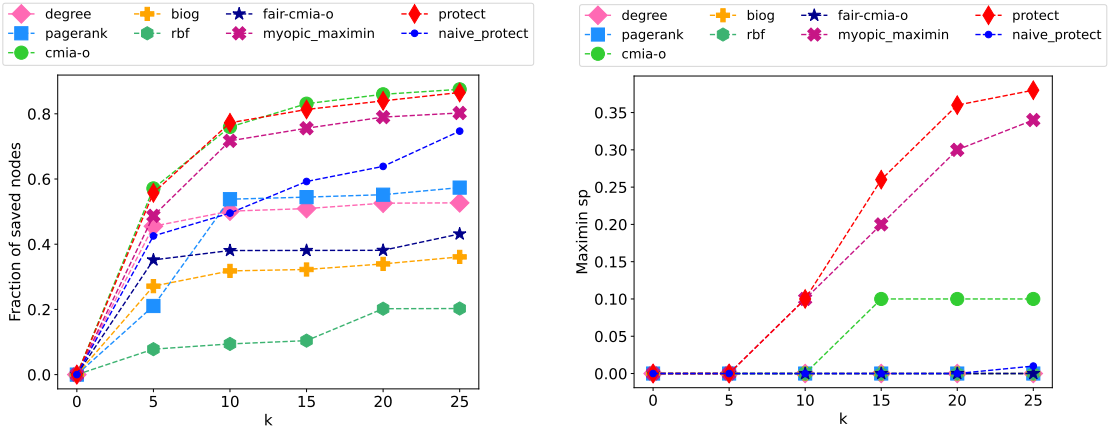
In the MCICM with high effectiveness property, when a node that has not been influenced by any information receives true information, it has a 100% chance of being influenced by it. This represents a realistic scenario where the dissemination of accurate information, such as official records, is highly effective in countering misinformation. By incorporating this diffusion model into our experiments, we aim to evaluate the effectiveness of our proposed methods in mitigating the spread of misinformation when reliable information is shared.

In our experiments on the MCICM with high effectiveness property, we focus on two datasets: the Facebook dataset, which exhibits a high clustering coefficient, and the Arenas Email dataset, which has a low clustering coefficient. To ensure a consistent comparison, we maintain the same propagation probability for misinformation as in previous experiments. These values are carefully chosen to ensure that the initial outreach of misinformation is at least 70%, providing a challenging scenario for evaluating the effectiveness of different methods. The nodes with the highest degree are chosen as misinformation starters as the unfairness of the methods was more visible in this case in the previous experiments.

We present the results of on two datasets in Figure 7.18 and Figure 7.19. In the Facebook dataset, PROTECT matches the saved fraction of the best-performing method, CMIA-O, while surpassing CMIA-O by a significant margin in Maximin sp. The group-fairness method, Fair CMIA-O demonstrates a considerable drop in performance in this model, compared to its performance in COICM. CMIA-O, which uses most probable paths to compute influence, becomes particularly effective as all paths have a probability of 1. Myopic Welfare Maximin performs well due to its strategy of selecting nodes with the lowest save probability, which results in choosing nodes that are farther apart.

As discussed in section 5.2, Naive PROTECT initially operates in Basic Protection mode, aiming to have at least one truth starter close to all immediate neighbors of M. In this model, it is sufficient for a node to have at least one truth seed within a distance equal to or less than the node's distance from the misinformation seed to be saved. Hence, Naive PROTECT demonstrates better performance in terms of both saved fraction and Maximin sp compared to COICM.

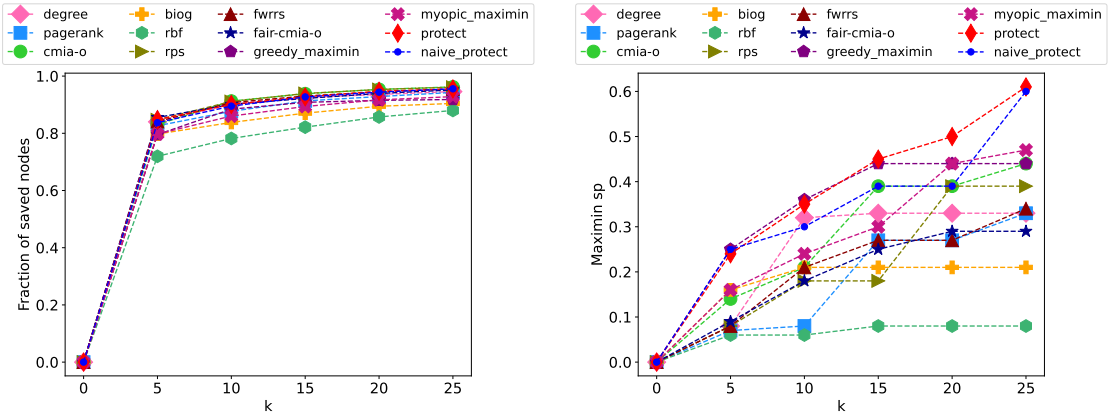
Moving to the Arenas Email dataset, we observe a significant increase in the saved fraction with just 5 seeds for most methods. The Maximin sp of Naive PROTECT matches the Maximin



(a) Evaluation on Saved Fraction, with varying k values

(b) Evaluation on Maximin value, with varying k values

Figure 7.18: Comparison of results on Facebook dataset with Highest Degree M under MCICM with high effectiveness



(a) Evaluation on Saved Fraction, with varying k values

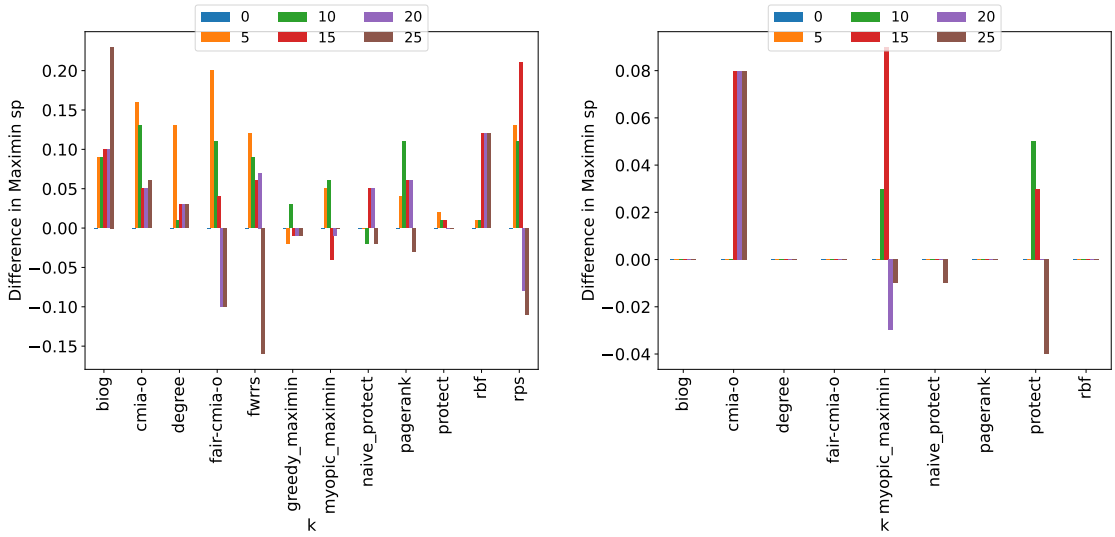
(b) Evaluation on Maximin value, with varying k values

Figure 7.19: Comparison of results on Arenas Email dataset with Highest Degree M under MCICM with high effectiveness

sp achieved by PROTECT as k increases to 25. Due to the small size of the network, a budget of $k = 25$ is sufficient to have at least one node close to all immediate neighbors of M. Therefore, the strong performance of Naive PROTECT can be attributed to the network's size.

As we noted earlier, CMIA-O achieves a high Maximin sp in this setting, matching the Maximin sp achieved by the Greedy Heuristic in the Arenas Email dataset. To further evaluate the fairness of these methods, we examine the welfare gap, which measures the disparity in the network. It is calculated by randomly dividing the network into two partitions and computing the Maximin sp for each partition (see Figure 7.20). We compare the welfare gap of CMIA-O and the Greedy Heuristic for the Arenas Email dataset. Both these methods achieve the same Maximin sp at $k = 25$. We find that CMIA-O has a higher welfare gap, indicating greater disparity. In contrast, the choice made by the Greedy Heuristic contributes to minimal disparity in the network, and therefore a fairer outcome. The fairness of the selecting mechanism in the PROTECT method is evident in the minimal welfare gap observed. This is also visible in the Facebook dataset, where

PROTECT maintains a lower welfare gap than CMIA-O while achieving a higher Maximin sp.



(a) Arenas Email dataset

(b) Facebook dataset

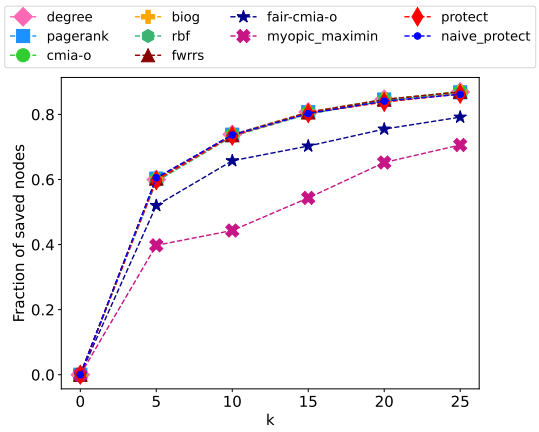
Figure 7.20: Comparison of the Welfare Gap on datasets with Highest Degree M under MCICM with high effectiveness

7.4 Experiments on Synthetic datasets

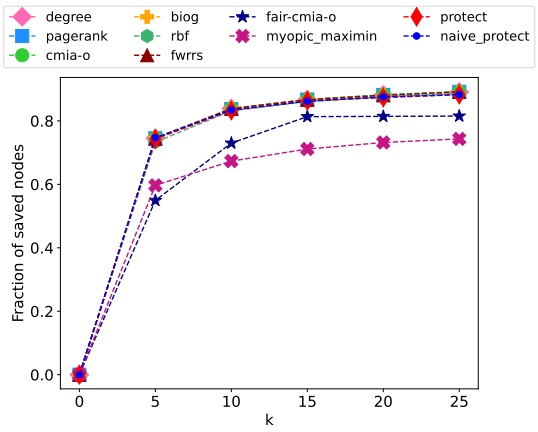
In this section, we conduct experiments on the synthetic datasets introduced in Section 6.1.2. These datasets exhibit variations in clustering coefficient, average degree, average shortest path, and diameter. Additionally, we investigate the impact of changes in the propagation probability (α) by running experiments with two different values: 0.1 and 0.3. This allows us to examine the impact of varying information spread levels on the effectiveness of our methods in mitigating misinformation propagation and ensuring protection for all nodes in the network.

SN1 has a high clustering coefficient of 0.56 similar to the Facebook dataset. However, it has a mediocre average degree and a lower shortest path. In this setting, most methods perform similarly in terms of saved fraction (see Figure 7.21) as the lower shortest path and high clustering coefficient is an ideal setting for nodes with overall influence minimization objective. PROTECT and Naive PROTECT matches the saved fraction of the best-performing methods though the group fairness baseline (Fair CMIA-O) and Individual fairness baseline (Myopic Welfare Maximin) fall behind. As α increases, the saved fraction increases more rapidly as the selected nodes have a greater potential to spread true information now. On the other hand, the Maximin sp values drop as the misinformation starters have a greater influence when α is high. We see that PROTECT outperforms other methods in terms of Maximin sp for both settings of α. Comparing the Maximin sp at α=0.1, we see that PROTECT has the highest Maximin sp, though other methods closely follow it. When α is low, the spread of misinformation would be limited and there are lesser nodes to be saved. We see that CMIA-O outperforms the Maximin sp of our Naive PROTECT in this case. But as the α increases, the Maximin sp of fairness-agnostic methods plateau at 0.10 while Naive PROTECT and PROTECT reach values 0.28 and 0.36 respectively. This demonstrates the superior performance of our heuristics when the influence spread is significant. PROTECT achieves a three-fold increase in Maximin sp compared to fairness-agnostic methods while saving the same number of nodes.

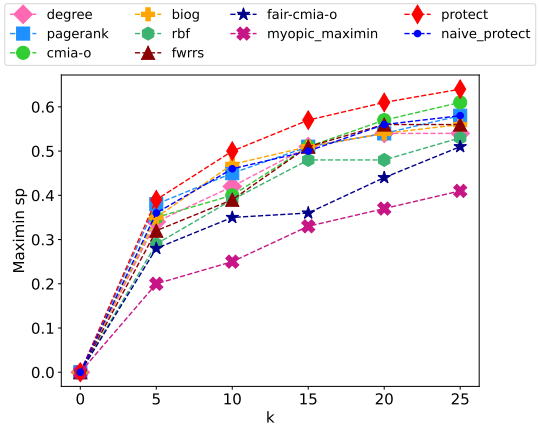
The synthetic network SN2 has a low clustering coefficient, very low average degree, long diameter, and long average shortest paths. The results from this network are shown in Figure



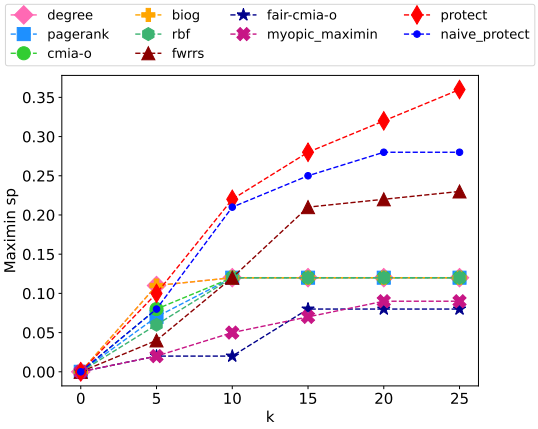
(a) Evaluation on Saved Fraction, with varying k value, and propagation probability 0.1



(b) Evaluation on Saved Fraction, with varying k value, and propagation probability 0.3



(c) Evaluation on Maximin value, with varying k values, and propagation probability 0.1

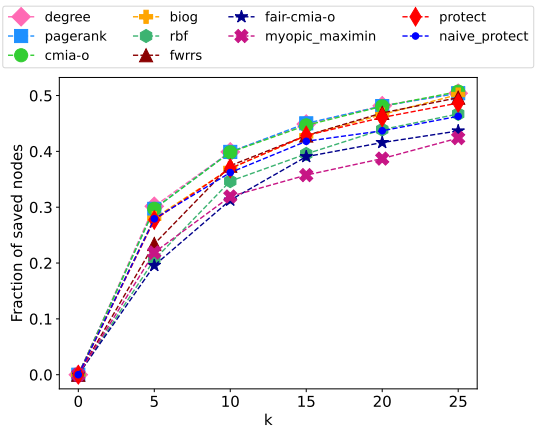


(d) Evaluation on Maximin value, with varying k values, and propagation probability 0.3

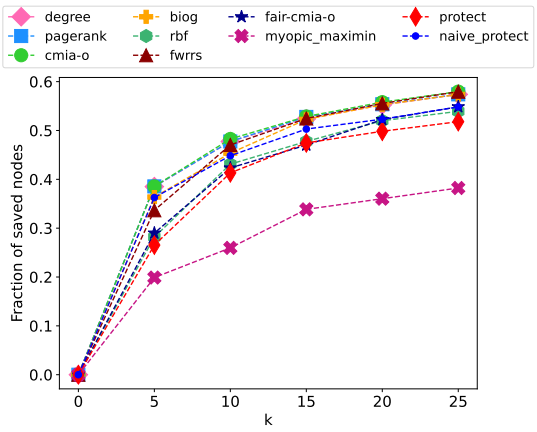
Figure 7.21: Comparison of results on synthetic dataset SN1 with Highest Degree M when the propagation probability is 0.1 (left) and 0.3 (right)

7.22. In terms of saved fraction, fairness-agnostic methods outperform PROTECT in this network. Interestingly, Naive PROTECT achieves a higher saved fraction in this case. Naive PROTECT minimizes the difference in the number of closest misinformation starters and truth seeds for nodes at distance 1 from M. With a low average degree, the number of paths in the network is limited, and the proximity-based condition becomes sufficient to effectively reduce the influence of misinformation. This behavior is also observed in the CA-AstroPh dataset, which has a long diameter. However, PROTECT, being aware of the actual save probabilities, performs better at maximizing the minimum save probability of the network. The Myopic Heuristic performs well in terms of Maximin sp in this network. Due to the low average degree, long diameter, and negative degree assortativity, influence tends to be in small clusters within the network. The Myopic Heuristic selects nodes with the minimum save probability, effectively extending its reach to these clusters.

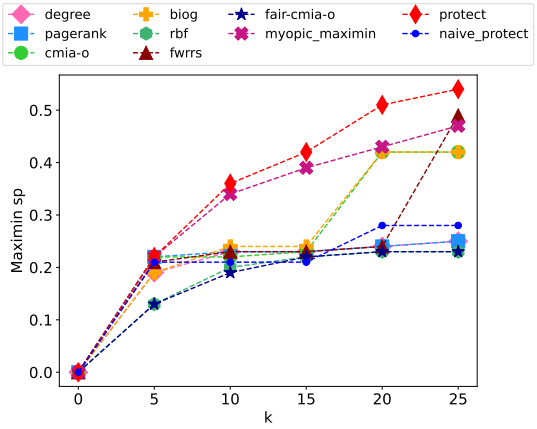
On the other hand, group fairness methods FWRRS and Fair CMIA-O aims to maximize the minimum saved fraction within each community. We observe that they perform poorly when α is 0.3. The subpar performance of these methods in this particular setting indicates that this objective may lead to the suboptimal distribution of resources, these methods save over 50% of the nodes in the network, but, their Maximin sp does not improve beyond 0.06. Thus, we observe



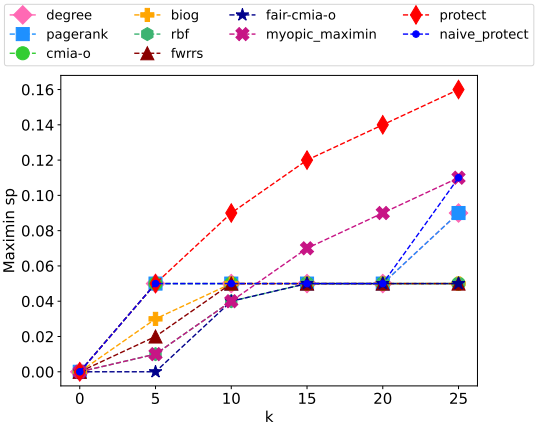
(a) Evaluation on Saved Fraction, with varying k value, and propagation probability 0.1



(b) Evaluation on Saved Fraction, with varying k value, and propagation probability 0.3



(c) Evaluation on Maximin value, with varying k values, and propagation probability 0.1



(d) Evaluation on Maximin value, with varying k values, and propagation probability 0.3

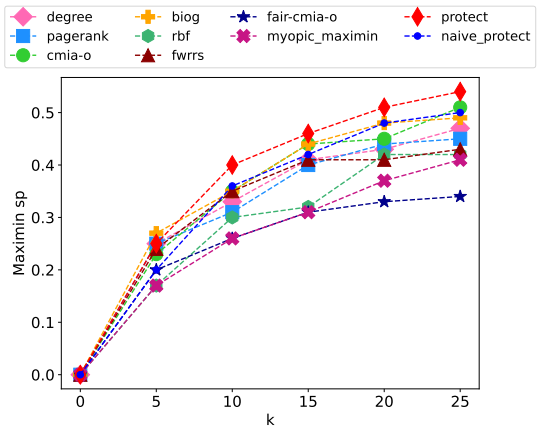
Figure 7.22: Comparison of results on synthetic dataset SN2 with Highest Degree M when the propagation probability is 0.1 (left) and 0.3 (right)

that when influence is dispersed throughout the network, our proposed methods prove to be more effective.

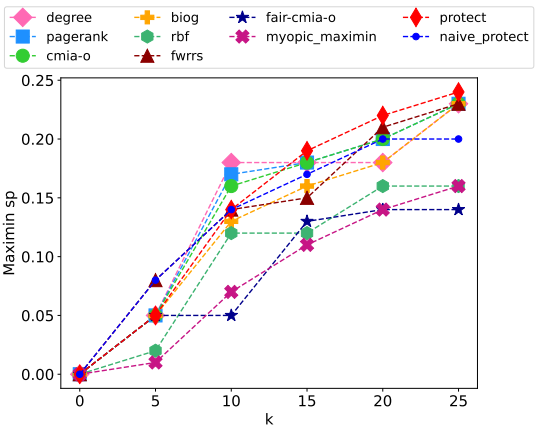
SN3, SN4, and SN5 are synthetic networks with low average degrees, where SN3 has the longest shortest path, followed by SN4 and SN5. The Maximin sp of these networks for α values 0.1 and 0.3 are shown in figure 7.23. At α=0.1, PROTECT has the highest Maximin sp in all these networks.

SN3 has a relatively high clustering coefficient (0.48) among the three networks. The low degree assortativity indicates that there is a low tendency for nodes of similar degree to share an edge. This, with the longer average shortest path value, indicates that higher degree nodes are not connected to each other. Degree Heuristic, PageRank Heuristic and CMIA-O momentarily surpass the Maximin sp of PROTECT at $k = 10$. We observe that all these methods pick the nodes with the highest degree in the network as their initial seeds. At $k = 10$, this strategy proves to be most effective initially, as they also achieve a saved fraction of over 70%. This could be because these scattered high degree nodes would provide good coverage over the network. However, with higher k values, PROTECT, which selects nodes based on their ability to enhance the most vulnerable nodes, surpasses these methods, albeit with a close performance.

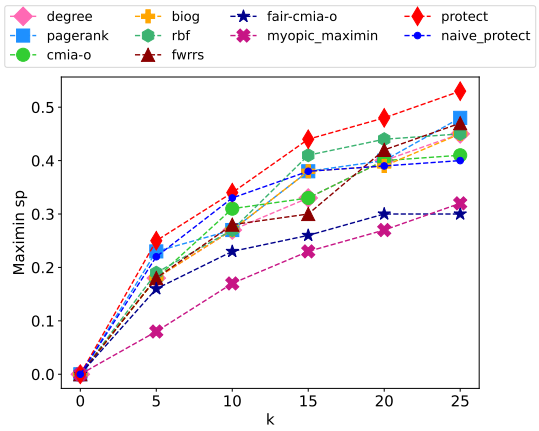
SN4, with low p_T configuration, has a low clustering coefficient. We observe that as α increases,



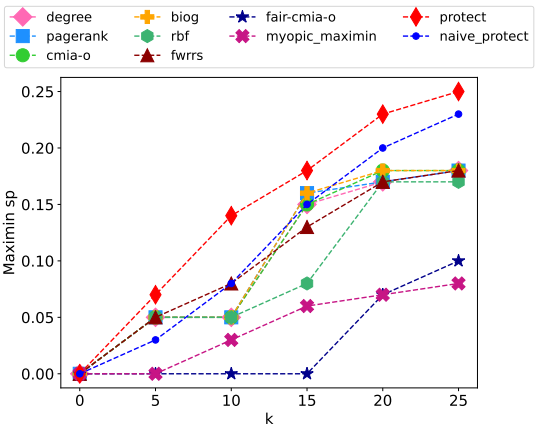
(a) Evaluation on Maximin value on SN3, with varying k value, and propagation probability 0.1



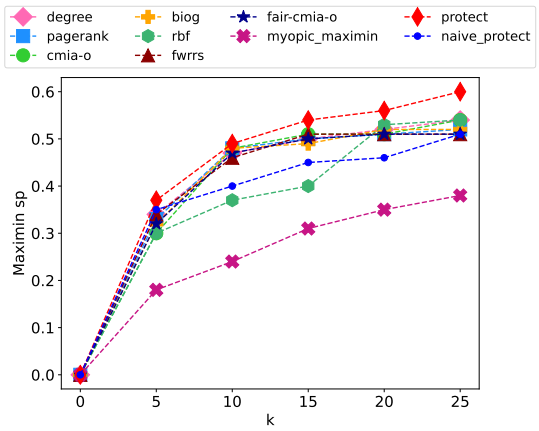
(b) Evaluation on Maximin value on SN3, with varying k value, and propagation probability 0.3



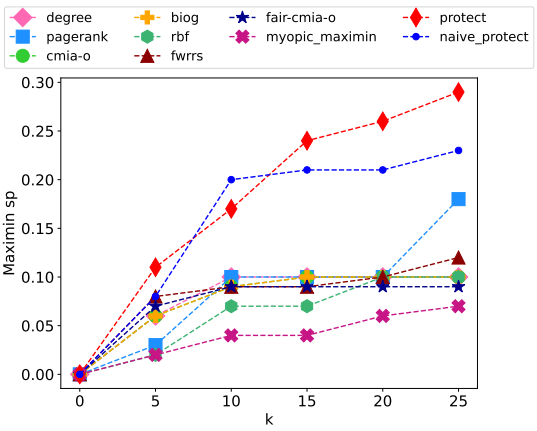
(c) Evaluation on Maximin value on SN4, with varying k value, and propagation probability 0.1



(d) Evaluation on Maximin value on SN4, with varying k value, and propagation probability 0.3

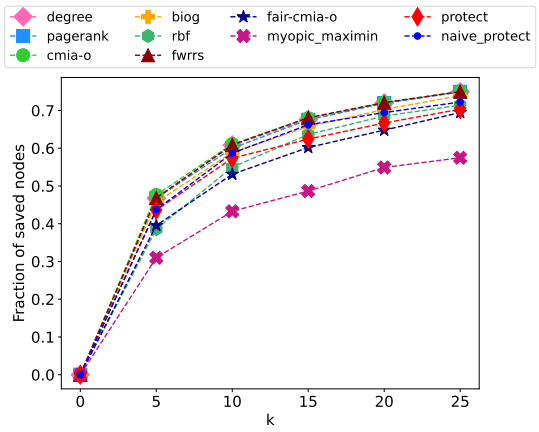


(e) Evaluation on Maximin value on SN5, with varying k value, and propagation probability 0.1

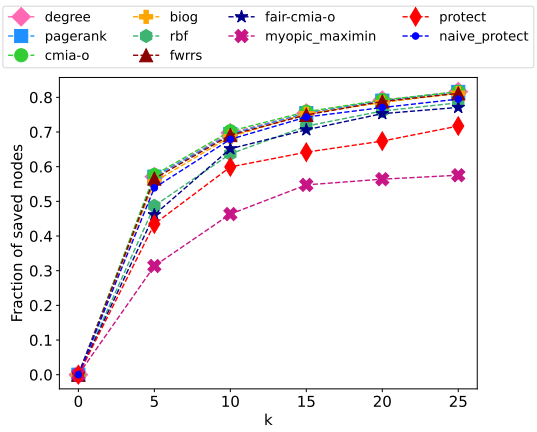


(f) Evaluation on Maximin value on SN5, with varying k value, and propagation probability 0.3

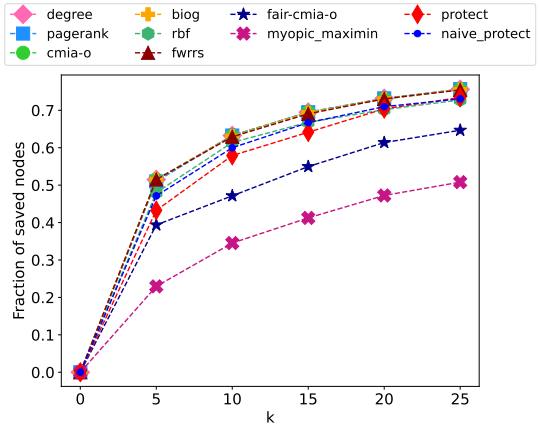
Figure 7.23: Comparison of Maximin sp on synthetic datasets SN3, SN4 and SN5 with Highest Degree M when the propagation probability is 0.1 (left) and 0.3 (right)



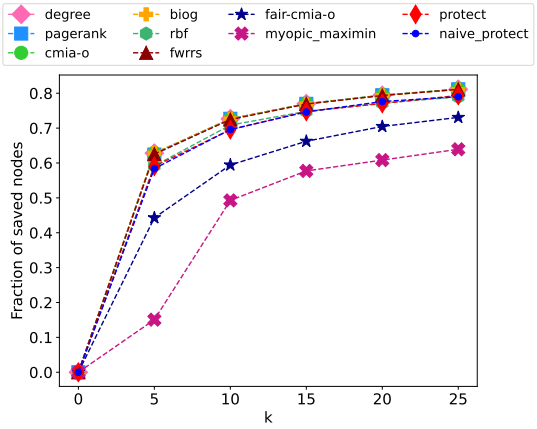
(a) Evaluation on Saved Fraction on SN3, with varying k values, and propagation probability 0.1



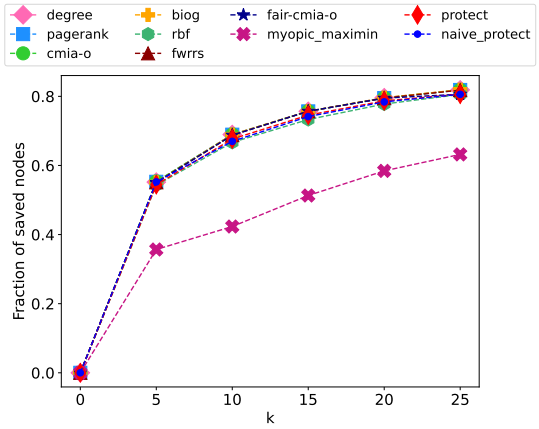
(b) Evaluation on Saved Fraction on SN3, with varying k values, and propagation probability 0.3



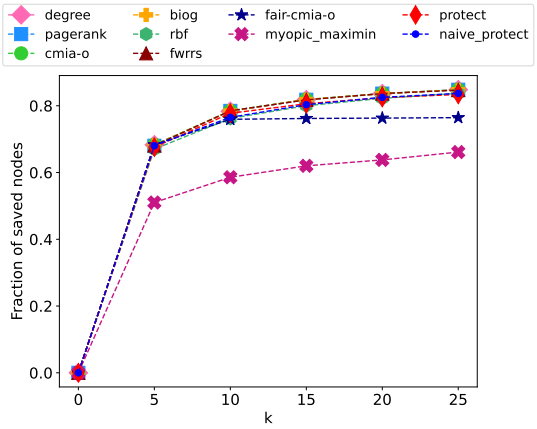
(c) Evaluation on Saved Fraction on SN4, with varying k values, and propagation probability 0.1



(d) Evaluation on Saved Fraction on SN4, with varying k values, and propagation probability 0.3



(e) Evaluation on Saved Fraction on SN5, with varying k values, and propagation probability 0.1

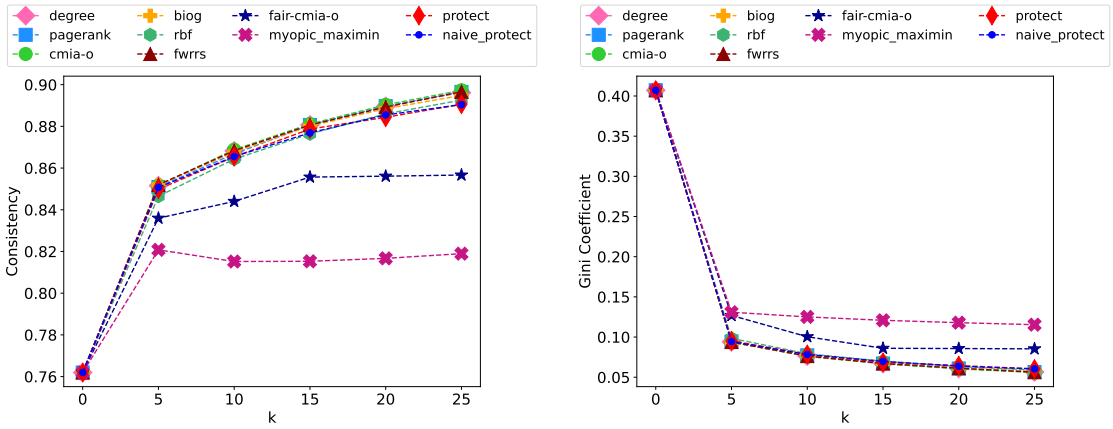


(f) Evaluation on Saved Fraction on SN5, with varying k values, and propagation probability 0.3

Figure 7.24: Comparison of Saved Fraction on synthetic datasets SN3, SN4 and SN5 with Highest Degree M when the propagation probability is 0.1 (left) and 0.3 (right)

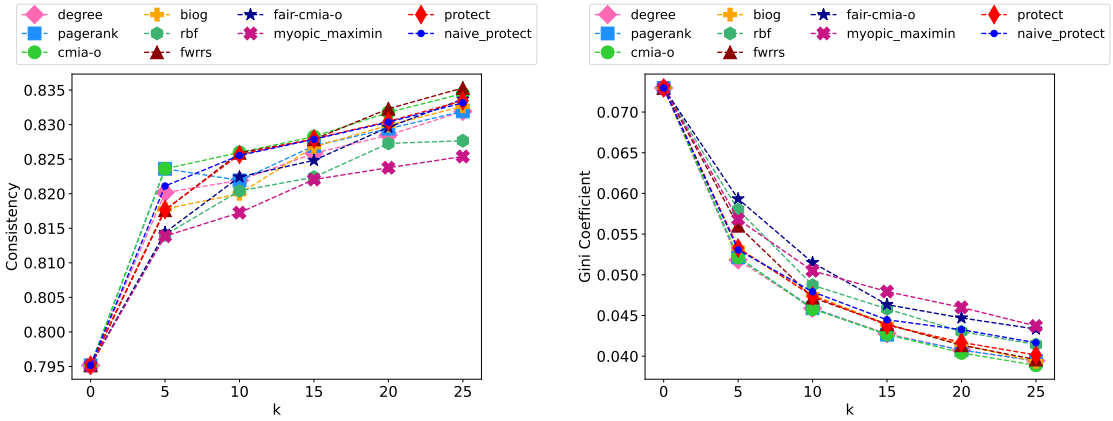
Naive PROTECT has a higher performance than all other baseline methods. This relatively reduced performance of other methods is even more evident in SN5. We observe that as the average shortest path value reduces, this disparity increases.

Comparing the saved fraction of these networks (see Figure 7.24), we see that Myopic underperforms in all networks as expected, as this heuristic does not aim at maximizing the number of saved nodes. In SN3, fairness-agnostic methods achieve a higher saved fraction than PROTECT. As we observed, in this method, higher degree nodes have a low tendency to connect to each other. As a result, picking these nodes gives good coverage of the network. Naive PROTECT also tends to pick the nodes with highest influence. This may be the reason for it achieving a higher Saved Fraction value than PROTECT. Fairness-agnostic methods, which select nodes with the most influence (highest degree), are successful in effectively saving nodes in this setting. On the other hand, PROTECT selects nodes that can reach the most vulnerable nodes, which may be suboptimal in terms of maximizing the number of nodes saved. In SN3, Naive PROTECT surpasses the saved fraction of PROTECT, consistently observed when the average shortest paths of the networks are long. Most methods achieve similar saved fractions for SN4 and SN5. Despite strong homophily, low clustering coefficient seems to impact the performance of group fairness method Fair CMIA-O as it achieves lower saved fraction in SN4.



(a) Evaluation on Consistency on SN1, with varying k values

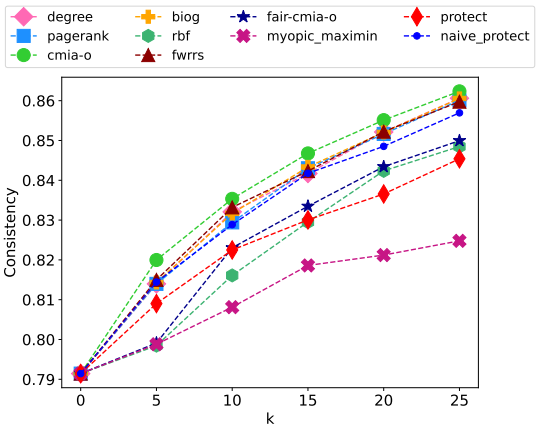
(b) Evaluation on Gini coefficient on SN1, with varying k values



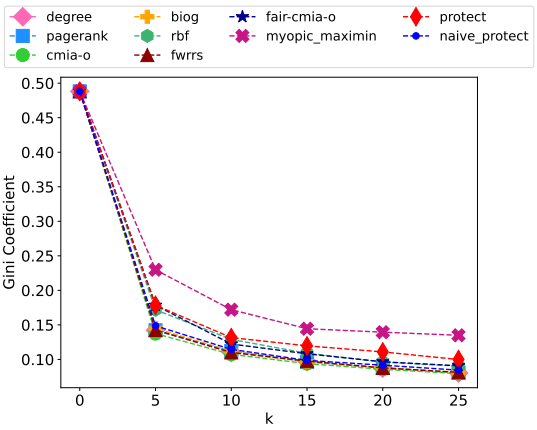
(c) Evaluation on Consistency on SN2, with varying k values

(d) Evaluation on Gini coefficient on SN2, with varying k values

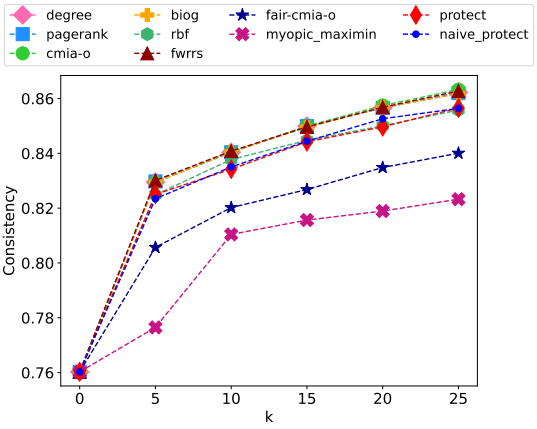
Figure 7.25: Comparison of Consistency and Gini coefficient on synthetic datasets SN1 and SN2 with Highest Degree M when the propagation probability is 0.3



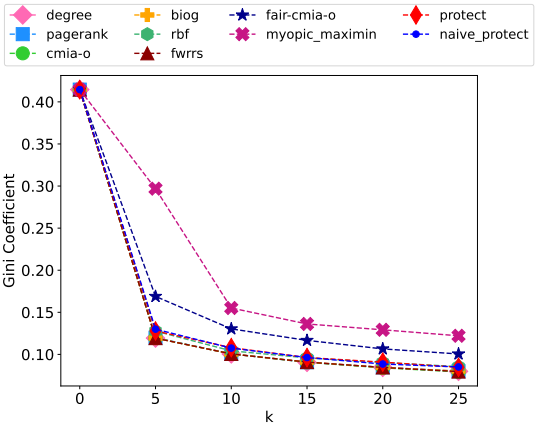
(a) Evaluation on Consistency on SN3, with varying k values



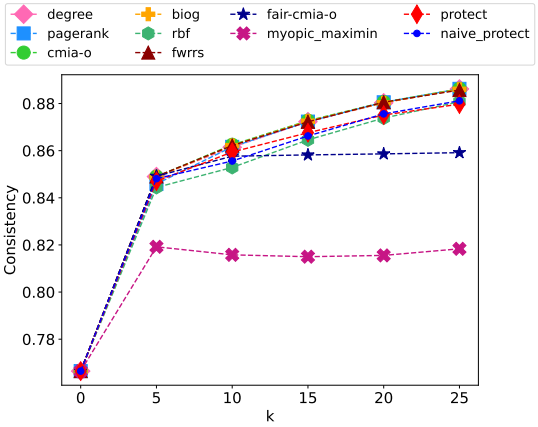
(b) Evaluation on Gini coefficient on SN3, with varying k values



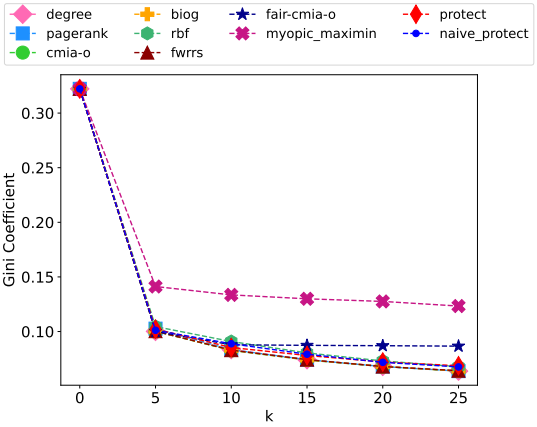
(c) Evaluation on Consistency on SN4, with varying k values



(d) Evaluation on Gini coefficient on SN4, with varying k values



(e) Evaluation on Consistency on SN5, with varying k values



(f) Evaluation on Gini coefficient on SN5, with varying k values

Figure 7.26: Comparison of Consistency and Gini coefficient on synthetic datasets SN3, SN4 and SN5 with Highest Degree M when the propagation probability is 0.3

Figure 7.25 and Figure 7.26 presents the Consistency and Gini coefficient obtained on the synthetic networks when α is 0.3. As previously noted, methods with higher saved fractions tend to exhibit higher consistency. Thus we observe that Myopic Maximin method has a low consistency in all networks. This shows that maximizing the number of saved nodes is important to achieve equity fairness and lower inequality. All the remaining methods achieve similar consistency and Gini coefficient in all networks except SN3.

In SN3, Naive Protect has a higher Saved Fraction value than PROTECT. Thus, it also has a higher consistency. The Gini coefficient is also slightly higher for PROTECT than the average trend in this method. For SN3, we noted earlier that the difference in Maximin sp between PROTECT and other fairness-agnostic methods such as CMIA-O minimal. This network has low average degree, small diameter, longer shortest paths and low positive degree assortativity. However, in other settings with a higher diameter (SN2) or lower average shortest paths (SN4, SN5, Arenas Email), the saved fraction may be lower, but the Maximin sp achieved by our method is several folds better than fairness-agnostic methods. Thus, we see that low preferential attachment has an impact on performance. Overall, our proposed heuristics are fairer in terms of Consistency and Gini coefficient than the individual fairness baseline in all cases.

In summary, our PROTECT heuristic performs well in all networks. In most scenarios, it guarantees a minimum save probability for all nodes, which is several folds better than what the baselines promise. However, in networks with specific combinations of characteristics, its advantage over fairness-agnostic methods is less significant. Nevertheless, the fairness-efficiency balance of our approach seems justified.

Chapter 8

Conclusions

Our research aimed to address the challenge of achieving feature-blind fairness in negative influence minimization, with a specific focus on individual fairness. A central objective was to define individual fairness within the context of negative influence minimization, leading us to formulate the novel Maximin protection problem as a means to achieve fairness. Furthermore, we encountered the challenge of efficiently solving this problem, which is both NP-hard and non-submodular. In the following section, we present a concise overview of our significant findings and contributions. Additionally, we reflect on the limitations of our work and propose potential avenues for future research.

8.1 Main Contributions

In our research, we have addressed the central research question of achieving fairness in NIM. To answer this question, we conducted an extensive study exploring different fairness notions and definitions, which provided us with a comprehensive understanding of the fairness challenges associated with NIM.

After careful consideration, we proposed an individual fairness objective based on social welfare, where the utility is the save probability (sp) and the welfare function is minimum ($\mu_{-\infty}$). This objective identifies truth starters that maximize the minimum chance of being shielded from misinformation for all nodes in the network, and we refer to this as the Maximin Protection problem. This also allowed us to utilize the minimum value as a comparative measure of fairness across different methods. By formulating this objective, we provided a practical and effective approach to achieving individual fairness in NIM and evaluating the fairness outcomes of various algorithms.

In our research, we established the relationship between two objectives: maximizing the average save probability and maximizing the number of saved nodes. This finding clarifies the inherent connection between the maximization of saved fraction and various fairness notions. Empirically, we observed that prioritizing the number of saved nodes is crucial for achieving equity fairness and reducing inequality. This finding deepens our understanding of the relationship between equity fairness and the fairness-agnostic objective of NIM.

Furthermore, in addition to defining individual fairness and exploring fairness objectives, we introduced specific metrics to measure Equity fairness and Inequality within the context of NIM. We devised a similarity metric to compare nodes and adapted fairness metrics such as Consistency and the Gini coefficient for the NIM problem. These metrics provide valuable insights into the fairness properties of different methods, facilitating a more comprehensive evaluation of fairness outcomes.

By proving the NP-hardness and non-submodularity of the Maximin Protection problem, we revealed the computational challenges involved in this problem. We demonstrated the effectiveness of utilizing shortest path information to improve the save probability of nodes. To compute the

shortest distances efficiently, we utilized a matrix-based method, which significantly improves the scalability and computational efficiency of solving the Maximin Protection problem. This approach facilitates the practical application of our proposed fairness-aware methods in real-world scenarios.

To provide practical solutions for fairness-aware negative influence minimization, we introduced two heuristic algorithms: PROTECT and Naive PROTECT. These heuristics address the Maximin Protection problem and achieve individual fairness in NIM. They offer efficient and effective approaches for mitigating the spread of misinformation while ensuring that no node is left behind.

To evaluate the performance of our methods, we conducted extensive experiments on real-world and synthetic datasets, comparing our several other baselines. The results consistently demonstrated that our PROTECT heuristic outperforms other methods in terms of the Maximin fairness objective. These findings validate the effectiveness of our proposed methods in achieving individual fairness in negative influence minimization scenarios.

Through our extensive experiments, we gained valuable insights into the factors that impact the performance of our methods. Notably, when testing on larger real-world datasets, our approach, PROTECT, outperformed other baselines by several folds in terms of Maximin sp. These compelling results underscore the effectiveness of our approach in effectively countering the spread of misinformation in social networks. Additionally, we identified certain network conditions where there was no significant improvement in Maximin sp compared to fairness-agnostic methods that achieved higher saved fractions.

Specifically, we observed that as the average shortest path increased, PROTECT maintained a higher Maximin sp, albeit with a diminishing difference compared to the fairness-agnostic method (CMIA-O). In such scenarios, CMIA-O tended to achieve a higher saved fraction. However, it is important to note that social networks commonly exhibit shorter average shortest paths. Consequently, our findings suggest that our method is well-suited for effectively addressing the dissemination of fake news and misinformation within social networks, aligning perfectly with the problem we aimed to solve.

By identifying the network characteristics that influence the performance of our method, we gain a better understanding of its adaptability and suitability for real-world scenarios. Our results highlight the significant potential of our method to mitigate negative influence, promote fairness, and ultimately contribute to a healthier information dissemination within social networks.

Moreover, the fairness objective we have developed is feature-blind, which is a unique contribution to the field of Social Network Analysis (SNA) problems. Feature-blind fairness is a relatively rare approach in the context of SNA, where the focus is often on utilizing specific features or attributes to achieve fairness objectives. This novel approach expands the scope of fairness considerations in SNA problems and opens up new possibilities for addressing fairness concerns in various social network contexts.

Overall, our research contributes novel insights and practical solutions to the challenge of fairness-aware NIM. By delving into fairness notions, formulating the Maximin Protection problem, and developing efficient algorithms, we pave the way for further advancements in fair negative influence minimization and social network analysis. Furthermore, by presenting a feature-blind fairness solution for an SNA problem, we hope to motivate Feature-Blind fairness approaches for other social network problems as well.

8.2 Limitations and Future Work

In this section, we acknowledge the limitations of our research and outline potential directions for future work.

First and foremost, our current approach relies on Monte-Carlo simulations to estimate the spread of influence. However, for larger networks, this approach can be computationally intensive. Exploring more efficient methods, particularly for identifying nodes with the minimum save probability in each iteration, would significantly enhance the scalability of our approach.

Secondly, we observed a cost associated with fairness, albeit a small one in certain experiments. Exploring ways to minimize this cost and incorporating a trade-off parameter to control it would be

a valuable direction for future work. Furthermore, our observations indicate that as the average shortest path in the network increases, the performance of our method in terms of Maximin sp maintains a higher value, although the difference compared to the fairness-agnostic method diminishes. It would be worthwhile to focus on enhancing the performance of our method in such networks as well.

Moreover, for this study, we selected the Campaign Oblivious Independent Cascade model with Uniform Cascade as the diffusion model. While this choice allowed us to evaluate our methods effectively, exploring their performance in other popular propagation models, such as the Linear Threshold Model and the Susceptible Infected Recovered/Removed (SIR) model, would provide a more comprehensive evaluation across different diffusion scenarios.

Furthermore, the Uniform Cascade Model assumes that all nodes spread information at the same rate, disregarding the nuanced dynamics observed in real-world information diffusion. Extending our study to other models that better capture the actual propagation dynamics of networks would be a valuable extension of our work.

Additionally, it would be worthwhile to consider variants of the NIM problem that incorporate temporal and variable budget considerations. By incorporating temporal constraints, we can address the urgency of minimizing negative influence within specific timeframes, which is relevant in many real-world applications. Moreover, in practice, the cost associated with each node may not be uniform. Thus, incorporating variable costs would also be a valuable extension to consider.

In future work, an interesting direction would be to explore is a feature-aware fairness solution. This involves considering relevant parameters, such as user demographics, interests, and contextual factors, to compute each individual's vulnerability to specific misinformation. For instance, in the context of misinformation related to an election campaign, factors like geographical location, age, and political affiliation can be taken into account to develop targeted mitigation strategies.

Addressing these limitations and pursuing the avenues for future work outlined above would contribute to a more comprehensive understanding of fairness in NIM and open up practical applications for mitigating the spread of negative influence in social networks.

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Appendix A

Shortest Distance Computation

To calculate the shortest distance between all pairs of nodes in the network, we utilize the concepts of paths and walks within a graph. A path represents a subgraph consisting of a sequence of nodes and edges, with the length of a path defined as the number of edges it contains. A simple path is a path where no nodes are repeated, while a simple cycle is a simple path where the first and last nodes are the same.

Given an adjacency matrix A that represents a directed graph G, with $A_{ij} = 1$ indicating the existence of an edge from node i to node j in G, we can determine the number of walks from node i to node j using the entry A_{ij} [42]. This allows us to compute the shortest path between two nodes in the graph, which is equivalent to the shortest walk.

Our objective is to find the length of the shortest path between all pairs of nodes in the network, up to a length not greater than the length of the longest shortest path from any node in the set of misinformation starters M to any other node in the network. The resulting solution will be stored in a matrix D. Algorithm 6 outlines the steps.

Algorithm 6: Shortest Distance Computation

input : $G(V, E), M$
output: D
$A \leftarrow$ Adjacency Matrix of G
$A[i][i] \leftarrow 0 \forall i \in V$
$A[i][m] \leftarrow 0 \forall i \in V, m \in M$
Initialize $D[i][j] \leftarrow 0 \forall i, j \in V$
Initialize $d \leftarrow 1$
Initialize $Q \leftarrow A$
Initialize $D_M[i] \leftarrow 0 \forall i \in V \setminus M$
while $D_M[i] \neq 0 \forall i \in V$ **do**
  if $d \neq 1$ **then**
    $Q \leftarrow Q \times A$
    $Q[i][i] \leftarrow 0 \forall i \in V$
  end
  $D[i][j] \leftarrow 1 if D[i][j] = 0 or Q[i][j] > 0$
  $D_M[i] \leftarrow D[m][i] \forall i \in V, m \in M$
end
Return D

To begin, we set the diagonal terms of the adjacency matrix to 0, as we are only interested in simple paths and do not consider self-loops. Additionally, we are not interested in paths that pass through nodes in M, so we remove all incoming edges to M from the adjacency matrix before starting the computation. This modified matrix provides the number of simple paths of length 1

between every pair of nodes in the network.

We then raise the adjacency matrix to higher powers, recording the length of the shortest path between each pair of nodes in D. The algorithm terminates when at least one path is found from any $m \in M$ to all $v \in V \setminus M$. Thus, the maximum number of iterations the algorithm would execute for is the diameter of the network.

In summary, the process of computing the shortest distance between all pairs of nodes involves iteratively computing the powers of the adjacency matrix A and updating the distance matrix D until all the shortest distances are obtained. This algorithm efficiently determines the lengths of the shortest paths in the network, which is crucial for our proposed solution.